Register Specification for ATMx2xx-x1x

Revision History

Date	Version	Description
March 29, 2021	0.50	For x1x release version.
April 14, 2021	0.51	Updated format, no content change.
June 30, 2021	0.52	Update release

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- TIMER0 –
  - CTRL
  - VALUE
  - RELOAD
  - INTSTATUS_CLEAR
  - PID4
  - PID5
  - PID6
  - PID7
  - PID0
  - PID1
  - PID2
  - PID3
  - CID0
  - CID1
  - CID2
  - CID3
- TIMER1 –
  - CTRL
  - VALUE
  - RELOAD
  - INTSTATUS_CLEAR
  - PID4
  - PID5
  - PID6
  - PID7
  - PID0
  - PID1
  - PID2
  - PID3
  - CID0
  - CID1
  - CID2
  - CID3
- DUALTIMER2 –
  - TIMER1LOAD
  - TIMER1VALUE
  - TIMER1CONTROL
  - TIMER1INTCLR
  - TIMER1RIS
  - TIMER1MIS
  - TIMER1BGLOAD
  - TIMER2LOAD
  - TIMER2VALUE
  - TIMER2CONTROL
  - TIMER2INTCLR
  - TIMER2RIS
  - TIMER2MIS
  - TIMER2BGLOAD
  - TIMERITCR
  - TIMERITOP

- TIMERPERIPHID4
- TIMERPERIPHID5
- TIMERPERIPHID6
- TIMERPERIPHID7
- TIMERPERIPHID0
- TIMERPERIPHID1
- TIMERPERIPHID2
- TIMERPERIPHID3
- TIMERPCELLID0
- TIMERPCELLID1
- TIMERPCELLID2
- TIMERPCELLID3

- WRPR –
  - APB0_CTRL
  - APB1_CTRL
  - APB2_CTRL
  - APB3_CTRL
  - APB4_CTRL
  - APB5_CTRL
  - APB6_CTRL
  - APB7_CTRL
  - APB8_CTRL
  - APB9_CTRL
  - APB10_CTRL
  - APB11_CTRL
  - APB12_CTRL
  - APB13_CTRL
  - APB14_CTRL
  - APB15_CTRL
  - PIN_SELECTION_A
  - PIN_SELECTION_B
  - PIN_SELECTION_C
  - PIN_SELECTION_D
  - PIN_SELECTION_E
  - PIN_SELECTION_F
  - PIN_PU_A
  - PIN_PU_B
  - PIN_PU_C
  - PIN_PU_D
  - PIN_PU_E
  - INTRPT_CFG_0
  - INTRPT_CFG_1
  - INTRPT_CFG_2
  - INTRPT_CFG_3
  - INTRPT_CFG_4
  - INTRPT_CFG_5
  - INTRPT_CFG_6
  - INTRPT_CFG_7
  - INTRPT_CFG_8
  - INTRPT_CFG_9
  - INTRPT_CFG_10
  - INTRPT_CFG_11
  - INTRPT_CFG_12
  - INTRPT_CFG_13
  - INTRPT_CFG_14
  - INTRPT_CFG_15
  - REMAP
  - AHB_INTERPOSERS
  - CHIPID1
  - CHIPID2
  - CHIPREV
  - CORE_ID
- UART0 –
  - DATA
  - STATE

CONFIDENTIAL

- CTRL
- INTSTATUS_CLEAR
- BAUDDIV
- RX_LWM
- TX_LWM
- RX_FIFO_SPACES
- TX_FIFO_SPACES
- HW_FLOW_OVRD
- PID4
- PID5
- PID6
- PID7
- PID0
- PID1
- PID2
- PID3
- CID0
- CID1
- CID2
- CID3

- UART1 −
  - DATA
  - STATE
  - CTRL
  - INTSTATUS_CLEAR
  - BAUDDIV
  - RX_LWM
  - TX_LWM
  - RX_FIFO_SPACES
  - TX_FIFO_SPACES
  - HW_FLOW_OVRD
  - PID4
  - PID5
  - PID6
  - PID7
  - PID0
  - PID1
  - PID2
  - PID3
  - CID0
  - CID1
  - CID2
  - CID3
- PWM −
  - PWM0_CTRL
  - PWM1_CTRL
  - PWM2_CTRL
  - PWM3_CTRL
  - PWM4_CTRL
  - PWM5_CTRL
  - PWM6_CTRL
  - PWM7_CTRL
  - CORE_ID
- SPI0 −
  - TRANSACTION_SETUP
  - TRANSACTION_STATUS
  - DATA_BYTES_LOWER
  - DATA_BYTES_UPPER
  - INTERRUPT_MASK
  - INTERRUPT_STATUS
  - SET_INTERRUPT
  - RESET_INTERRUPT
  - CORE_ID
- WATCHDOG −
  - WDOGLOAD

CONFIDENTIAL

- WDOGVALUE
- WDOGCONTROL
- WDOGINTCLR
- WDOGRIS
- WDOGMIS
- WDOGLOCK
- WDOGITCR
- WDOGITOP
- WDOGPERIPHID4
- WDOGPERIPHID5
- WDOGPERIPHID6
- WDOGPERIPHID7
- WDOGPERIPHID0
- WDOGPERIPHID1
- WDOGPERIPHID2
- WDOGPERIPHID3
- WDOGPCELLID0
- WDOGPCELLID1
- WDOGPCELLID2
- WDOGPCELLID3

- KSM –
  - CTRL0
  - TIME_PARAM0
  - TIME_PARAM1
  - MATRIX_SIZE
  - KEYBOARD_PACKET
  - STATUS
  - INTERRUPTS
  - INTERRUPT_MASK
  - INTERRUPT_CLEAR
  - CORE_ID
- SPI2 –
  - TRANSACTION_SETUP
  - TRANSACTION_STATUS
  - DATA_BYTES_LOWER
  - DATA_BYTES_UPPER
  - INTERRUPT_MASK
  - INTERRUPT_STATUS
  - SET_INTERRUPT
  - RESET_INTERRUPT
  - CORE_ID
- QDEC –
  - SETUP_FOR_AXIS_0
  - ACCUM_PURGE_STATUS_FOR_AXIS_0
  - ACCUMULATOR_VALUE_FOR_AXIS_0
  - ERR_PURGE_STATUS_FOR_AXIS_0
  - ERROR_COUNT_FOR_AXIS_0
  - SETUP_FOR_AXIS_1
  - ACCUM_PURGE_STATUS_FOR_AXIS_1
  - ACCUMULATOR_VALUE_FOR_AXIS_1
  - ERR_PURGE_STATUS_FOR_AXIS_1
  - ERROR_COUNT_FOR_AXIS_1
  - SETUP_FOR_AXIS_2
  - ACCUM_PURGE_STATUS_FOR_AXIS_2
  - ACCUMULATOR_VALUE_FOR_AXIS_2
  - ERR_PURGE_STATUS_FOR_AXIS_2
  - ERROR_COUNT_FOR_AXIS_2
  - CORE_ID
- SPI1 –
  - TRANSACTION_SETUP
  - TRANSACTION_STATUS
  - DATA_BYTES_LOWER
  - DATA_BYTES_UPPER
  - INTERRUPT_MASK
  - INTERRUPT_STATUS

CONFIDENTIAL

    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- SLWTIMER –
    - CONTROL
    - INIT_LOW
    - INIT_HIGH
    - CNT_LOW
    - CNT_HIGH
    - THRES0_LOW
    - THRES0_HIGH
    - THRES1_LOW
    - THRES1_HIGH
    - THRES2_LOW
    - THRES2_HIGH
    - STATUS
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - CORE_ID
- QSPI –
    - TRANSACTION_SETUP
    - READ_DATA
    - OVERRIDE_DIN
    - MODE
    - REMOTE_AHB_SETUP
    - REMOTE_AHB_SETUP_2
    - REMOTE_AHB_SETUP_3
    - REMOTE_AHB_SETUP_4
    - REMOTE_AHB_SETUP_5
    - REMOTE_AHB_SETUP_6
    - CORE_ID
- GPIO0 –
    - DATA
    - DATAOUT
    - PULLUP_ENABLE_SET
    - PULLUP_ENABLE_CLR
    - INENABLE_SET
    - INENABLE_CLR
    - OUTENSET
    - OUTENCLR
    - ALTFUNCSET
    - ALTFUNCCLR
    - INTENSET
    - INTENCLR
    - INTTYPESET
    - INTTYPECLR
    - INTPOLSET
    - INTPOLCLR
    - INTSTATUS_CLEAR
    - MASKLOWBYTE
    - MASKHIGHBYTE
- GPIO1 –
    - DATA
    - DATAOUT
    - PULLUP_ENABLE_SET
    - PULLUP_ENABLE_CLR
    - INENABLE_SET
    - INENABLE_CLR
    - OUTENSET
    - OUTENCLR
    - ALTFUNCSET
    - ALTFUNCCLR
    - INTENSET
    - INTENCLR
    - INTTYPESET

CONFIDENTIAL

    - INTTYPECLR
    - INTPOLSET
    - INTPOLCLR
    - INTSTATUS_CLEAR
    - MASKLOWBYTE
    - MASKHIGHBYTE
- SYSCTRL −
    - MEM_CTRL
    - SYS_CTRL
    - RESET_INFORMATION
- I2C0 −
    - TRANSACTION_SETUP
    - CLOCK_CONTROL
    - OUTGOING_DATA
    - IO_CTRL
    - INCOMING_DATA
    - TRANSACTION_STATUS
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - ID
- I2C1 −
    - TRANSACTION_SETUP
    - CLOCK_CONTROL
    - OUTGOING_DATA
    - IO_CTRL
    - INCOMING_DATA
    - TRANSACTION_STATUS
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - ID
- NVM −
    - OPMODE
    - ADDRESS
    - STATUS
    - READ_DATA
    - T_SP_PG_AVDD
    - T_SP_PGM
    - T_PGM
    - T_HP_PGM
    - T_HP_PG_AVDD
    - T_SR_RD
    - T_RD
    - T_HR_RD
    - T_SETUP
    - T_HOLD
    - T_ADDR
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - EFUSE_AUTOREAD
    - ID
- WRPR1 −
    - APB0_CTRL
    - APB1_CTRL
    - APB2_CTRL
    - APB3_CTRL
    - APB4_CTRL
    - APB5_CTRL
    - APB6_CTRL
    - APB7_CTRL

CONFIDENTIAL

- APB8_CTRL
- APB9_CTRL
- APB10_CTRL
- APB11_CTRL
- APB12_CTRL
- APB13_CTRL
- APB14_CTRL
- APB15_CTRL
- CORE_ID

- RADIO –
  - TRANSACTION_SETUP
  - TRANSACTION_STATUS
  - DATA_BYTES_LOWER
  - DATA_BYTES_UPPER
  - INTERRUPT_MASK
  - INTERRUPT_STATUS
  - SET_INTERRUPT
  - RESET_INTERRUPT
  - CORE_ID
- PMU –
  - TRANSACTION_SETUP
  - TRANSACTION_STATUS
  - DATA_BYTES_LOWER
  - DATA_BYTES_UPPER
  - INTERRUPT_MASK
  - INTERRUPT_STATUS
  - SET_INTERRUPT
  - RESET_INTERRUPT
  - CORE_ID
- PSEQ –
  - CTRL0
  - RETAIN_ALL_WAKE_MASK
  - HIB_WAKE_MASK
  - GPIO_WAKE_MASK
  - GPIO_WAKE_TYPE
  - GPIO_WAKE_POL
  - GPIO_WAKE_BOTH_EDGES
  - WURX_CONFIG
  - GADC_CONFIG
  - COUNT0
  - COUNT1
  - COUNT2
  - COUNT3
  - COUNT4
  - COUNT5
  - COUNT6
  - COUNT7
  - COUNT8
  - COUNT9
  - COUNT0_RADIO
  - COUNT1_RADIO
  - COUNT2_RADIO
  - COUNT3_RADIO
  - COUNT4_RADIO
  - COUNT5_RADIO
  - COUNT6_RADIO
  - COUNT7_RADIO
  - COUNT8_RADIO
  - COUNT9_RADIO
  - XTAL_BITS0
  - XTAL_BITS1
  - OVERRIDES
  - OVERRIDES2
  - OVERRIDES3
  - OVERRIDES4

CONFIDENTIAL

- OVERRIDES5
- SYSRAM_OVERRIDES
- SYSRAM_OVERRIDES2
- SYSRAM_OVERRIDES3
- SYSRAM_OVERRIDES4
- SYSRAM_OVERRIDES5
- SYSRAM_OVERRIDES6
- SYSRAM_OVERRIDES7
- SYSRAM_OVERRIDES8
- EMRAM_OVERRIDES
- EMRAM_OVERRIDES2
- EMRAM_OVERRIDES3
- EMRAM_OVERRIDES4
- EMRAM_OVERRIDES5
- EMRAM_OVERRIDES6
- EMRAM_OVERRIDES7
- EMRAM_OVERRIDES8
- COUNTER_CONTROL
- CURRENT_REAL_TIME
- CURRENT_COUNT_DOWN_TIME
- INIT_COUNT_DOWN
- INST_PENDING
- STATUS
- PERSISTENT0
- SENSOR_HUB_CONTROL
- KSMQDEC_CONTROL
- PMU_STATUS
- FLASH_CONTROL
- PMU_INTERRUPT
- INTERRUPT_STATUS
- INTERRUPT_MASK
- RESET_INTERRUPT
- CORE_ID

- GADC –
  - CTRL
  - DATAPATH_PARAMETERS
  - DATAPATH_OUTPUT
  - FINAL_INVERSION
  - INTERRUPTS
  - INTERRUPT_MASK
  - INTERRUPT_CLEAR
  - CORE_ID
- TRNG –
  - CONTROL
  - STATUS
  - INTERRUPT_STATUS
  - INTERRUPT_MASK
  - SET_INTERRUPT
  - RESET_INTERRUPT
  - TRNG
  - CORE_ID
- RCOS_CAL –
  - CAL_CTRL
  - SLOW_CLK_SET
  - SLOW_CLK_CNT
  - FAST_CLK_CNT
  - STAT
  - CORE_ID
- SHUB –
  - SETUP0
  - SETUP1
  - PORT0_SETUP
  - PORT0_SENSOR0_SETUP
  - PORT0_SENSOR1_SETUP
  - PORT0_SENSOR2_SETUP

CONFIDENTIAL

- PORT0_SENSOR3_SETUP
- PORT0_SENSOR4_SETUP
- PORT0_SENSOR5_SETUP
- PORT0_SENSOR6_SETUP
- PORT0_SENSOR7_SETUP
- PORT0_SPI_ENG0_CTRL
- PORT0_I2C_IDW_0
- PORT0_I2C_IDW_1
- PORT0_I2C_IDW_2
- PORT0_I2C_IDW_3
- PORT0_I2C_IDW_4
- PORT0_I2C_IDW_5
- PORT0_I2C_IDW_6
- PORT0_I2C_IDW_7
- PORT0_I2C_ADDR_0
- PORT0_I2C_ADDR_1
- PORT0_I2C_ADDR_2
- PORT0_I2C_ADDR_3
- PORT0_I2C_ADDR_4
- PORT0_I2C_ADDR_5
- PORT0_I2C_ADDR_6
- PORT0_I2C_ADDR_7
- PORT0_I2C_IDR_0
- PORT0_I2C_IDR_1
- PORT0_I2C_IDR_2
- PORT0_I2C_IDR_3
- PORT0_I2C_IDR_4
- PORT0_I2C_IDR_5
- PORT0_I2C_IDR_6
- PORT0_I2C_IDR_7
- PORT0_I2C_DAT_CTRL
- PORT0_I2C_DAT_LAST_CTRL
- PORT0_ALM0_QUAN_CTRL
- PORT0_ALM0_TRIG_CTRL
- PORT0_ALM0_THRHLD_MAX_0
- PORT0_ALM0_THRHLD_MAX_1
- PORT0_ALM0_THRHLD_MAX_2
- PORT0_ALM0_THRHLD_MIN_0
- PORT0_ALM0_THRHLD_MIN_1
- PORT0_ALM0_THRHLD_MIN_2
- PORT0_STATUS
- PORT1_SETUP
- PORT1_SENSOR0_SETUP
- PORT1_SPI_ENG0_CTRL
- PORT1_I2C_IDW_0
- PORT1_I2C_ADDR_0
- PORT1_I2C_IDR_0
- PORT1_I2C_DAT_CTRL
- PORT1_I2C_DAT_LAST_CTRL
- PORT1_ALM0_QUAN_CTRL
- PORT1_ALM0_TRIG_CTRL
- PORT1_ALM0_THRHLD_MAX_0
- PORT1_ALM0_THRHLD_MAX_1
- PORT1_ALM0_THRHLD_MAX_2
- PORT1_ALM0_THRHLD_MIN_0
- PORT1_ALM0_THRHLD_MIN_1
- PORT1_ALM0_THRHLD_MIN_2
- PORT1_STATUS
- FLASH_CTRL0
- FLASH_CTRL1
- FLASH_CTRL2
- FLASH_CTRL3
- FLASH_CTRL4
- FLASH_STATUS0
- FLASH_STATUS1

CONFIDENTIAL

    - SHUB_STATUS0
    - ID
- PDM −
    - CONTROL
    - IIR_CONFIG
    - GAIN_CONTROL_CONFIG
    - PCM_SAMPLE
    - INTERRUPTS
    - INTERRUPT_MASK
    - INTERRUPT_CLEAR
    - CORE_ID
- AT_DMA −
    - OPMODE
    - CONST_WDATA
    - SRC_ADDR
    - TAR_ADDR
    - SIZE
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - ERR_STAT
    - ID
- PMU_TOP −
    - PMU0
    - PMU1
    - PMU11
    - CORE_ID
- PMU_WURX −
    - WURX_GENERAL
    - WURX0_0
    - WURX0_1
    - WURX0_2
    - WURX0_3
    - WURX0_4
    - WURX1_0
    - WURX1_1
    - WURX1_RB
    - WURX_SPARE

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TIMER0

Name CTRL
Address 0x40000000
Description timer register
Reset 0x00000000

Name	RSVD	timer_int_en	ext_in_clk	ext_in_en	timer_en
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R/W	timer_int_en	Timer interrupt enable.	1'b0
2	R/W	ext_in_clk	Select external input as clock.	1'b0

1	R/W	ext_in_en	Select external input as enable.	1'b0
0	R/W	timer_en	Enable.	1'b0

Back to top

Name VALUE
Address 0x40000004
Description timer register
Reset 0x00000000

Name	value																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	value	current value of timer.	32'b0

Back to top

Name RELOAD
Address 0x40000008
Description timer register
Reset 0x00000000

Name	reload																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	reload	Reload value. A write to this register sets the current value.	32'b0

Back to top

Name INTSTATUS_CLEAR
Address 0x4000000c
Description timer register
Reset 0x00000000

Name	RSVD																													intstatus_clear
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	intstatus_clear	Timer interupt. Write one to clear.	1'b0

Back to top

CONFIDENTIAL

Name PID4
Address 0x40000fd0
Description timer register
Reset 0x00000004

Name	RSVD	block_count	jep106_c_code
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4	3 2 1 0

Bits	SW	Name	Description	Reset
7:4	R	block_count	Block count	4'b0
3:0	R	jep106_c_code	jep106_c_code	4'd4

Back to top

Name PID5
Address 0x40000fd4
Description timer register
Reset 0x00000000

Name	RSVD	pid5
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
7:0	R	pid5	Peripheral ID Register 5.	8'b0

Back to top

Name PID6
Address 0x40000fd8
Description timer register
Reset 0x00000000

Name	RSVD	pid6
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
7:0	R	pid6	Peripheral ID Register 6.	8'b0

Back to top

Name PID7
Address 0x40000fdc
Description timer register
Reset 0x00000000

Name	R	p

CONFIDENTIAL

	RSVD	id7
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
7:0	R	pid7	Peripheral ID Register 7.	8'b0

Back to top

Name PID0
Address 0x40000fe0
Description timer register
Reset 0x00000022

Name	RSVD	part_num
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
7:0	R	part_num	Part number[7:0].	8'h22

Back to top

Name PID1
Address 0x40000fe4
Description timer register
Reset 0x000000b8

Name	RSVD	jep106_id_3_0	part_num
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4	3 2 1 0

Bits	SW	Name	Description	Reset
7:4	R	jep106_id_3_0	jep106_id_3_0	4'hb
3:0	R	part_num	Part number[11:8].	4'h8

Back to top

Name PID2
Address 0x40000fe8
Description timer register
Reset 0x0000001b

Name	RSVD	rev	jedec_	jep106

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
																													used	id_6_4		

Bits	SW	Name	Description	Reset
7:4	R	rev	Revision	4'h1
3	R	jedec_used	jedec_used	1'h1
2:0	R	jep106_id_6_4	jep106_id_6_4	3'h3

Back to top

Name PID3
Address 0x40000fec
Description timer register
Reset 0x00000000

Name	RSVD																								ecorevnum				mod_num			
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:4	R	ecorevnum	ECO revision number.	4'h0
3:0	R	mod_num	modification number.	4'h0

Back to top

Name CID0
Address 0x40000ff0
Description timer register
Reset 0x0000000d

Name	RSVD																								cid0							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	cid0	Component ID Register 0	8'h0d

Back to top

Name CID1
Address 0x40000ff4
Description timer register
Reset 0x000000f0

Name	RSVD																								cid1							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

CONFIDENTIAL

Bits	SW	Name	Description	Reset
7:0	R	cid1	Component ID Register 1	8'hf0

Back to top

Name CID2
Address 0x40000ff8
Description timer register
Reset 0x00000005

Name	RSVD																						cid2							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid2	Component ID Register 2	8'h05

Back to top

Name CID3
Address 0x40000ffc
Description timer register
Reset 0x000000b1

Name	RSVD																						cid3							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid3	Component ID Register 3	8'hb1

Back to top

TIMER1

Name CTRL
Address 0x40001000
Description timer register
Reset 0x00000000

Name	RSVD																										timer_int_en	ext_in_clk	ext_in_en	timer_en
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R/W	timer_int_en	Timer interrupt enable.	1'b0
2	R/W	ext_in_clk	Select external input as clock.	1'b0
1	R/W	ext_in_en	Select external input as enable.	1'b0
0	R/W	timer_en	Enable.	1'b0

CONFIDENTIAL

Back to top

Name VALUE
Address 0x40001004
Description timer register
Reset 0x00000000

Name	value																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	value	current value of timer.	32'b0

Back to top

Name RELOAD
Address 0x40001008
Description timer register
Reset 0x00000000

Name	reload																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	reload	Reload value. A write to this register sets the current value.	32'b0

Back to top

Name INTSTATUS_CLEAR
Address 0x4000100c
Description timer register
Reset 0x00000000

Name	RSVD																													intstatus_clear
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	intstatus_clear	Timer interupt. Write one to clear.	1'b0

Back to top

Name PID4

CONFIDENTIAL

Address 0x40001fd0
Description timer register
Reset 0x00000004

Name	RSVD																						block_count				jep106_c_code			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	block_count	Block count	4'b0
3:0	R	jep106_c_code	jep106_c_code	4'd4

Back to top

Name PID5
Address 0x40001fd4
Description timer register
Reset 0x00000000

Name	RSVD																								pid5							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	pid5	Peripheral ID Register 5.	8'b0

Back to top

Name PID6
Address 0x40001fd8
Description timer register
Reset 0x00000000

Name	RSVD																								pid6							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	pid6	Peripheral ID Register 6.	8'b0

Back to top

Name PID7
Address 0x40001fdc
Description timer register
Reset 0x00000000

Name	RS																								pi							

CONFIDENTIAL

	RSVD																						pid7							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	pid7	Peripheral ID Register 7.	8'b0

Back to top

Name PID0
Address 0x40001fe0
Description timer register
Reset 0x00000022

Name	RSVD																						part_num							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	part_num	Part number[7:0].	8'h22

Back to top

Name PID1
Address 0x40001fe4
Description timer register
Reset 0x000000b8

Name	RSVD																								jep106_id_3_0				part_num			
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:4	R	jep106_id_3_0	jep106_id_3_0	4'hb
3:0	R	part_num	Part number[11:8].	4'h8

Back to top

Name PID2
Address 0x40001fe8
Description timer register
Reset 0x0000001b

Name	RSVD																								rev				jedec_u	jep106_

CONFIDENTIAL

																											s e d		i d _ 6 _ 4	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	rev	Revision	4'h1
3	R	jedec_used	jedec_used	1'h1
2:0	R	jep106_id_6_4	jep106_id_6_4	3'h3

Back to top

Name PID3
Address 0x40001fec
Description timer register
Reset 0x00000000

Name	R S V D																								e c o r e v n u m				m o d _ n u m			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	ecorevnum	ECO revision number.	4'h0
3:0	R	mod_num	modification number.	4'h0

Back to top

Name CID0
Address 0x40001ff0
Description timer register
Reset 0x0000000d

Name	R S V D																								c i d 0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid0	Component ID Register 0	8'h0d

Back to top

Name CID1
Address 0x40001ff4
Description timer register
Reset 0x000000f0

Name	R S V D																								c i d 1							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

7:0	R	cid1	Component ID Register 1	8'hf0

Back to top

Name CID2
Address 0x40001ff8
Description timer register
Reset 0x00000005

Name	RSVD																						cid2							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid2	Component ID Register 2	8'h05

Back to top

Name CID3
Address 0x40001ffc
Description timer register
Reset 0x000000b1

Name	RSVD																						cid3							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid3	Component ID Register 3	8'hb1

Back to top

DUALTIMER2

Name TIMER1LOAD
Address 0x40002000
Description timer1 reload value
Reset 0x00000000

Name	reload																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	reload	The TIMERXLOAD Register contains the value from which the counter is to decrement. This is the value used to reload the counter when Periodic mode is enabled, and the current count reaches 0. When this register is written to directly, the current count is immediately reset to the new value at the next rising edge of timer clock that TIMCLKEN enables. The value in this register is also overwritten if the TIMERXBGLOAD Register is written to, but the current count is not immediately affected. If values are written to both the TIMERXLOAD and TIMERXBGLOAD registers before an enabled rising edge on timer clock, the following occurs: 1. On the next enabled timer clock edge, the value written to the TIMERXLOAD value replaces the current count value.	32'b0

CONFIDENTIAL

			2. Every time the counter reaches 0, the current count value is reset to the value written to TIMERXBGLOAD. Reading from the TIMERXLOAD Register at any time after the two writes have occurred retrieves the value written to TIMERXBGLOAD. That is, the value read from TIMERXLOAD is always the value that takes effect for Periodic mode after the next time the counter reaches 0.	

Back to top

Name TIMER1VALUE
Address 0x40002004
Description current value of timer1
Reset 0x00000000

Name	value																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	value	current value of timer.	32'b0

Back to top

Name TIMER1CONTROL
Address 0x40002008
Description timer1 control
Reset 0x00000020

Name	RSVD																								timclken	timer_mode	int_en	RSVD	TimerPre		Timer_Size	One_shot_Count
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7	R/W	timclken	Enable bit 0 Timer disabled, default. 1 Timer enabled.	1'b0
6	R/W	timer_mode	Mode bit 0 Timer is in free-running mode, default. 1 Timer is in periodic mode.	1'b0
5	R/W	int_en	Interrupt Enable bit 0 Timer interrupt disabled. 1 Timer interrupt enabled, default.	1'b1
3:2	R/W	TimerPre	Prescale bits 00 0 stages of prescale, clock is divided by 1, default. 01 4 stages of prescale, clock is divided by 16. 10 8 stages of prescale, clock is divided by 256. 11 undefined, do not use.	2'b0
1	R/W	Timer_Size	Selects 16 bit or 32-bit counter operation 0 16-bit counter, default. 1 32-bit counter.	1'b0
0	R/W	One_shot_Count	Selects one-shot or wrapping counter mode 0 Wrapping mode, default. 1 One-shot mode.	1'b0

CONFIDENTIAL

Back to top

Name TIMER1INTCLR
Address 0x4000200c
Description clear timer1 interrupt
Reset 0x00000000

Name	RSVD	timerlintclr
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	timer1intclr	Any write to the TIMERXINTCLR Register clears the interrupt output from the counter.	1'b0

Back to top

Name TIMER1RIS
Address 0x40002010
Description raw interrupt for timer1
Reset 0x00000000

Name	RSVD	timerlris
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R	timer1ris	The TIMERXRIS Register indicates the raw interrupt status from the counter. This value is ANDed with the timer interrupt enable bit from the Timer Control Register to create the masked interrupt, that is passed to the interrupt output pin	1'b0

Back to top

Name TIMER1MIS
Address 0x40002014
Description masked interrupt for timer1
Reset 0x00000000

Name	RSVD	timerlmis
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
0	R	timer1mis	The TIMERXMIS Register indicates the masked interrupt status from the counter. This value is the logical AND of the raw interrupt status with the timer interrupt enable bit from the Timer Control Register, and is the same value that is passed to the interrupt output pin.	1'b0

Back to top

Name TIMER1BGLOAD
Address 0x40002018
Description timer1 alternate reload value
Reset 0x00000000

Name	timer1bgload																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	timer1bgload	The TIMERXBGLOAD Register contains the value from which the counter is to decrement. This is the value used to reload the counter when Periodic mode is enabled, and the current count reaches 0. This register provides an alternative method of accessing the TIMERXLOAD Register. The difference is that writes to TIMERXBGLOAD do not cause the counter to immediately restart from the new value.	32'h0

Back to top

Name TIMER2LOAD
Address 0x40002020
Description timer2 reload value
Reset 0x00000000

Name	reload																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	reload	The TIMERXLOAD Register contains the value from which the counter is to decrement. This is the value used to reload the counter when Periodic mode is enabled, and the current count reaches 0. When this register is written to directly, the current count is immediately reset to the new value at the next rising edge of timer clock that TIMCLKEN enables. The value in this register is also overwritten if the TIMERXBGLOAD Register is written to, but the current count is not immediately affected. If values are written to both the TIMERXLOAD and TIMERXBGLOAD registers before an enabled rising edge on timer clock, the following occurs: 1. On the next enabled timer clock edge, the value written to the TIMERXLOAD value replaces the current count value. 2. Every time the counter reaches 0, the current count value is reset to the value written to TIMERXBGLOAD. Reading from the TIMERXLOAD Register at any time after the two writes have occurred retrieves the value written to TIMERXBGLOAD. That is, the value read from TIMERXLOAD is always the value that takes effect for Periodic mode after the next time the counter reaches 0.	32'b0

Back to top

CONFIDENTIAL

Name TIMER2VALUE
Address 0x40002024
Description current value of timer2
Reset 0x00000000

Name	value																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	value	current value of timer.	32'b0

Back to top

Name TIMER2CONTROL
Address 0x40002028
Description timer2 control
Reset 0x00000020

Name	RSVD																								timclken	timer_mode	int_en	RSVD	TimerPre		Timer_Size	One_shot_Count
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7	R/W	timclken	Enable bit 0 Timer disabled, default. 1 Timer enabled.	1'b0
6	R/W	timer_mode	Mode bit 0 Timer is in free-running mode, default. 1 Timer is in periodic mode.	1'b0
5	R/W	int_en	Interrupt Enable bit 0 Timer interrupt disabled. 1 Timer interrupt enabled, default.	1'b1
3:2	R/W	TimerPre	Prescale bits 00 0 stages of prescale, clock is divided by 1, default. 01 4 stages of prescale, clock is divided by 16. 10 8 stages of prescale, clock is divided by 256. 11 undefined, do not use.	2'b0
1	R/W	Timer_Size	Selects 16 bit or 32-bit counter operation 0 16-bit counter, default. 1 32-bit counter.	1'b0
0	R/W	One_shot_Count	Selects one-shot or wrapping counter mode 0 Wrapping mode, default. 1 One-shot mode.	1'b0

Back to top

Name TIMER2INTCLR
Address 0x4000202c
Description clear timer2 interrupt
Reset 0x00000000

CONFIDENTIAL

Name	RSVD	timer2intclr

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
0	R/W	timer2intclr	Any write to the TIMERXINTCLR Register clears the interrupt output from the counter.	1'b0

Back to top

Name TIMER2RIS
Address 0x40002030
Description raw interrupt for timer2
Reset 0x00000000

Name	RSVD	timer2ris

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
0	R	timer2ris	The TIMERXRIS Register indicates the raw interrupt status from the counter. This value is ANDed with the timer interrupt enable bit from the Timer Control Register to create the masked interrupt, that is passed to the interrupt output pin.	1'b0

Back to top

Name TIMER2MIS
Address 0x40002034
Description masked interrupt for timer2
Reset 0x00000000

Name	RSVD	timer2mis

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
0	R	timer2mis	The TIMERXMIS Register indicates the masked interrupt status from the counter. This value is the logical AND of the raw interrupt status with the timer interrupt enable bit from the Timer Control Register, and is the same value that is passed to the interrupt output pin.	1'b0

Back to top

Name TIMER2BGLOAD

CONFIDENTIAL

Address 0x40002038
Description timer2 alternate reload value
Reset 0x00000000

Name	timer2bgload																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	timer2bgload	The TIMERXBGLOAD Register contains the value from which the counter is to decrement. This is the value used to reload the counter when Periodic mode is enabled, and the current count reaches 0. This register provides an alternative method of accessing the TIMERXLOAD Register. The difference is that writes to TIMERXBGLOAD do not cause the counter to immediately restart from the new value.	32'h0

Back to top

Name TIMERITCR
Address 0x40002f00
Description enables interrupt test mode
Reset 0x00000000

Name	RSVD																													timeritcr
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	timeritcr	The TIMERITCR Register enables integration test mode. When in this mode, the Integration Test Output Set Register directly controls the masked interrupt outputs. The combined interrupt output, TIMINTC, then becomes the logical OR of the bits set in the Integration Test Output Set Register.	1'b0

Back to top

Name TIMERITOP
Address 0x40002f04
Description drives the test mode of the interrupts
Reset 0x00000000

Name	RSVD																													timeritop
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

1:0	W	timeritop	When in integration test mode, the values in this write-only register, TIMERITOP, directly drive the enabled interrupt outputs	2'b0

Back to top

Name TIMERPERIPHID4
Address 0x40002fd0
Description Peripheral ID Register 4
Reset 0x00000004

Name	RSVD																						block_count				jep106_c_code			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	block_count	Block count	4'b0
3:0	R	jep106_c_code	jep106_c_code	4'd4

Back to top

Name TIMERPERIPHID5
Address 0x40002fd4
Description Peripheral ID Register 5
Reset 0x00000000

Name	RSVD																								pid5							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	pid5	Peripheral ID Register 5.	8'b0

Back to top

Name TIMERPERIPHID6
Address 0x40002fd8
Description Peripheral ID Register 6
Reset 0x00000000

Name	RSVD																								pid6							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	pid6	Peripheral ID Register 6.	8'b0

Back to top

Name TIMERPERIPHID7
Address 0x40002fdc

CONFIDENTIAL

Description Peripheral ID Register 7
Reset 0x00000000

Name	RSVD																						pid7							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	pid7	Peripheral ID Register 7.	8'b0

Back to top

Name TIMERPERIPHID0
Address 0x40002fe0
Description Peripheral ID Register 0
Reset 0x00000023

Name	RSVD																						part_num							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	part_num	Part number[7:0].	8'h23

Back to top

Name TIMERPERIPHID1
Address 0x40002fe4
Description Peripheral ID Register 1
Reset 0x000000b8

Name	RSVD																						jep106_id_3_0				part_num			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	jep106_id_3_0	jep106_id_3_0	4'hb
3:0	R	part_num	Part number[11:8].	4'h8

Back to top

Name TIMERPERIPHID2
Address 0x40002fe8
Description Peripheral ID Register 2
Reset 0x0000001b

Name	R																						r				j		j	

CONFIDENTIAL

	SVD																						rev				jedec_used	jep106_id_6_4		
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:4	R	rev	Revision	4'h1
3	R	jedec_used	jedec_used	1'h1
2:0	R	jep106_id_6_4	jep106_id_6_4	3'h3

Back to top

Name TIMERPERIPHID3
Address 0x40002fec
Description Peripheral ID Register 3
Reset 0x00000000

Name	RSVD																						ecorevnum				mod_num			
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:4	R	ecorevnum	ECO revision number.	4'h0
3:0	R	mod_num	modification number.	4'h0

Back to top

Name TIMERPCELLID0
Address 0x40002ff0
Description Component ID Register 0
Reset 0x0000000d

Name	RSVD																						cid0							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	cid0	Component ID Register 0	8'h0d

Back to top

Name TIMERPCELLID1
Address 0x40002ff4
Description Component ID Register 1
Reset 0x000000f0

Name	RS																						ci							

CONFIDENTIAL

	RSVD																						cid1							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid1	Component ID Register 1	8'hf0

Back to top

Name TIMERPCELLID2
Address 0x40002ff8
Description Component ID Register 2
Reset 0x00000005

Name	RSVD																						cid2							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid2	Component ID Register 2	8'h05

Back to top

Name TIMERPCELLID3
Address 0x40002ffc
Description Component ID Register 3
Reset 0x000000b1

Name	RSVD																						cid3							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid3	Component ID Register 3	8'hb1

Back to top

WRPR

Name apb0_ctrl
Address 0x40003000
Description Contains register fields associated with apb0_ctrl.
Reset 0x00000002

Name	RSVD																											sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 0 1 = reset	1'b1

CONFIDENTIAL

			0 = not reset	
0	R/W	clk_enable	enable clock to peripheral in APB slot 0 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb1_ctrl
Address 0x40003004
Description Contains register fields associated with apb1_ctrl.
Reset 0x00000002

Name	RSVD																												sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 1 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 1 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb2_ctrl
Address 0x40003008
Description Contains register fields associated with apb2_ctrl.
Reset 0x00000002

Name	RSVD																												sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 2 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 2 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb3_ctrl
Address 0x4000300c
Description Contains register fields associated with apb3_ctrl.
Reset 0x00000002

Name	RS																												sr	cl

CONFIDENTIAL

	VD																											reset	clk_enable	
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 3 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 3 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb4_ctrl
Address 0x40003010
Description Contains register fields associated with apb4_ctrl.
Reset 0x00000002

Name	RSVD																											reset	clk_enable	
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 4 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 4 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb5_ctrl
Address 0x40003014
Description Contains register fields associated with apb5_ctrl.
Reset 0x00000002

Name	RSVD																											reset	clk_enable	
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 5 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 5	1'b0

CONFIDENTIAL

			1 = clock enabled 0 = clock not enabled	

Back to top

Name apb6_ctrl
Address 0x40003018
Description Contains register fields associated with apb6_ctrl.
Reset 0x00000002

Name	RSVD																												sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 6 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 6 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb7_ctrl
Address 0x4000301c
Description Contains register fields associated with apb7_ctrl.
Reset 0x00000002

Name	RSVD																												sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 7 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 7 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb8_ctrl
Address 0x40003020
Description Contains register fields associated with apb8_ctrl.
Reset 0x00000002

Name	RSVD	sres	clk_

CONFIDENTIAL

																													reset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 8 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 8 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb9_ctrl
Address 0x40003024
Description Contains register fields associated with apb9_ctrl.
Reset 0x00000002

Name	RSVD																												reset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 9 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 9 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb10_ctrl
Address 0x40003028
Description Contains register fields associated with apb10_ctrl.
Reset 0x00000002

Name	RSVD																												reset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 10 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 10 1 = clock enabled 0 = clock not enabled	1'b0

CONFIDENTIAL

Back to top

Name apb11_ctrl
Address 0x4000302c
Description Contains register fields associated with apb11_ctrl.
Reset 0x00000002

Name	RSVD																												reset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 11 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 11 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb12_ctrl
Address 0x40003030
Description Contains register fields associated with apb12_ctrl.
Reset 0x00000002

Name	RSVD																												reset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 12 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 12 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb13_ctrl
Address 0x40003034
Description Contains register fields associated with apb13_ctrl.
Reset 0x00000002

Name	RSVD																												reset	clk_en

CONFIDENTIAL

																														able
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 13 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 13 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb14_ctrl
Address 0x40003038
Description Contains register fields associated with apb14_ctrl.
Reset 0x00000002

Name	RSVD																											reset	clk_enable	
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 14 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 14 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb15_ctrl
Address 0x4000303c
Description Contains register fields associated with apb15_ctrl.
Reset 0x00000002

Name	RSVD																											reset	clk_enable	
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 15 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 15 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

CONFIDENTIAL

Name pin_selection_A
Address 0x40003050
Description Contains register fields associated with pin_selection_A.
Reset 0x00000000

Name	p0_sel					p1_sel					p2_sel					p3_sel					p4_sel					p5_sel					RSVD	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:27	R/W	p0_sel	pin selection setting for P0 P0_GPIO_0_ 0 P0_KSI_19_ 1 P0_QDZ_1_ 2 P0_KSO_19_ 3 P0_KSO_8_ 4 P0_SHUB1_CLK 5 P0_XPAON 6 P0_SPI1_CLK 7	5'b0
26:22	R/W	p1_sel	pin selection setting for P1 P1_GPIO_1_ 0 P1_KSI_18_ 1 P1_QDZ_0_ 2 P1_KSO_18_ 3 P1_KSO_9_ 4 P1_SHUB1_CS 5 P1_SPI1_CS 7	5'b0
21:17	R/W	p2_sel	pin selection setting for P2 P2_GPIO_2_ 0 P2_KSI_17_ 1 P2_QDY_1_ 2 P2_KSO_17_ 3 P2_KSO_10_ 4 P2_KSO_8_ 5 P2_SHUB1_MOSI 6 P2_SPI1_MOSI 8	5'b0
16:12	R/W	p3_sel	pin selection setting for P3 P3_GPIO_3_ 0 P3_KSI_16_ 1 P3_QDY_0_ 2 P3_UART1_RTS 3 P3_I2C0_SCK 4 P3_KSO_16_ 6 P3_KSO_11_ 7 P3_KSO_9_ 8 P3_SHUB0_SCK 9	5'b0
11:7	R/W	p4_sel	pin selection setting for P4 P4_GPIO_4_ 0 P4_KSI_15_ 1 P4_UART1_RX 2 P4_I2C0_SDA 3 P4_KSO_15_ 5 P4_KSO_12_ 6 P4_KSO_10_ 7 P4_SHUB0_SDA 8 P4_QSPI_D_0_ 10	5'b0
6:2	R/W	p5_sel	pin selection setting for P5 P5_GPIO_5_ 0 P5_KSI_14_ 1 P5_UART1_TX 2 P5_PWM_0_ 3 P5_KSO_14_ 7 P5_KSO_13_ 8 P5_KSO_11_ 9 P5_BLE_IN_PROCESS 10 P5_QSPI_CS 11	5'b0

CONFIDENTIAL

Back to top

Name pin_selection_B

Address 0x40003054
Description Contains register fields associated with pin_selection_B.
Reset 0x00000000

Name	p6_sel					p7_sel					p8_sel					p9_sel					p10_sel					p11_sel					RSVD	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:27	R/W	p6_sel	pin selection setting for P6 P6_GPIO_6_ 0 P6_KSI_13_ 1 P6_UART1_CTS 2 P6_SPI1_CLK 3 P6_KSO_13_ 8 P6_KSO_14_ 9 P6_KSO_12_ 10 P6_SHUB1_CLK 11 P6_BLE_TX 12 P6_QSPI_D_1_ 13	5'b0
26:22	R/W	p7_sel	pin selection setting for P7 P7_GPIO_7_ 0 P7_KSI_12_ 1 P7_PWM_0_ 2 P7_SPI1_CS 3 P7_KSO_12_ 8 P7_KSO_15_ 9 P7_KSO_13_ 10 P7_SHUB1_CS 11 P7_QSPI_D_2_ 13	5'b0
21:17	R/W	p8_sel	pin selection setting for P8 P8_GPIO_8_ 0 P8_KSI_11_ 1 P8_PWM_1_ 2 P8_SPI1_MOSI 3 P8_KSO_11_ 8 P8_KSO_16_ 10 P8_KSO_14_ 11 P8_SHUB1_MOSI 12	5'b0
16:12	R/W	p9_sel	pin selection setting for P9 P9_GPIO_9_ 0 P9_KSI_10_ 1 P9_PWM_2_ 2 P9_SPI1_MISO 3 P9_KSO_10_ 7 P9_SPI0_CLK 8 P9_I2C0_SCK 9 P9_WURX_DET_0_ 11 P9_KSO_17_ 13 P9_KSO_15_ 14 P9_SHUB0_CLK 15 P9_SHUB0_SCK 16 P9_SHUB1_MISO 17 P9_WLAN_RX 18 P9_PDM_CLK 19	5'b0
11:7	R/W	p10_sel	pin selection setting for P10 P10_GPIO_10_ 0 P10_KSI_9_ 1 P10_SPI0_CLK 2 P10_PWM_1_ 3 P10_UART0_RX 7 P10_KSO_9_ 8 P10_SPI0_CS 9 P10_I2C0_SDA 10 P10_WURX_DET_1_ 12 P10_KSO_18_ 14 P10_KSO_16_ 15 P10_SHUB0_CS 16 P10_SHUB0_SDA 17 P10_SHUB0_CLK 18 P10_PDM_IN 20 P10_XPAON 21	5'b0

CONFIDENTIAL

6:2	R/W	p11_sel	pin selection setting for P11 P11_GPIO_11_ 0 P11_KSI_8_ 1 P11_SPI0_CS 2 P11_PWM_2_ 3 P11_UART0_TX 7 P11_KSO_8_ 8 P11_SPI0_MOSI 9 P11_I2C1_SCK 10 P11_KSO_19_ 13 P11_KSO_17_ 14 P11_UART0_CTS 15 P11_SHUB0_MOSI 16 P11_SHUB1_SCK 17 P11_SHUB0_CS 18 P11_XPAON 19	5'b0

Back to top

Name pin_selection_C
Address 0x40003058
Description Contains register fields associated with pin_selection_C.
Reset 0x00000000

Name	p12_sel					p13_sel					p14_sel					p15_sel					p16_sel					p17_sel					RSVD	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:27	R/W	p12_sel	pin selection setting for P12 P12_GPIO_12_ 0 P12_KSI_7_ 1 P12_QDZ_1_ 2 P12_SPI0_MOSI 3 P12_I2C1_SCK 4 P12_KSO_18_ 9 P12_UART0_CTS 10 P12_SHUB0_MOSI 11 P12_SHUB1_SCK 12	5'b0
26:22	R/W	p13_sel	pin selection setting for P13 P13_GPIO_13_ 0 P13_KSI_6_ 1 P13_QDZ_0_ 2 P13_SPI0_MISO 3 P13_I2C1_SDA 4 P13_KSO_19_ 9 P13_UART0_RTS 10 P13_SHUB0_MISO 11 P13_SHUB1_SDA 12	5'b0
21:17	R/W	p14_sel	pin selection setting for P14 P14_QSPI_CLK 0 P14_QSPI_D_3_ 1	5'b0
16:12	R/W	p15_sel	pin selection setting for P15 P15_QSPI_CS 0	5'b0
11:7	R/W	p16_sel	pin selection setting for P16 P16_QSPI_D_0_ 0	5'b0
6:2	R/W	p17_sel	pin selection setting for P17 P17_QSPI_D_1_ 0 P17_QSPI_CLK 1 P17_QSPI_D_0_ 2	5'b0

Back to top

Name pin_selection_D
Address 0x4000305c
Description Contains register fields associated with pin_selection_D.
Reset 0x00000000

CONFIDENTIAL

Name	p18_sel					p19_sel					p20_sel					p21_sel					p22_sel					p23_sel					RSVD	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:27	R/W	p18_sel	pin selection setting for P18 P18_QSPI_D_2_ 0 P18_KSI_5_ 1 P18_GPIO_14_ 3	5'b0
26:22	R/W	p19_sel	pin selection setting for P19 P19_QSPI_D_3_ 0 P19_KSI_4_ 1 P19_GPIO_15_ 3 P19_QSPI_CS 4 P19_QSPI_CLK 6	5'b0
21:17	R/W	p20_sel	pin selection setting for P20 P20_GPIO_16_ 0 P20_KSI_3_ 1 P20_PWM_3_ 2 P20_SPI0_CLK 3 P20_BLE_DBG_0_ 6 P20_QSPI_D_0_ 9 P20_SHUB0_CLK 12	5'b0
16:12	R/W	p21_sel	pin selection setting for P21 P21_GPIO_17_ 0 P21_KSI_2_ 1 P21_PWM_4_ 2 P21_SPI0_CS 3 P21_BLE_DBG_1_ 6 P21_UART0_CTS 10 P21_SHUB0_CS 11	5'b0
11:7	R/W	p22_sel	pin selection setting for P22 P22_GPIO_18_ 0 P22_KSI_1_ 1 P22_PWM_5_ 2 P22_SPI0_MOSI 3 P22_UART0_RX 9 P22_QSPI_D_1_ 10 P22_SHUB0_MOSI 13	5'b0
6:2	R/W	p23_sel	pin selection setting for P23 P23_GPIO_19_ 0 P23_KSI_0_ 1 P23_PWM_6_ 2 P23_SPI0_MISO 3 P23_BLE_DBG_3_ 6 P23_UART0_TX 9 P23_QSPI_D_2_ 10 P23_SHUB0_MISO 13 P23_BLE_IN_PROCESS 14	5'b0

CONFIDENTIAL

Back to top

Name pin_selection_E
Address 0x40003060
Description Contains register fields associated with pin_selection_E.
Reset 0x00000000

Name	p24_sel					p25_sel					p26_sel					p27_sel					p28_sel					p29_sel					RSVD	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

31:27	R/W	p24_sel	pin selection setting for P24 P24_GPIO_20_0 P24_KSO_7_1 P24_QDY_1_2 P24_SPI1_CLK 3 P24_PWM_3_4 P24_MDM_OUT_4_8 P24_QSPI_D_3_10 P24_UART0_RTS 13 P24_SHUB1_CLK 14 P24_BLE_TX 15 P24_PDM_CLK 16	5'b0
26:22	R/W	p25_sel	pin selection setting for P25 P25_GPIO_21_0 P25_KSO_6_1 P25_QDY_0_2 P25_SPI1_CS 3 P25_PWM_4_4 P25_BLE_DBG_5_7 P25_UART0_RX 10 P25_SHUB1_CS 12 P25_PDM_IN 14 P25_XPAON 16	5'b0
21:17	R/W	p26_sel	pin selection setting for P26 P26_GPIO_22_0 P26_KSO_5_1 P26_QDX_1_2 P26_SPI1_MOSI 3 P26_PWM_5_4 P26_SHUB1_MOSI 11	5'b0
16:12	R/W	p27_sel	pin selection setting for P27 P27_GPIO_23_0 P27_KSO_4_1 P27_QDX_0_2 P27_SPI1_MISO 3 P27_PWM_6_4 P27_MDM_OUT_7_8 P27_SHUB1_MISO 11 P27_WLAN_RX 12	5'b0
11:7	R/W	p28_sel	pin selection setting for P28 P28_GPIO_24_0 P28_KSO_3_1 P28_PWM_7_2	5'b0
6:2	R/W	p29_sel	pin selection setting for P29 P29_GPIO_25_0 P29_KSO_2_1 P29_I2C0_SCK 2 P29_UART1_RTS 3 P29_SHUB0_SCK 6	5'b0

CONFIDENTIAL

Back to top

Name pin_selection_F
Address 0x40003064
Description Contains register fields associated with pin_selection_F.
Reset 0x00000000

Name	p30_sel					p31_sel					p32_sel					p33_sel					RSVD											
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:27	R/W	p30_sel	pin selection setting for P30 P30_GPIO_26_0 P30_KSO_1_1 P30_I2C0_SDA 2 P30_UART1_TX 3 P30_UART0_TX 6 P30_SHUB0_SDA 8	5'b0
26:22	R/W	p31_sel	pin selection setting for P31	5'b0

			P31_GPIO_27_ 0 P31_KSO_0_ 1 P31_I2C1_SCK 2 P31_UART1_RX 3 P31_SHUB1_SCK 6	
21:17	R/W	p32_sel	pin selection setting for P32 P32_GPIO_28_ 0 P32_UART1_RX 1 P32_QDX_1_ 2 P32_KSI_17_ 3 P32_I2C1_SDA 4 P32_UART1_CTS 5 P32_KSO_17_ 7 P32_SHUB1_SDA 8	5'b0
16:12	R/W	p33_sel	pin selection setting for P33 P33_GPIO_29_ 0 P33_UART1_TX 1 P33_QDX_0_ 2 P33_SHUB1_MISO 7 P33_SPI1_MISO 9	5'b0

Back to top

Name pin_pu_A
Address 0x40003068
Description Contains register fields associated with pin_pu_A.
Reset 0x00000000

Name	RSVD			p0_pu	RSVD			p1_pu	RSVD			p2_pu	RSVD			p3_pu	RSVD			p4_pu	RSVD			p5_pu	RSVD			p6_pu	RSVD			p7_pu
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
28	R/W	p0_pu	pull-up enable for P0 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
24	R/W	p1_pu	pull-up enable for P1 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
20	R/W	p2_pu	pull-up enable for P2 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
16	R/W	p3_pu	pull-up enable for P3 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
12	R/W	p4_pu	pull-up enable for P4 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
8	R/W	p5_pu	pull-up enable for P5 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
4	R/W	p6_pu	pull-up enable for P6 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
0	R/W	p7_pu	pull-up enable for P7 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0

Back to top

Name pin_pu_B
Address 0x4000306c
Description Contains register fields associated with pin_pu_B.
Reset 0x00000000

Name	RS	p8	RS	p9	RS	p1	RS	p1	RS	p1	RS	p1	RS	p1	RS	p1

CONFIDENTIAL

	VD			_pu	VD			_pu	VD			0_pu	VD			1_pu	VD			2_pu	VD			3_pu	VD			4_pu	VD			5_pu
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
28	R/W	p8_pu	pull-up enable for P8 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
24	R/W	p9_pu	pull-up enable for P9 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
20	R/W	p10_pu	pull-up enable for P10 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
16	R/W	p11_pu	pull-up enable for P11 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
12	R/W	p12_pu	pull-up enable for P12 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
8	R/W	p13_pu	pull-up enable for P13 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
4	R/W	p14_pu	pull-up enable for P14 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
0	R/W	p15_pu	pull-up enable for P15 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0

Back to top

Name pin_pu_C
Address 0x40003070
Description Contains register fields associated with pin_pu_C.
Reset 0x00000000

Name	RSVD			p16_pu	RSVD			p17_pu	RSVD			p18_pu	RSVD			p19_pu	RSVD			p20_pu	RSVD			p21_pu	RSVD			p22_pu	RSVD			p23_pu
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
28	R/W	p16_pu	pull-up enable for P16 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
24	R/W	p17_pu	pull-up enable for P17 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
20	R/W	p18_pu	pull-up enable for P18 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
16	R/W	p19_pu	pull-up enable for P19 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
12	R/W	p20_pu	pull-up enable for P20 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
8	R/W	p21_pu	pull-up enable for P21 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
4	R/W	p22_pu	pull-up enable for P22 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
0	R/W	p23_pu	pull-up enable for P23	1'b0

CONFIDENTIAL

			0 = pull-up resistor not enabled 1 = pull-up resistor enabled	

Back to top

Name pin_pu_D
Address 0x40003074
Description Contains register fields associated with pin_pu_D.
Reset 0x00000000

Name	RSVD			p24_pu	RSVD			p25_pu	RSVD			p26_pu	RSVD			p27_pu	RSVD			p28_pu	RSVD			p29_pu	RSVD			p30_pu	RSVD			p31_pu
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
28	R/W	p24_pu	pull-up enable for P24 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
24	R/W	p25_pu	pull-up enable for P25 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
20	R/W	p26_pu	pull-up enable for P26 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
16	R/W	p27_pu	pull-up enable for P27 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
12	R/W	p28_pu	pull-up enable for P28 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
8	R/W	p29_pu	pull-up enable for P29 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
4	R/W	p30_pu	pull-up enable for P30 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
0	R/W	p31_pu	pull-up enable for P31 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0

Back to top

Name pin_pu_E
Address 0x40003078
Description Contains register fields associated with pin_pu_E.
Reset 0x00000000

Name	RSVD			p32_pu	RSVD			p33_pu	RSVD																							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
28	R/W	p32_pu	pull-up enable for P32 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0
24	R/W	p33_pu	pull-up enable for P33 0 = pull-up resistor not enabled 1 = pull-up resistor enabled	1'b0

Back to top

CONFIDENTIAL

Name intrpt_cfg_0
Address 0x4000307c
Description Contains register fields associated with intrpt_cfg_0.
Reset 0x00000001

Name	intrpt0_mask																												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt0_mask	interrupt mask for interrupt[0] going into Cortex-M0	32'b00000000000000000000000000000001

Back to top

Name intrpt_cfg_1
Address 0x40003080
Description Contains register fields associated with intrpt_cfg_1.
Reset 0x00000002

Name	intrpt1_mask																												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt1_mask	interrupt mask for interrupt[1] going into Cortex-M0	32'b00000000000000000000000000000010

Back to top

Name intrpt_cfg_2
Address 0x40003084
Description Contains register fields associated with intrpt_cfg_2.
Reset 0x00000004

Name	intrpt2_mask																												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

31:0	R/W	intrpt2_mask	interrupt mask for interrupt[2] going into Cortex-M0	32'b00000000000000000000000000000100

Back to top

Name intrpt_cfg_3
Address 0x40003088
Description Contains register fields associated with intrpt_cfg_3.
Reset 0x00000008

Name	intrpt3_mask																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt3_mask	interrupt mask for interrupt[3] going into Cortex-M0	32'b00000000000000000000000000001000

Back to top

Name intrpt_cfg_4
Address 0x4000308c
Description Contains register fields associated with intrpt_cfg_4.
Reset 0x00002000

Name	intrpt4_mask																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt4_mask	interrupt mask for interrupt[4] going into Cortex-M0	32'b00000000000000000010000000000000

Back to top

Name intrpt_cfg_5
Address 0x40003090
Description Contains register fields associated with intrpt_cfg_5.
Reset 0x00000000

Name	intrpt5_mask

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt5_mask	interrupt mask for interrupt[5] going into Cortex-M0	32'b00000000000000000000000000000000

Back to top

Name intrpt_cfg_6
Address 0x40003094
Description Contains register fields associated with intrpt_cfg_6.
Reset 0x01ff0000

Name	intrpt6_mask																															
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt6_mask	interrupt mask for interrupt[6] going into Cortex-M0	32'b00000001111111110000000000000000

Back to top

Name intrpt_cfg_7
Address 0x40003098
Description Contains register fields associated with intrpt_cfg_7.
Reset 0x02000010

Name	intrpt7_mask																															
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt7_mask	interrupt mask for interrupt[7] going into Cortex-M0	32'b00000010000000000000000000010000

Back to top

Name intrpt_cfg_8
Address 0x4000309c
Description Contains register fields associated with intrpt_cfg_8.
Reset 0x00000020

Name	intrpt8																															

CONFIDENTIAL

																	_	m	a	s	k									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt8_mask	interrupt mask for interrupt[8] going into Cortex-M0	32'b00000000000000000000000000100000

Back to top

Name intrpt_cfg_9
Address 0x400030a0
Description Contains register fields associated with intrpt_cfg_9.
Reset 0x00000040

Name																	i	n	t	r	p	t	9	_	m	a	s	k		
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt9_mask	interrupt mask for interrupt[9] going into Cortex-M0	32'b00000000000000000000000001000000

Back to top

Name intrpt_cfg_10
Address 0x400030a4
Description Contains register fields associated with intrpt_cfg_10.
Reset 0x00000080

Name																	i	n	t	r	p	t	1	0	_	m	a	s	k	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt10_mask	interrupt mask for interrupt[10] going into Cortex-M0	32'b00000000000000000000000010000000

Back to top

Name intrpt_cfg_11
Address 0x400030a8
Description Contains register fields associated with intrpt_cfg_11.
Reset 0x00000100

Name																	i														

CONFIDENTIAL

																	intrpt11_mask													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt11_mask	interrupt mask for interrupt[11] going into Cortex-M0	32'b00000000000000000000000100000000

Back to top

Name intrpt_cfg_12
Address 0x400030ac
Description Contains register fields associated with intrpt_cfg_12.
Reset 0x00000200

Name																intrpt12_mask														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt12_mask	interrupt mask for interrupt[12] going into Cortex-M0	32'b00000000000000000000001000000000

Back to top

Name intrpt_cfg_13
Address 0x400030b0
Description Contains register fields associated with intrpt_cfg_13.
Reset 0x00000400

Name																intrpt13_mask														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt13_mask	interrupt mask for interrupt[13] going into Cortex-M0	32'b00000000000000000000010000000000

Back to top

CONFIDENTIAL

Name intrpt_cfg_14
Address 0x400030b4
Description Contains register fields associated with intrpt_cfg_14.
Reset 0x00000800

Name	intrpt14_mask																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt14_mask	interrupt mask for interrupt[14] going into Cortex-M0	32'b00000000000000000000100000000000

Back to top

Name intrpt_cfg_15
Address 0x400030b8
Description Contains register fields associated with intrpt_cfg_15.
Reset 0x3c001000

Name	intrpt15_mask																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	intrpt15_mask	interrupt mask for interrupt[15] going into Cortex-M0	32'b00111100000000000001000000000000

Back to top

Name remap
Address 0x400030bc
Description Contains register fields associated with remap.
Reset 0x00000000

Name	RSVD																													qspi_drop
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	qspi_drop	Drops the at_ahb_qspi address start from 0x10000000 to 0x0. No change in	1'b0

CONFIDENTIAL

			range.	

Back to top

Name ahb_interposers
Address 0x400030c0
Description Contains register fields associated with ahb_interposers.
Reset 0x00000001

Name	RSVD	qspi_intp_enable
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	qspi_intp_enable	If set, interposers are inserted between backplane and peripherals. If peripherals do not respond within allocated time, the interposer returns error on bus. Prevents the software from hanging on bad bus transactions such as read or write to invalid address.	1'b1

Back to top

Name chipid1
Address 0x40003ff0
Description Contains register fields associated with chipid1.
Reset 0x5379646e

Name	char0	char1	char2	char3
Bits	**31 30 29 28 27 26 25 24**	**23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8**	**7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
31:24	R	char0	S	8'h53
23:16	R	char1	y	8'h79
15:8	R	char2	d	8'h64
7:0	R	char3	n	8'h6e

Back to top

Name chipid2
Address 0x40003ff4
Description Contains register fields associated with chipid2.
Reset 0x65790000

Name	char4	char5	char6	char7
Bits	**31 30 29 28 27 26 25 24**	**23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8**	**7 6 5 4 3 2 1 0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
31:24	R	char4	e	8'h65
23:16	R	char5	y	8'h79
15:8	R	char6	nul	8'h0
7:0	R	char7	nul	8'h0

Back to top

Name chiprev
Address 0x40003ff8
Description Contains register fields associated with chiprev.
Reset 0x00000202

Name	RSVD	running_off_32khz_xtal	major	minor
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17	16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
16	R	running_off_32khz_xtal	Is clk_lpc running off crystal? 0 = oscillator based 1 = crystal based	1'b0
15:8	R	major	Major revision number.	8'd2
7:0	R	minor	Minor revision number.	8'd2

Back to top

Name core_id
Address 0x40003ffc
Description Contains register fields associated with core_id.
Reset 0x57525052

Name	id
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
31:0	R	id	WRPR in ASCII	32'h57525052

Back to top

UART0

Name DATA
Address 0x40004000
Description DATA

CONFIDENTIAL

Reset 0x00000000

Name	RSVD																								Data							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R/W	Data	Read Received data. Write Transmit data.	8'b0

Back to top

Name STATE
Address 0x40004004
Description STATE
Reset 0x00000000

Name	RSVD																										nUARTCTS	nUARTRTS	rx_overrun	tx_overrun	rx_buf_not_empty	tx_buf_full
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R/W	nUARTCTS	current status of nUARTCTS (Read only) // (Atmosic)	1'b0
4	R/W	nUARTRTS	current status of nUARTRTS (Read only) // (Atmosic)	1'b0
3	R/W	rx_overrun	RX buffer overrun, write 1 to clear.	1'b0
2	R/W	tx_overrun	TX buffer overrun, write 1 to clear.	1'b0
1	R/W	rx_buf_not_empty	1=RX buffer is not empty, 0= RX buffer is empty read-only.	1'b0
0	R/W	tx_buf_full	TX buffer full, read-only.	1'b0

Back to top

Name CTRL
Address 0x40004008
Description CTRL
Reset 0x00000000

Name	RSVD																										hs_testmode	rx_overrun_int_en	tx_overrun_int_en	RxIntEn	TxIntEn	RxEn	TxEn
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
6	R/W	hs_testmode	High-speed test mode for TX only.	1'b0
5	R/W	rx_overrun_int_en	RX overrun interrupt enable.	1'b0
4	R/W	tx_overrun_int_en	TX overrun interrupt enable.	1'b0
3	R/W	RxIntEn	RX interrupt enable.	1'b0
2	R/W	TxIntEn	TX interrupt enable.	1'b0
1	R/W	RxEn	RX enable.	1'b0
0	R/W	TxEn	TX enable.	1'b0

Back to top

Name INTSTATUS_CLEAR
Address 0x4000400c
Description INTSTATUS_CLEAR
Reset 0x00000000

Name	RSVD	rx_overflow_intr	tx_overflow_intr	rxintr	txintr
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4	3	2	1	0

Bits	SW	Name	Description	Reset
3	R/W	rx_overflow_intr	RX overrun interrupt. Write 1 to clear.	1'b0
2	R/W	tx_overflow_intr	TX overrun interrupt. Write 1 to clear.	1'b0
1	R/W	rxintr	RX interrupt. Write 1 to clear.	1'b0
0	R/W	txintr	TX interrupt. Write 1 to clear.	1'b0

Back to top

Name BAUDDIV
Address 0x40004010
Description BAUDDIV
Reset 0x00000000

Name	RSVD	baud_div
Bits	31 30 29 28 27 26 25 24 23 22 21 20	19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
19:0	R/W	baud_div	Baud rate divider. The minimum number is 16.	20'b0

Back to top

Name RX_LWM
Address 0x40004014

CONFIDENTIAL

Description RX_LWM
Reset 0x00000001

Name	RSVD																											Rx_LWM				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	Rx_LWM	low water mark of RX FIFO; a trigger is issued when (Atmosic) RX FIFO has waited long enough to have received these many items, starting from the first item. Constraint: (1<= RX_LWM <= 14)	5'd1

Back to top

Name TX_LWM
Address 0x40004018
Description TX_LWM
Reset 0x00000010

Name	RSVD																											Tx_LWM				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	Tx_LWM	low water mark of TX FIF0; a trigger is issued when (Atmosic) tX FIFO is drained to these many unoccupied slots. Constraint (1<= TX_LWM <= 16)	5'd16

Back to top

Name RX_FIFO_SPACES
Address 0x4000401c
Description RX_FIFO_SPACES
Reset 0x00000000

Name	RSVD																											rx_fifo_spaces				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R	rx_fifo_spaces	unoccupied spaces in RX_FIFO (Atmosic)	5'b0

Back to top

Name TX_FIFO_SPACES

CONFIDENTIAL

Address 0x40004020
Description TX_FIFO_SPACES
Reset 0x00000000

Name	RSVD																											tx_fifo_spaces				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R	tx_fifo_spaces	unoccupied spaces in TX_FIFO (Atmosic)	5'b0

Back to top

Name HW_FLOW_OVRD
Address 0x40004024
Description HW_FLOW_OVRD
Reset 0x0000000a

Name	RSVD																											nCTS_OVRD	nCTS_VAL	nRTS_OVRD	nRTS_VAL
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R/W	nCTS_OVRD	0= nUARTCTS override override disabled; 1(default)= nUARTCTS override enabled (Atmosic)	1'b1
2	R/W	nCTS_VAL	0(default) = nCTS is asserted to enable TX; 1= nCTS is deasserted	1'b0
1	R/W	nRTS_OVRD	0= nUARTRTS override override disabled; 1(default)= nUARTRTS override enabled ;(Atmosic)	1'b1
0	R/W	nRTS_VAL	0(default) = nRTS is asserted to enable external TX; 1= nRTS is deasserted	1'b0

Back to top

Name PID4
Address 0x40004fd0
Description PID4
Reset 0x00000004

Name	RSVD																											block_count			jep106_c_code	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:4	R	block_count	Block count	4'b0
3:0	R	jep106_c_code	jep106_c_code	4'd4

Back to top

Name PID5
Address 0x40004fd4
Description PID5
Reset 0x00000000

Name	RSVD																								pid5							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	pid5	Peripheral ID Register 5.	8'b0

Back to top

Name PID6
Address 0x40004fd8
Description PID6
Reset 0x00000000

Name	RSVD																								pid6							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	pid6	Peripheral ID Register 6.	8'b0

Back to top

Name PID7
Address 0x40004fdc
Description PID7
Reset 0x00000000

Name	RSVD																								pid7							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	pid7	Peripheral ID Register 7.	8'b0

Back to top

Name PID0
Address 0x40004fe0
Description PID0
Reset 0x00000021

Name	RS																								pa							

CONFIDENTIAL

	VD																							part_num						
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	part_num	Part number[7:0].	8'h21

Back to top

Name PID1
Address 0x40004fe4
Description PID1
Reset 0x000000b8

Name	RSVD																								jep106_id_3_0				part_num			
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:4	R	jep106_id_3_0	jep106_id_3_0	4'hb
3:0	R	part_num	Part number[11:8].	4'h8

Back to top

Name PID2
Address 0x40004fe8
Description PID2
Reset 0x0000001b

Name	RSVD																								rev				jedec_used	jep106_id_6_4		
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:4	R	rev	Revision	4'h1
3	R	jedec_used	jedec_used	1'h1
2:0	R	jep106_id_6_4	jep106_id_6_4	3'h3

Back to top

Name PID3
Address 0x40004fec

CONFIDENTIAL

Description PID3
Reset 0x00000000

Name	RSVD																						ecorevnum				mod_num			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	ecorevnum	ECO revision number.	4'h0
3:0	R	mod_num	modification number.	4'h0

Back to top

Name CID0
Address 0x40004ff0
Description CID0
Reset 0x0000000d

Name	RSVD																						cid0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid0	Component ID Register 0	8'h0d

Back to top

Name CID1
Address 0x40004ff4
Description CID1
Reset 0x000000f0

Name	RSVD																						cid1							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid1	Component ID Register 1	8'hf0

Back to top

Name CID2
Address 0x40004ff8
Description CID2
Reset 0x00000005

Name	RSVD																						cid2							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

7:0	R	cid2	Component ID Register 2	8'h05

Back to top

Name CID3
Address 0x40004ffc
Description CID3
Reset 0x000000b1

Name	RSVD																						cid3							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid3	Component ID Register 3	8'hb1

Back to top

UART1

Name DATA
Address 0x40005000
Description DATA
Reset 0x00000000

Name	RSVD																						Data							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R/W	Data	Read Received data. Write Transmit data.	8'b0

Back to top

Name STATE
Address 0x40005004
Description STATE
Reset 0x00000000

Name	RSVD																								nUARTCTS	nUARTRTS	rx_overrun	tx_overrun	rx_buf_not_empty	tx_buf_full
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R/W	nUARTCTS	current status of nUARTCTS (Read only) // (Atmosic)	1'b0

CONFIDENTIAL

4	R/W	nUARTRTS	current status of nUARTRTS (Read only) // (Atmosic)	1'b0
3	R/W	rx_overrun	RX buffer overrun, write 1 to clear.	1'b0
2	R/W	tx_overrun	TX buffer overrun, write 1 to clear.	1'b0
1	R/W	rx_buf_not_empty	1=RX buffer is not empty, 0= RX buffer is empty read-only.	1'b0
0	R/W	tx_buf_full	TX buffer full, read-only.	1'b0

Back to top

Name CTRL
Address 0x40005008
Description CTRL
Reset 0x00000000

Name	RSVD	hs_testmode	rx_overrun_int_en	tx_overrun_int_en	RxIntEn	TxIntEn	RxEn	TxEn
Bits	**31:7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
6	R/W	hs_testmode	High-speed test mode for TX only.	1'b0
5	R/W	rx_overrun_int_en	RX overrun interrupt enable.	1'b0
4	R/W	tx_overrun_int_en	TX overrun interrupt enable.	1'b0
3	R/W	RxIntEn	RX interrupt enable.	1'b0
2	R/W	TxIntEn	TX interrupt enable.	1'b0
1	R/W	RxEn	RX enable.	1'b0
0	R/W	TxEn	TX enable.	1'b0

Back to top

Name INTSTATUS_CLEAR
Address 0x4000500c
Description INTSTATUS_CLEAR
Reset 0x00000000

Name	RSVD	rx_overflow_intr	tx_overflow_intr	rxintr	txintr
Bits	**31:4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R/W	rx_overflow_intr	RX overrun interrupt. Write 1 to clear.	1'b0

CONFIDENTIAL

2	R/W	tx_overflow_intr	TX overrun interrupt. Write 1 to clear.	1'b0
1	R/W	rxintr	RX interrupt. Write 1 to clear.	1'b0
0	R/W	txintr	TX interrupt. Write 1 to clear.	1'b0

Back to top

Name BAUDDIV
Address 0x40005010
Description BAUDDIV
Reset 0x00000000

Name	RSVD												baud_div																	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
19:0	R/W	baud_div	Baud rate divider. The minimum number is 16.	20'b0

Back to top

Name RX_LWM
Address 0x40005014
Description RX_LWM
Reset 0x00000001

Name	RSVD																									Rx_LWM				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	Rx_LWM	low water mark of RX FIFO; a trigger is issued when (Atmosic) RX FIFO has waited long enough to have received these many items, starting from the first item. Constraint: (1<= RX_LWM <= 14)	5'd1

Back to top

Name TX_LWM
Address 0x40005018
Description TX_LWM
Reset 0x00000010

Name	RSVD																									Tx_LWM				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	Tx_LWM	low water mark of TX FIF0; a trigger is issued when (Atmosic) tX FIFO is drained to these many unoccupied slots. Constraint (1<= TX_LWM <= 16)	5'd16

CONFIDENTIAL

Back to top

Name RX_FIFO_SPACES
Address 0x4000501c
Description RX_FIFO_SPACES
Reset 0x00000000

Name	RSVD																									rx_fifo_spaces				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R	rx_fifo_spaces	unoccupied spaces in RX_FIFO (Atmosic)	5'b0

Back to top

Name TX_FIFO_SPACES
Address 0x40005020
Description TX_FIFO_SPACES
Reset 0x00000000

Name	RSVD																									tx_fifo_spaces				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R	tx_fifo_spaces	unoccupied spaces in TX_FIFO (Atmosic)	5'b0

Back to top

Name HW_FLOW_OVRD
Address 0x40005024
Description HW_FLOW_OVRD
Reset 0x0000000a

Name	RSVD																									nCTS_OVRD	nCTS_VAL	nRTS_OVRD	nRTS_VAL
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
3	R/W	nCTS_OVRD	0= nUARTCTS override override disabled; 1(default)= nUARTCTS override enabled (Atmosic)	1'b1
2	R/W	nCTS_VAL	0(default) = nCTS is asserted to enable TX; 1= nCTS is deasserted	1'b0
1	R/W	nRTS_OVRD	0= nUARTRTS override override disabled; 1(default)= nUARTRTS override enabled ;(Atmosic)	1'b1
0	R/W	nRTS_VAL	0(default) = nRTS is asserted to enable external TX; 1= nRTS is deasserted	1'b0

Back to top

Name PID4
Address 0x40005fd0
Description PID4
Reset 0x00000004

Name	RSVD																								block_count				jep106_c_code			
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:4	R	block_count	Block count	4'b0
3:0	R	jep106_c_code	jep106_c_code	4'd4

Back to top

Name PID5
Address 0x40005fd4
Description PID5
Reset 0x00000000

Name	RSVD																								pid5							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	pid5	Peripheral ID Register 5.	8'b0

Back to top

Name PID6
Address 0x40005fd8
Description PID6
Reset 0x00000000

Name	RSVD																								pid6							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

CONFIDENTIAL

Bits	SW	Name	Description	Reset
7:0	R	pid6	Peripheral ID Register 6.	8'b0

Back to top

Name PID7
Address 0x40005fdc
Description PID7
Reset 0x00000000

Name	RSVD																						pid7							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	pid7	Peripheral ID Register 7.	8'b0

Back to top

Name PID0
Address 0x40005fe0
Description PID0
Reset 0x00000021

Name	RSVD																						part_num							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	part_num	Part number[7:0].	8'h21

Back to top

Name PID1
Address 0x40005fe4
Description PID1
Reset 0x000000b8

Name	RSVD																								jep106_id_3_0				part_num			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	jep106_id_3_0	jep106_id_3_0	4'hb
3:0	R	part_num	Part number[11:8].	4'h8

Back to top

CONFIDENTIAL

Name PID2
Address 0x40005fe8
Description PID2
Reset 0x0000001b

Name	RSVD																						rev				jedec_used	jep106_id_6_4		
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	rev	Revision	4'h1
3	R	jedec_used	jedec_used	1'h1
2:0	R	jep106_id_6_4	jep106_id_6_4	3'h3

Back to top

Name PID3
Address 0x40005fec
Description PID3
Reset 0x00000000

Name	RSVD																						ecorevnum				mod_num			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	ecorevnum	ECO revision number.	4'h0
3:0	R	mod_num	modification number.	4'h0

Back to top

Name CID0
Address 0x40005ff0
Description CID0
Reset 0x0000000d

Name	RSVD																						cid0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid0	Component ID Register 0	8'h0d

Back to top

CONFIDENTIAL

Name CID1
Address 0x40005ff4
Description CID1
Reset 0x000000f0

Name	RSVD																						cid1							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid1	Component ID Register 1	8'hf0

Back to top

Name CID2
Address 0x40005ff8
Description CID2
Reset 0x00000005

Name	RSVD																						cid2							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid2	Component ID Register 2	8'h05

Back to top

Name CID3
Address 0x40005ffc
Description CID3
Reset 0x000000b1

Name	RSVD																						cid3							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid3	Component ID Register 3	8'hb1

Back to top

PWM

Name pwm0_ctrl
Address 0x40006000
Description control signals for pwm0 subcore
Reset 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | |

12:7	R/W	lo_dur	duration of low portion of period in clk_mpc clock cycles	6'b0
6:1	R/W	tot_dur	total duration of period in clk_mpc clock cycles	6'b0
0	R/W	ok_to_run	start pwm subcore	1'b0

Back to top

Name pwm3_ctrl
Address 0x4000600c
Description control signals for pwm3 subcore
Reset 0x00000000

Name	RSVD	invert	lo_dur	tot_dur	ok_to_run
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14**	**13**	**12 11 10 9 8 7**	**6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
13	R/W	invert	invert output polarity	1'b0
12:7	R/W	lo_dur	duration of low portion of period in clk_mpc clock cycles	6'b0
6:1	R/W	tot_dur	total duration of period in clk_mpc clock cycles	6'b0
0	R/W	ok_to_run	start pwm subcore	1'b0

Back to top

Name pwm4_ctrl
Address 0x40006010
Description control signals for pwm4 subcore
Reset 0x00000000

Name	RSVD	invert	lo_dur	tot_dur	ok_to_run
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14**	**13**	**12 11 10 9 8 7**	**6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
13	R/W	invert	invert output polarity	1'b0
12:7	R/W	lo_dur	duration of low portion of period in clk_mpc clock cycles	6'b0
6:1	R/W	tot_dur	total duration of period in clk_mpc clock cycles	6'b0
0	R/W	ok_to_run	start pwm subcore	1'b0

Back to top

Name pwm5_ctrl
Address 0x40006014
Description control signals for pwm5 subcore
Reset 0x00000000

Name	RSVD	invert	lo_dur	tot_dur	ok_to_r

CONFIDENTIAL

																														un
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
13	R/W	invert	invert output polarity	1'b0
12:7	R/W	lo_dur	duration of low portion of period in clk_mpc clock cycles	6'b0
6:1	R/W	tot_dur	total duration of period in clk_mpc clock cycles	6'b0
0	R/W	ok_to_run	start pwm subcore	1'b0

Back to top

Name pwm6_ctrl
Address 0x40006018
Description control signals for pwm6 subcore
Reset 0x00000000

Name	RSVD																		invert	lo_dur						tot_dur						ok_to_run
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
13	R/W	invert	invert output polarity	1'b0
12:7	R/W	lo_dur	duration of low portion of period in clk_mpc clock cycles	6'b0
6:1	R/W	tot_dur	total duration of period in clk_mpc clock cycles	6'b0
0	R/W	ok_to_run	start pwm subcore	1'b0

Back to top

Name pwm7_ctrl
Address 0x4000601c
Description control signals for pwm7 subcore
Reset 0x00000000

Name	RSVD																		invert	lo_dur						tot_dur						ok_to_run
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
13	R/W	invert	invert output polarity	1'b0
12:7	R/W	lo_dur	duration of low portion of period in clk_mpc clock cycles	6'b0
6:1	R/W	tot_dur	total duration of period in clk_mpc clock cycles	6'b0
0	R/W	ok_to_run	start pwm subcore	1'b0

Back to top

Name core_id
Address 0x40006ffc
Description core id
Reset 0x50574d20

CONFIDENTIAL

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	PWM in ASCII	32'h50574d20

Back to top

SPI0

Name transaction_setup
Address 0x40007000
Description transaction setup
Reset 0x00000000

Name	RSVD		dummy_cycles				csn_stays_low	loopback	opcode								clkdiv										rwb	num_data_bytes				start
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
29:26	R/W	dummy_cycles	number of extra shift cycles to help with turn-around	4'd0
25	R/W	csn_stays_low	at end of transaction, leave csn low	1'b0
24	R/W	loopback	if 1, mosi is internally connected to miso	1'b0
23:16	R/W	opcode	opcode	8'd0
15:6	R/W	clkdiv	freq_sclk = f_pclk/(2*(clkdiv+1))	10'd0
5	R/W	rwb	1=read 0=write; note it's possibly redundant with opcode	1'b0
4:1	R/W	num_data_bytes	just the data; don't add 1 for opcode	4'd0
0	R/W	start	FSM needs to see low to high transition to start; not auto cleared	1'b0

Back to top

Name transaction_status
Address 0x40007004
Description transaction status
Reset 0x00000000

Name	RSVD																opcode_status								RSVD						running	done
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

15:8	R	opcode_status	opcode status	8'b0
1	R	running	transaction is running	1'b0
0	R	done	transaction is done	1'b0

Back to top

Name data_bytes_lower
Address 0x40007008
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte3								byte2								byte1								byte0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte3	data byte3	8'b0
23:16	R/W	byte2	data byte2	8'b0
15:8	R/W	byte1	data byte1	8'b0
7:0	R/W	byte0	data byte0	8'b0

Back to top

Name data_bytes_upper
Address 0x4000700c
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte7								byte6								byte5								byte4							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte7	data byte7	8'b0
23:16	R/W	byte6	data byte6	8'b0
15:8	R/W	byte5	data byte5	8'b0
7:0	R/W	byte4	data byte4	8'b0

Back to top

Name interrupt_mask
Address 0x40007010
Description interrupt mask
Reset 0x00000000

Name	RSVD																															passthru0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	passthru0	1=allow interrupt0 to be OR'ed into core interrupt	1'b0

CONFIDENTIAL

Back to top

Name interrupt_status
Address 0x40007014
Description interrupt status
Reset 0x00000000

Name	RSVD	interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R	interrupt0	saw a rising edge on done; independent of mask	1'b0

Back to top

Name set_interrupt
Address 0x40007018
Description interrupt set
Reset 0x00000000

Name	RSVD	set_interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	set_interrupt0	set interrupt0; not auto cleared	1'b0

Back to top

Name reset_interrupt
Address 0x4000701c
Description interrupt clear
Reset 0x00000000

Name	RSVD	reset_interru

CONFIDENTIAL

																													pt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	reset_interrupt0	clear interrupt0; not auto cleared	1'b0

Back to top

Name core_id
Address 0x40007ffc
Description core ID
Reset 0x53504920

Name	id																												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	SPI in ASCII	32'h53504920

Back to top

WATCHDOG

Name WDOGLOAD
Address 0x40008000
Description Watchdog LOAD Register
Reset 0xffffffff

Name	wdogload																												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	wdogload	The WDOGLOAD Register contains the value from which the counter is to decrement. When this register is written to, the count is immediately restarted from the new value. The minimum value for WDOGLOAD is 1.	32'hffffffff

Back to top

Name WDOGVALUE
Address 0x40008004
Description Watchdog value Register
Reset 0xffffffff

Name	wdogvalue																												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
31:0	R	wdogvalue	The WDOGVALUE Register gives the current value of the decrementing counter.	32'hffffffff

Back to top

Name WDOGCONTROL
Address 0x40008008
Description Watchdog Control Register
Reset 0x00000000

Name	RSVD																													wdogcontrol	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1:0	R/W	wdogcontrol	The WDOGCONTROL Register enables the software to control the watchdog unit.	2'b0

Back to top

Name WDOGINTCLR
Address 0x4000800c
Description Watchdog Interrupt Clear
Reset 0x00000000

Name	wdogintclr																														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	W	wdogintclr	A write of any value to the WDOGINTCLR Register clears the watchdog interrupt, and reloads the counter from the value in WDOGLOAD.	32'b0

Back to top

Name WDOGRIS
Address 0x40008010
Description Watchdog Raw Interrupt Status Register
Reset 0x00000000

Name	RSVD																													wdogris	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
0	R	wdogris	The WDOGRIS Register indicates the raw interrupt status from the counter. This value is ANDed with the interrupt enable bit from the control register to create the masked interrupt, that is passed to the interrupt output pin	1'b0

Back to top

Name WDOGMIS
Address 0x40008014
Description Watchdog Masked Interrupt Status Register
Reset 0x00000000

Name	RSVD																													wdogmis
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R	wdogmis	The WDOGMIS Register indicates the masked interrupt status from the counter. This value is the logical AND of the raw interrupt status with the INTEN bit from the control register, and is the same value that is passed to the interrupt output pin	1'd0

Back to top

Name WDOGLOCK
Address 0x40008c00
Description Watchdog Lock Register
Reset 0x00000000

Name	wdoglock																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	wdoglock	The WDOGLOCK Register disables write accesses to all other registers. This is to prevent rogue software from disabling the watchdog functionality. Writing a value of 0x1ACCE551 enables write access to all other registers. Writing any other value disables write accesses. A read from this register returns only the bottom bit: 0 Indicates that write access is enabled, not locked. 1 Indicates that write access is disabled, locked.	32'h0

Back to top

Name WDOGITCR
Address 0x40008f00
Description Watchdog Integration Test Control Register
Reset 0x00000000

Name	RSVD																													wdogit

CONFIDENTIAL

																														cr
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	wdogitcr	The WDOGITCR Register enables integration test mode. When in this mode, the test output register directly controls the masked interrupt output, WDOGINT, and reset output, WDOGRES.	1'b0

Back to top

Name WDOGITOP
Address 0x40008f04
Description Watchdog Integration Test Output Set Register
Reset 0x00000000

Name	RSVD																												wdogitop	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1:0	W	wdogitop	When the WDOGITOP Register is in integration test mode, the values in this register directly drive the enabled interrupt output and reset output	2'b0

Back to top

Name WDOGPERIPHID4
Address 0x40008fd0
Description Peripheral ID Register 4
Reset 0x00000004

Name	RSVD																						block_count				jep106_c_code			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	block_count	Block count	4'b0
3:0	R	jep106_c_code	jep106_c_code	4'd4

Back to top

Name WDOGPERIPHID5
Address 0x40008fd4
Description Peripheral ID Register 5
Reset 0x00000000

Name	RSVD	pid5

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	pid5	Peripheral ID Register 5.	8'b0

Back to top

Name WDOGPERIPHID6
Address 0x40008fd8
Description Peripheral ID Register 6
Reset 0x00000000

Name	RSVD																								pid6							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	pid6	Peripheral ID Register 6.	8'b0

Back to top

Name WDOGPERIPHID7
Address 0x40008fdc
Description Peripheral ID Register 7
Reset 0x00000000

Name	RSVD																								pid7							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	pid7	Peripheral ID Register 7.	8'b0

Back to top

Name WDOGPERIPHID0
Address 0x40008fe0
Description Peripheral ID Register 0
Reset 0x00000024

Name	RSVD																								part_num							
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
7:0	R	part_num	Part number[7:0].	8'h24

Back to top

Name WDOGPERIPHID1
Address 0x40008fe4
Description Peripheral ID Register 1
Reset 0x000000b8

CONFIDENTIAL

Name	RSVD																							jep106_id_3_0				part_num			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	jep106_id_3_0	jep106_id_3_0	4'hb
3:0	R	part_num	Part number[11:8].	4'h8

Back to top

Name WDOGPERIPHID2
Address 0x40008fe8
Description Peripheral ID Register 2
Reset 0x0000001b

Name	RSVD																							rev				jedec_used	jep106_id_6_4			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	rev	Revision	4'h1
3	R	jedec_used	jedec_used	1'h1
2:0	R	jep106_id_6_4	jep106_id_6_4	3'h3

Back to top

Name WDOGPERIPHID3
Address 0x40008fec
Description Peripheral ID Register 3
Reset 0x00000000

Name	RSVD																							ecorevnum				mod_num			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:4	R	ecorevnum	ECO revision number.	4'h0
3:0	R	mod_num	modification number.	4'h0

CONFIDENTIAL

Back to top

Name WDOGPCELLID0
Address 0x40008ff0
Description Component ID Register 0
Reset 0x0000000d

Name	RSVD																						cid0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid0	Component ID Register 0	8'h0d

Back to top

Name WDOGPCELLID1
Address 0x40008ff4
Description Component ID Register 1
Reset 0x000000f0

Name	RSVD																						cid1							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid1	Component ID Register 1	8'hf0

Back to top

Name WDOGPCELLID2
Address 0x40008ff8
Description Component ID Register 2
Reset 0x00000005

Name	RSVD																						cid2							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cid2	Component ID Register 2	8'h05

Back to top

Name WDOGPCELLID3
Address 0x40008ffc
Description Component ID Register 3
Reset 0x000000b1

Name	RSVD																						cid3							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

7:0	R	cid3	Component ID Register 3	8'hb1

Back to top

KSM

Name ctrl0
Address 0x4000a000
Description control signals
Reset 0x00000800

Name	RSVD	ksm_not_empty_to_intrpt	consecscan	RSVD	flush	pop	loopback	go
Bits	**31:13**	**12**	**11:8**	**7:4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
12	R/W	ksm_not_empty_to_intrpt	in addition to fifo_write, /not empty/ signal will also feed into setting intrpt[0]	1'b0
11:8	R/W	consecscan	how many consecutive scans are needed before key press or key release event?	4'd8
3	R/W	flush	flushes the key packet fifo	1'b0
2	R/W	pop	read from keyboard event fifo; not self reseting	1'b0
1	R/W	loopback	enable lpback_ksi (see below)	1'b0
0	R/W	go	start the keyboard scanning; will scan while go is high; not self reseting	1'b0

CONFIDENTIAL

Back to top

Name time_param0
Address 0x4000a004
Description timing parameters
Reset 0x00100010

Name	t1	t2
Bits	**31:16**	**15:0**

Bits	SW	Name	Description	Reset
31:16	R/W	t1	from col falling to sampling first row	16'd16
15:0	R/W	t2	from sampling last row to col rising	16'd16

Back to top

Name time_param1
Address 0x4000a008
Description timing parameters

Reset 0x00080040

Name	t3																t4														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:16	R/W	t3	gap between adjacent row samplings	16'd8
15:0	R/W	t4	gap between col rising and next col falling	16'd64

Back to top

Name matrix_size
Address 0x4000a00c
Description maxtrix size
Reset 0x00001307

Name	RSVD																			num_row					RSVD			num_col				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
12:8	R/W	num_row	the actual number of rows is num_row + 1	5'd19
4:0	R/W	num_col	the actual number of cols is num_col + 1	5'd7

Back to top

Name keyboard_packet
Address 0x4000a040
Description a read of this register pops the data FIFO
Reset 0x00000000

Name	RSVD																				empty	pressed_released_n							col									row					
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11	R	empty	if 1, ignore this packet	1'd0
10	R	pressed_released_n	1=pressed 0=released	1'b0
9:5	R	col	the column of the pressed or released key; 0-indexed	5'd0
4:0	R	row	the row of the pressed or released key; 0-indexed	5'd0

Back to top

Name status

CONFIDENTIAL

Address 0x4000a080
Description status information
Reset 0x00000000

Name	RSVD	command_taken
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R	command_taken	fsm in on 32kHz clock; it takes two 32kHz cycles for the command to be picked up	1'b0

Back to top

Name interrupts
Address 0x4000a0c0
Description interrupt status
Reset 0x00000000

Name	RSVD	intrpt3	intrpt2	intrpt1	intrpt0
Bits	**31:4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R	intrpt3	flush read is done	1'b0
2	R	intrpt2	pop read is done	1'b0
1	R	intrpt1	keyboard event fifo overran	1'b0
0	R	intrpt0	hardware wrote packet to keyboard event fifo	1'b0

Back to top

Name interrupt_mask
Address 0x4000a0c4
Description interrupt mask
Reset 0x0000000f

Name	RSVD	mask_intrpt3	mask_intrpt2	mask_intrpt1	mask_intrpt0
Bits	**31:4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R/W	mask_intrpt3	allow intrpt3 to propogate to core's single output interrupt	1'b1

CONFIDENTIAL

2	R/W	mask_intrpt2	allow intrpt2 to propogate to core's single output interrupt	1'b1
1	R/W	mask_intrpt1	allow intrpt1 to propogate to core's single output interrupt	1'b1
0	R/W	mask_intrpt0	allow intrpt0 to propogate to core's single output interrupt	1'b1

Back to top

Name interrupt_clear
Address 0x4000a0c8
Description interrupt clear
Reset 0x00000000

Name	RSVD																											clear_intrpt3	clear_intrpt2	clear_intrpt1	clear_intrpt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R/W	clear_intrpt3	clear interrupt 3; not self reseting	1'b0
2	R/W	clear_intrpt2	clear interrupt 2; not self reseting	1'b0
1	R/W	clear_intrpt1	clear interrupt 1; not self reseting	1'b0
0	R/W	clear_intrpt0	clear interrupt 0; not self reseting	1'b0

Back to top

Name core_id
Address 0x4000affc
Description core ID
Reset 0x4b534d20

Name	id																														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	KSM in ASCII	32'h4b534d20

Back to top

SPI2

Name transaction_setup
Address 0x4000b000
Description transaction setup
Reset 0x00000000

Name	RSVD	dummy_cyc	csn_stays	loopback	opcode	clkdiv	rwb	num_data_	start

			les				_low																			bytes				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
29:26	R/W	dummy_cycles	number of extra shift cycles to help with turn-around	4'd0
25	R/W	csn_stays_low	at end of transaction, leave csn low	1'b0
24	R/W	loopback	if 1, mosi is internally connected to miso	1'b0
23:16	R/W	opcode	opcode	8'd0
15:6	R/W	clkdiv	freq_sclk = f_pclk/(2*(clkdiv+1))	10'd0
5	R/W	rwb	1=read 0=write; note it's possibly redundant with opcode	1'b0
4:1	R/W	num_data_bytes	just the data; don't add 1 for opcode	4'd0
0	R/W	start	FSM needs to see low to high transition to start; not auto cleared	1'b0

Back to top

Name transaction_status
Address 0x4000b004
Description transaction status
Reset 0x00000000

Name	RSVD																opcode_status								RSVD						running	done
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:8	R	opcode_status	opcode status	8'b0
1	R	running	transaction is running	1'b0
0	R	done	transaction is done	1'b0

Back to top

Name data_bytes_lower
Address 0x4000b008
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte3								byte2								byte1								byte0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte3	data byte3	8'b0
23:16	R/W	byte2	data byte2	8'b0
15:8	R/W	byte1	data byte1	8'b0
7:0	R/W	byte0	data byte0	8'b0

Back to top

CONFIDENTIAL

Name data_bytes_upper
Address 0x4000b00c
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte7								byte6								byte5								byte4							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte7	data byte7	8'b0
23:16	R/W	byte6	data byte6	8'b0
15:8	R/W	byte5	data byte5	8'b0
7:0	R/W	byte4	data byte4	8'b0

Back to top

Name interrupt_mask
Address 0x4000b010
Description interrupt mask
Reset 0x00000000

Name	RSVD																															passthru0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	passthru0	1=allow interrupt0 to be OR'ed into core interrupt	1'b0

Back to top

Name interrupt_status
Address 0x4000b014
Description interrupt status
Reset 0x00000000

Name	RSVD																															interrupt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R	interrupt0	saw a rising edge on done; independent of mask	1'b0

Back to top

Name set_interrupt
Address 0x4000b018

CONFIDENTIAL

Description interrupt set
Reset 0x00000000

Name	RSVD	set_interrupt0
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	set_interrupt0	set interrupt0; not auto cleared	1'b0

Back to top

Name reset_interrupt
Address 0x4000b01c
Description interrupt clear
Reset 0x00000000

Name	RSVD	reset_interrupt0
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	reset_interrupt0	clear interrupt0; not auto cleared	1'b0

Back to top

Name core_id
Address 0x4000bffc
Description core ID
Reset 0x53504920

Name	id
Bits	**31:0**

Bits	SW	Name	Description	Reset
31:0	R	id	SPI in ASCII	32'h53504920

Back to top

CONFIDENTIAL

QDEC

Name setup_for_axis_0
Address 0x4000c000
Description axis 0
Reset 0x00000030

Name	RSVD	purge_err_post_read	purge_accum_post_read	reverse_polarity	qsig_a_loopback_val	qsig_b_loopback_val	enable_loopback
Bits	**31:6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R/W	purge_err_post_read	purge the error count after a read to its register	1'b1
4	R/W	purge_accum_post_read	purge the accumulator count after a read to its register	1'b1
3	R/W	reverse_polarity	invert both phase signals before any processing	1'b0
2	R/W	qsig_a_loopback_val	the first phase signal if in loopback	1'b0
1	R/W	qsig_b_loopback_val	the second phase signal if in loopback	1'b0
0	R/W	enable_loopback	enable loopback	1'b0

Back to top

Name accum_purge_status_for_axis_0
Address 0x4000c004
Description axis 0
Reset 0x00000000

Name	RSVD	accum_purge_sts
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R	accum_purge_sts	indicates if purge is in progress; purge signal needs to be synch'ed to 32kHz domain; purge is done when bit makes high to low transition; alternatively software can either wait or not issue new read for 5 LPC cycles	1'b0

Back to top

CONFIDENTIAL

Name accumulator_value_for_axis_0
Address 0x4000c008
Description axis 0
Reset 0x00000000

Name	accum_val																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	accum_val	accumulator count	32'b0

Back to top

Name err_purge_status_for_axis_0
Address 0x4000c00c
Description axis 0
Reset 0x00000000

Name	RSVD																													err_purge_sts
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R	err_purge_sts	indicates if purge is in progress; purge signal needs to be synch'ed to 32kHz domain; purge is done when bit makes high to low transition; alternatively software can either wait or not issue new read for 5 LPC cycles	1'b0

Back to top

Name error_count_for_axis_0
Address 0x4000c010
Description axis 0
Reset 0x00000000

Name	err_cnt																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	err_cnt	error count	32'b0

Back to top

CONFIDENTIAL

Name setup_for_axis_1
Address 0x4000c014
Description axis 1
Reset 0x00000030

Name	RSVD	purge_err_post_read	purge_accum_post_read	reverse_polarity	qsig_a_loopback_val	qsig_b_loopback_val	enable_loopback
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R/W	purge_err_post_read	purge the error count after a read to its register	1'b1
4	R/W	purge_accum_post_read	purge the accumulator count after a read to its register	1'b1
3	R/W	reverse_polarity	invert both phase signals before any processing	1'b0
2	R/W	qsig_a_loopback_val	the first phase signal if in loopback	1'b0
1	R/W	qsig_b_loopback_val	the second phase signal if in loopback	1'b0
0	R/W	enable_loopback	enable loopback	1'b0

Back to top

Name accum_purge_status_for_axis_1
Address 0x4000c018
Description axis 1
Reset 0x00000000

Name	RSVD	accum_purge_sts
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R	accum_purge_sts	indicates if purge has completed	1'b0

Back to top

Name accumulator_value_for_axis_1
Address 0x4000c01c
Description axis 1
Reset 0x00000000

CONFIDENTIAL

Name	accum_val																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	accum_val	accumulator count	32'b0

Back to top

Name err_purge_status_for_axis_1
Address 0x4000c020
Description axis 1
Reset 0x00000000

Name	RSVD																													err_purge_sts
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R	err_purge_sts	indicates if purge has completed	1'b0

Back to top

Name error_count_for_axis_1
Address 0x4000c024
Description axis 1
Reset 0x00000000

Name	err_cnt																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	err_cnt	error count	32'b0

Back to top

Name setup_for_axis_2
Address 0x4000c028
Description axis 2
Reset 0x00000030

Name	R																												p	p	r	q	q	e

CONFIDENTIAL

	SVD																								purge_err_post_read	purge_accum_post_read	reverse_polarity	qsig_a_loopback_val	qsig_b_loopback_val	enable_loopback
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R/W	purge_err_post_read	purge the error count after a read to its register	1'b1
4	R/W	purge_accum_post_read	purge the accumulator count after a read to its register	1'b1
3	R/W	reverse_polarity	invert both phase signals before any processing	1'b0
2	R/W	qsig_a_loopback_val	the first phase signal if in loopback	1'b0
1	R/W	qsig_b_loopback_val	the second phase signal if in loopback	1'b0
0	R/W	enable_loopback	enable loopback	1'b0

Back to top

Name accum_purge_status_for_axis_2
Address 0x4000c02c
Description axis 2
Reset 0x00000000

Name	RSVD																													accum_purge_sts
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R	accum_purge_sts	indicates if purge has completed	1'b0

Back to top

Name accumulator_value_for_axis_2
Address 0x4000c030
Description axis 2
Reset 0x00000000

Name	accum_

CONFIDENTIAL

	val																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	accum_val	accumulator count	32'b0

Back to top

Name err_purge_status_for_axis_2
Address 0x4000c034
Description axis 2
Reset 0x00000000

Name	RSVD																													err_purge_sts
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R	err_purge_sts	indicates if purge has completed	1'b0

Back to top

Name error_count_for_axis_2
Address 0x4000c038
Description axis 2
Reset 0x00000000

Name	err_cnt																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	err_cnt	error count	32'b0

Back to top

Name core_id
Address 0x4000cffc
Description core ID
Reset 0x51444543

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	'QDEC' in ASCII	32'h51444543

CONFIDENTIAL

Back to top

SPI1

Name transaction_setup
Address 0x4000d000
Description transaction setup
Reset 0x00000000

Name	RSVD	dummy_cycles	csn_stays_low	loopback	opcode	clkdiv	rwb	num_data_bytes	start
Bits	**31 30**	**29 28 27 26**	**25**	**24**	**23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8 7 6**	**5**	**4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
29:26	R/W	dummy_cycles	number of extra shift cycles to help with turn-around	4'd0
25	R/W	csn_stays_low	at end of transaction, leave csn low	1'b0
24	R/W	loopback	if 1, mosi is internally connected to miso	1'b0
23:16	R/W	opcode	opcode	8'd0
15:6	R/W	clkdiv	freq_sclk = f_pclk/(2*(clkdiv+1))	10'd0
5	R/W	rwb	1=read 0=write; note it's possibly redundant with opcode	1'b0
4:1	R/W	num_data_bytes	just the data; don't add 1 for opcode	4'd0
0	R/W	start	FSM needs to see low to high transition to start; not auto cleared	1'b0

Back to top

Name transaction_status
Address 0x4000d004
Description transaction status
Reset 0x00000000

Name	RSVD	opcode_status	RSVD	running	done
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8**	**7 6 5 4 3 2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:8	R	opcode_status	opcode status	8'b0
1	R	running	transaction is running	1'b0
0	R	done	transaction is done	1'b0

Back to top

CONFIDENTIAL

Name data_bytes_lower
Address 0x4000d008
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte3								byte2								byte1								byte0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte3	data byte3	8'b0
23:16	R/W	byte2	data byte2	8'b0
15:8	R/W	byte1	data byte1	8'b0
7:0	R/W	byte0	data byte0	8'b0

Back to top

Name data_bytes_upper
Address 0x4000d00c
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte7								byte6								byte5								byte4							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte7	data byte7	8'b0
23:16	R/W	byte6	data byte6	8'b0
15:8	R/W	byte5	data byte5	8'b0
7:0	R/W	byte4	data byte4	8'b0

Back to top

Name interrupt_mask
Address 0x4000d010
Description interrupt mask
Reset 0x00000000

Name	RSVD																															passthru0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	passthru0	1=allow interrupt0 to be OR'ed into core interrupt	1'b0

Back to top

Name interrupt_status
Address 0x4000d014
Description interrupt status

CONFIDENTIAL

Reset 0x00000000

Name	RSVD	interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R	interrupt0	saw a rising edge on done; independent of mask	1'b0

Back to top

Name set_interrupt
Address 0x4000d018
Description interrupt set
Reset 0x00000000

Name	RSVD	set_interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	set_interrupt0	set interrupt0; not auto cleared	1'b0

Back to top

Name reset_interrupt
Address 0x4000d01c
Description interrupt clear
Reset 0x00000000

Name	RSVD	reset_interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

0	R/W	reset_interrupt0	clear interrupt0; not auto cleared	1'b0

Back to top

Name core_id
Address 0x4000dffc
Description core ID
Reset 0x53504920

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	SPI in ASCII	32'h53504920

Back to top

SLWTIMER

Name control
Address 0x4000e000
Description control
Reset 0x00000000

Name	RSVD																								reset_thres2	reset_thres1	reset_thres0	reset_expired	load	clear
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R/W	reset_thres2	reset thres2 interrupt; not self clearing	1'b0
4	R/W	reset_thres1	reset thres1 interrupt; not self clearing	1'b0
3	R/W	reset_thres0	reset thres0 interrupt; not self clearing	1'b0
2	R/W	reset_expired	reset expired interrupt; not self clearing	1'b0
1	R/W	load	load slow timer	1'b0
0	R/W	clear	clear slow timer	1'b0

Back to top

Name init_low
Address 0x4000e004
Description initial count value
Reset 0x00000000

Name	init_low																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
31:0	R/W	init_low	lower 32b of the 40b initial count value	32'b0

Back to top

Name init_high
Address 0x4000e008
Description initial count value
Reset 0x00000000

Name	RSVD																						init_high							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R/W	init_high	upper 8b of the 40b initial count value	8'b0

Back to top

Name cnt_low
Address 0x4000e00c
Description current count value
Reset 0x00000000

Name	cnt_low																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	cnt_low	lower 32b of the 40b current count value	32'b0

Back to top

Name cnt_high
Address 0x4000e010
Description current count value
Reset 0x00000000

Name	RSVD																						cnt_high							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	cnt_high	upper 8b of the 40b current count value	8'b0

Back to top

CONFIDENTIAL

Name thres0_low
Address 0x4000e014
Description threshold 0 value
Reset 0x00000000

Name	thres0_low																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	thres0_low	lower 32b of the 40b threshold 0 value; when counter decrements from one above to this value, an optional interrupt triggers	32'b0

Back to top

Name thres0_high
Address 0x4000e018
Description threshold 0 value
Reset 0x00000000

Name	RSVD																								thres0_high							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R/W	thres0_high	upper 8b of the 40b theshold 0 value	8'b0

Back to top

Name thres1_low
Address 0x4000e01c
Description threshold 1 value
Reset 0x00000000

Name	thres1_low																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	thres1_low	lower 32b of the 40b threshold 1 value; when counter decrements from one above to this value, an optional interrupt triggers	32'b0

CONFIDENTIAL

Back to top

Name thres1_high
Address 0x4000e020
Description threshold 1 value
Reset 0x00000000

Name	RSVD																						thres1_high							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R/W	thres1_high	upper 8b of the 40b theshold 1 value	8'b0

Back to top

Name thres2_low
Address 0x4000e024
Description threshold 2 value
Reset 0x00000000

Name	thres2_low																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	thres2_low	lower 32b of the 40b threshold 2 value; when counter decrements from one above to this value, an optional interrupt triggers	32'b0

Back to top

Name thres2_high
Address 0x4000e028
Description threshold 2 value
Reset 0x00000000

Name	RSVD																						thres2_high							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

7:0	R/W	thres2_high	upper 8b of the 40b theshold 2 value	8'b0

Back to top

Name status
Address 0x4000e02c
Description status information
Reset 0x00000000

Name	RSVD																													op_running
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R	op_running	the operation (either load or clear) is still running; handoff to 32kHz domain takes time	1'b0

Back to top

Name interrupt_status
Address 0x4000e030
Description interrupt status
Reset 0x00000000

Name	RSVD																												hit_thres2	hit_thres1	hit_thres0	expired
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R	hit_thres2	threshold 2 interrupt triggered	1'b0
2	R	hit_thres1	threshold 1 interrupt triggered	1'b0
1	R	hit_thres0	threshold 0 interrupt triggered	1'b0
0	R	expired	counter transitioned from 1 to 0	1'b0

Back to top

Name interrupt_mask
Address 0x4000e034
Description interrupt mask
Reset 0x00000000

Name	RSVD																												block_hit_t	block_hit_t	block_hit_t	block_expir

CONFIDENTIAL

																													hres2	hres1	hres0	ed
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R/W	block_hit_thres2	prevents thres2 from generating core interrupt	1'b0
2	R/W	block_hit_thres1	prevents thres1 from generating core interrupt	1'b0
1	R/W	block_hit_thres0	prevents thres0 from generating core interrupt	1'b0
0	R/W	block_expired	prevents counter expiring from generating core interrupt	1'b0

Back to top

Name core_id
Address 0x4000effc
Description core ID
Reset 0x53544d52

Name	id																															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	'STMR' in hex	32'h53544d52

Back to top

QSPI

Name transaction_setup
Address 0x4000f000
Description transaction setup
Reset 0x22000000

Name	RSVD		remote_spi_has_control	remote_ahb_qspi_has_control	RSVD		csn_val	clk_val	RSVD	dout_3_ctrl		RSVD		dout_2_ctrl		RSVD		dout_1_ctrl		RSVD		dout_0_ctrl		sample_din							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

29	R/W	remote_spi_has_control	the core at_apb_spi has control of qaud SPI interface	1'b1
28	R/W	remote_ahb_qspi_has_control	the core at_ahb_qspi has control of qaud SPI interface	1'b0
25	R/W	csn_val	value to drive on spi_csn	1'b1
24	R/W	clk_val	value to drive on spi_clk	1'b0
21:20	R/W	dout_3_ctrl	value to drive on spi_d3; {oe val, drive val}	2'h0
17:16	R/W	dout_2_ctrl	value to drive on spi_d2; {oe val, drive val}	2'h0
13:12	R/W	dout_1_ctrl	value to drive on spi_d1; {oe val, drive val}	2'h0
9:8	R/W	dout_0_ctrl	value to drive on spi_d0; {oe val, drive val}	2'h0
7:0	R/W	sample_din	sample high nibble or low nibble on din	8'h0

Back to top

Name read_data
Address 0x4000f004
Description read data
Reset 0x00000000

Name	RSVD																								rdata							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	rdata	sampled input data	8'b0

Back to top

Name override_din
Address 0x4000f008
Description override din
Reset 0x00000000

Name	RSVD																											override_din	din_override_val			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4	R/W	override_din	override din; useful for testing	1'b0
3:0	R/W	din_override_val	din values to force in	4'b0

Back to top

Name mode
Address 0x4000f00c
Description mode
Reset 0x00000002

Name	RS																														is_	is_

CONFIDENTIAL

	VD																												quad	dual
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	is_quad	higher priority than is_dual	1'b1
0	R/W	is_dual	for debug	1'b0

Back to top

Name remote_ahb_setup
Address 0x4000f010
Description remote ahb setup
Reset 0x02030000

Name	RSVD	slip_half_cycle	hyper	skew_csn_act_wen	serialize_pp_address	invalidate_entire_cache	enable_fine_clock_gating	enable_cache	enable_clocks	wdata_word_swap	wdata_halfword_swap	wdata_byte_swap	is_macronix	rdata_byte_swap	clkdivsel	opcode	RSVD	is_opcode	mode	dummy_cycles
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17:16**	**15:8**	**7**	**6**	**5:4**	**3:0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
30	R/W	slip_half_cycle	sample hald cycle later; useful to meeting interface timing in hyper	1'b0
29	R/W	hyper	core runs at 32MHz instead of 16MHz	1'b0
28	R/W	skew_csn_act_wen	in a write sequence, controller stalls the start of the sequence; set if div-by-1	1'b0
27	R/W	serialize_pp_address	serialize page program address (even though operating in quad mode)	1'b0
26	R/W	invalidate_entire_cache	invalidate entire cache	1'b0
25	R/W	enable_fine_clock_gating	enable fine clock gating	1'b1
24	R/W	enable_cache	enable cache	1'b0
23	R/W	enable_clocks	for the core clocks not the qspi interface clock	1'b0
22	R/W	wdata_word_swap	if doing word write, swap bytes within word coming from AHB	1'b0
21	R/W	wdata_halfword_swap	if doing half word write, swap bytes within halfword coming from AHB	1'b0
20	R/W	wdata_byte_swap	if doing byte write, swap order in which bytes are pulled from AHB word	1'b0
19	R/W	is_macronix	otherwise it is assumed to be micron	1'b0
18	R/W	rdata_byte_swap	does a byte swap before passing rdata to backplane	1'b0
17:16	R/W	clkdivsel	0=1x 1=2x 2=4x 3=8x	2'd3
15:8	R/W	opcode	opcode for read	8'h0
6	R/W	is_opcode	Is an opcode required?	1'b0
5:4	R/W	mode	0,3=single 1=dual 2=quad	2'd0
3:0	R/W	dummy_cycles	number of dummy cycles	4'd0

Back to top

Name remote_ahb_setup_2
Address 0x4000f014

Description remote ahb setup
Reset 0x20050206

Name	opcode_se								opcode_wip								opcode_pp								opcode_we							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	opcode_se	opcode for sector erase	8'h20
23:16	R/W	opcode_wip	opcode for write in progress	8'h05
15:8	R/W	opcode_pp	opcode for page program	8'h02
7:0	R/W	opcode_we	opcode for write enable	8'h06

Back to top

Name remote_ahb_setup_3
Address 0x4000f018
Description remote ahb setup
Reset 0x01300000

Name	check_wle	wip_bit			RSVD			wip_polarity	wle_bit			wle_polarity	expm	enable_performance_mode	RSVD																	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R/W	check_wle	hardware controller should check state of 'write enable latch' bit before page program or sector erase	1'b0
30:28	R/W	wip_bit	specifies which bit is the 'wip' indicator	3'd0
24	R/W	wip_polarity	specifies polarity of 'wip' indicator	1'b1
23:21	R/W	wle_bit	specifies which bit is the 'wle' indicator	3'd1
20	R/W	wle_polarity	specifies polarity of 'wle' indicator	1'b1
19	R/W	expm	exit performance_enahace mode; if set, this code is executed prior to each pp/se transaction	1'b0
18	R/W	enable_performance_mode	if set, performance mode is enabled	1'b0

Back to top

Name remote_ahb_setup_4
Address 0x4000f01c
Description remote ahb setup

CONFIDENTIAL

Reset 0x00000000

Name	RSVD	invert_addr
Bits	**31 30 29 28 27 26 25 24**	**23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
23:0	R/W	invert_addr	invert address on a bit by bit basis	24'd0

Back to top

Name remote_ahb_setup_5
Address 0x4000f020
Description remote ahb setup
Reset 0x02008200

Name	RSVD	stall_we2pp	stall_wle	pp_stall_wip
Bits	**31 30**	**29 28 27 26 25 24 23 22 21 20**	**19 18 17 16 15 14 13 12 11 10**	**9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
29:20	R/W	stall_we2pp	wen or wle to pp depending on check_wle setting	10'd32
19:10	R/W	stall_wle	stall duration between each checking of wle if check_wle is set	10'd32
9:0	R/W	pp_stall_wip	When checking after page program for completion, specifies how long to wait between 'wip' checks. Large values mean less power consumption on interface but possibly longer than necessary wait.	10'd512

Back to top

Name remote_ahb_setup_6
Address 0x4000f024
Description remote ahb setup
Reset 0x00001fff

Name	RSVD	se_stall_wip
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
15:0	R/W	se_stall_wip	When checking after sector erase for completion, specifies how long to wait	16'd8191

CONFIDENTIAL

			between 'wip' checks. Large values mean less power consumption on interface but possibly longer than necessary wait.	

Back to top

Name core_id
Address 0x4000fffc
Description core ID
Reset 0x51535049

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	'QSPI' in ASCII	32'h51535049

Back to top

GPIO0

Name DATA
Address 0x40010000
Description Data value[15:0]
Reset 0x00000000

Name	RSVD																Data															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	Data	If doing a read, result contains values sampled at pin. Read back value goes through double flip-flop synchronization logic with a delay of two cycles. If doing a write, the data output register is loaded. Note how the output data can be set either through this register or with the DATAOUT register.	16'b0

Back to top

Name DATAOUT
Address 0x40010004
Description Data output Register value [15:0]
Reset 0x00000000

Name	RSVD																Dataout															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	Dataout	Read - Current value of data output register. Write - To data output register.	16'b0

Back to top

Name PULLUP_ENABLE_SET
Address 0x40010008

CONFIDENTIAL

Description pullup enable set [15:0]
Reset 0x00000000

Name	RSVD															pullup_enable_set															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	pullup_enable_set	Write 1 Sets the pullup enable bit. Write 0 No effect. Read back 0 Indicates the pin is not pulled up. Read back 1 Indicates the pin is pulled up.	16'b0

Back to top

Name PULLUP_ENABLE_CLR
Address 0x4001000c
Description pullup enable clear [15:0]
Reset 0x00000000

Name	RSVD															pullup_enable_clr															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	pullup_enable_clr	Write 1 Clears the pullup enable bit. Write 0 No effect. Read back 0 Indicates the pin is not pulled up. Read back 1 Indicates the pin is pulled up.	16'b0

Back to top

Name INENABLE_SET
Address 0x40010010
Description Input enable set [15:0]
Reset 0x00000000

Name	RS	ine

CONFIDENTIAL

	RSVD	inenable_set
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
15:0	R/W	inenable_set	Write 1 Sets the input enable bit. Write 0 No effect. Read back 0 Indicates the input signal is disabled. Read back 1 Indicates the input signal is enabled.	16'b0

Back to top

Name INENABLE_CLR
Address 0x40010014
Description Input enable clear [15:0]
Reset 0x00000000

Name	RSVD	inenable_clr
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
15:0	R/W	inenable_clr	Write 1 Clears the input enable bit. Write 0 No effect. Read back 0 Indicates the input signal is disabled. Read back 1 Indicates the input signal is enabled.	16'b0

Back to top

Name OUTENSET
Address 0x40010018
Description Output enable set [15:0]
Reset 0x00000000

Name	RSVD	outenset
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
15:0	R/W	outenset	Write 1 Sets the output enable bit. Write 0 No effect. Read back 0 Indicates the signal direction as input. Read back 1 Indicates the signal direction as output.	16'b0

Back to top

CONFIDENTIAL

Name OUTENCLR
Address 0x4001001c
Description Output enable clear [15:0]
Reset 0x00000000

Name	RSVD																outenclr															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	outenclr	Write 1 Clears the output enable bit. Write 0 No effect. Read back 0 Indicates the signal direction as input. Read back 1 Indicates the signal direction as output.	16'b0

Back to top

Name ALTFUNCSET
Address 0x40010020
Description Alternative function set [15:0]
Reset 0x00000000

Name	RSVD																altfuncset															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	altfuncset	Write 1 Sets the ALTFUNC bit. Write 0 No effect. Read back 0 For I/O. Read back 1 For an alternate function.	16'b0

Back to top

Name ALTFUNCCLR
Address 0x40010024
Description Alternative function clear [15:0]
Reset 0x00000000

Name	RSVD																altfuncclr															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	altfuncclr	Write 1 Clears the ALTFUNC bit. Write 0 No effect.	16'b0

CONFIDENTIAL

			Read back 0 For I/O. Read back 1 For an alternate function.	

Back to top

Name INTENSET
Address 0x40010028
Description Interrupt enable set [15:0]
Reset 0x00000000

Name	RSVD	intenset
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
15:0	R/W	intenset	Write 1 Sets the interrupt enable bit. Write 0 No effect. Read back 0 Interrupt disabled. Read back 1 Interrupt enabled.	16'b0

Back to top

Name INTENCLR
Address 0x4001002c
Description Interrupt enable clear [15:0]
Reset 0x00000000

Name	RSVD	intenclr
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
15:0	R/W	intenclr	Write 1 Clears the interrupt enable bit. Write 0 No effect. Read back 0 Interrupt disabled. Read back 1 Interrupt enabled.	16'b0

Back to top

Name INTTYPESET
Address 0x40010030
Description Interrupt type set [15:0]
Reset 0x00000000

Name	RSVD	inttypeset
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
15:0	R/W	inttypeset	Write 1 Sets the interrupt type bit. Write 0 No effect. Read back 0 For LOW or HIGH level. Read back 1 For falling edge or rising edge.	16'b0

Back to top

Name INTTYPECLR
Address 0x40010034
Description Interrupt type clear [15:0]
Reset 0x00000000

Name	RSVD																inttypeclr															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	inttypeclr	Write 1 Clears the interrupt enable bit. Write 0 No effect. Read back 0 For LOW or HIGH level. Read back 1 For falling edge or rising edge.	16'b0

Back to top

Name INTPOLSET
Address 0x40010038
Description Polarity-level, edge IRQ configuration [15:0]
Reset 0x00000000

Name	RSVD																intpolset															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	intpolset	Write 1 Sets the interrupt polarity bit. Write 0 No effect. Read back 0 For LOW level or falling edge. Read back 1 For HIGH level or rising edge.	16'b0

Back to top

Name INTPOLCLR
Address 0x4001003c
Description Interrupt type clear [15:0]
Reset 0x00000000

Name	RSVD																inttyp															

CONFIDENTIAL

																														e c l r
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	inttypeclr	Write 1 Clears the interrupt polarity bit. Write 0 No effect. Read back 0 For LOW level or falling edge. Read back 1 For HIGH level or rising edge.	16'b0

Back to top

Name INTSTATUS_CLEAR
Address 0x40010040
Description Write one to clear interrupt request:
Reset 0x00000000

Name	R S V D	i n t s t a t u s _ c l e a r
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16**	**15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
15:0	R/W	intstatus_clear	Write 1 clears the interrupt request. Write 0 No effect. Read back IRQ status Register.	16'b0

Back to top

Name MASKLOWBYTE
Address 0x40010400
Description Lower eight bits masked access. The register actually spans from 0x100 to 0x1FF
Reset 0x00000000

Name	R S V D	m a s k l o w b y t e
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8**	**7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
7:0	R/W	masklowbyte	Data for lower byte access, whith bits[9:2] of address value used as enable mask for each bit.	8'b0

Back to top

Name MASKHIGHBYTE
Address 0x40010800

CONFIDENTIAL

Description Higher eight bits masked access. The register actually spans from 0x200 to 0x2FF
Reset 0x00000000

Name	RSVD																maskhighbyte								RSVD							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:8	R/W	maskhighbyte	Data for higher byte access, whith bits[9:2] of address value used as enable mask for each bit.	8'b0

Back to top

GPIO1

Name DATA
Address 0x40011000
Description Data value[15:0]
Reset 0x00000000

Name	RSVD																Data															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	Data	If doing a read, result contains values sampled at pin. Read back value goes through double flip-flop synchronization logic with a delay of two cycles. If doing a write, the data output register is loaded. Note how the output data can be set either through this register or with the DATAOUT register.	16'b0

Back to top

Name DATAOUT
Address 0x40011004
Description Data output Register value [15:0]
Reset 0x00000000

Name	RSVD																Dataout															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	Dataout	Read - Current value of data output register. Write - To data output register.	16'b0

Back to top

Name PULLUP_ENABLE_SET

CONFIDENTIAL

Address 0x40011008
Description pullup enable set [15:0]
Reset 0x00000000

Name	RSVD															pullup_enable_set															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	pullup_enable_set	Write 1 Sets the pullup enable bit. Write 0 No effect. Read back 0 Indicates the pin is not pulled up. Read back 1 Indicates the pin is pulled up.	16'b0

Back to top

Name PULLUP_ENABLE_CLR
Address 0x4001100c
Description pullup enable clear [15:0]
Reset 0x00000000

Name	RSVD															pullup_enable_clr															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	pullup_enable_clr	Write 1 Clears the pullup enable bit. Write 0 No effect. Read back 0 Indicates the pin is not pulled up. Read back 1 Indicates the pin is pulled up.	16'b0

Back to top

Name INENABLE_SET
Address 0x40011010
Description Input enable set [15:0]
Reset 0x00000000

Name	RS															in															

CONFIDENTIAL

	RSVD	inenable_set
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
15:0	R/W	inenable_set	Write 1 Sets the input enable bit. Write 0 No effect. Read back 0 Indicates the input signal is disabled. Read back 1 Indicates the input signal is enabled.	16'b0

Back to top

Name INENABLE_CLR
Address 0x40011014
Description Input enable clear [15:0]
Reset 0x00000000

Name	RSVD	inenable_clr
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
15:0	R/W	inenable_clr	Write 1 Clears the input enable bit. Write 0 No effect. Read back 0 Indicates the input signal is disabled. Read back 1 Indicates the input signal is enabled.	16'b0

Back to top

Name OUTENSET
Address 0x40011018
Description Output enable set [15:0]
Reset 0x00000000

Name	RSVD	outenset
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
15:0	R/W	outenset	Write 1 Sets the output enable bit. Write 0 No effect. Read back 0 Indicates the signal direction as input. Read back 1 Indicates the signal direction as output.	16'b0

Back to top

CONFIDENTIAL

Name OUTENCLR
Address 0x4001101c
Description Output enable clear [15:0]
Reset 0x00000000

Name	RSVD															outenclr															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	outenclr	Write 1 Clears the output enable bit. Write 0 No effect. Read back 0 Indicates the signal direction as input. Read back 1 Indicates the signal direction as output.	16'b0

Back to top

Name ALTFUNCSET
Address 0x40011020
Description Alternative function set [15:0]
Reset 0x00000000

Name	RSVD															altfuncset															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	altfuncset	Write 1 Sets the ALTFUNC bit. Write 0 No effect. Read back 0 For I/O. Read back 1 For an alternate function.	16'b0

Back to top

Name ALTFUNCCLR
Address 0x40011024
Description Alternative function clear [15:0]
Reset 0x00000000

Name	RSVD															altfunccl r															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	altfuncclr	Write 1 Clears the ALTFUNC bit. Write 0 No effect.	16'b0

CONFIDENTIAL

			Read back 0 For I/O. Read back 1 For an alternate function.	

Back to top

Name INTENSET
Address 0x40011028
Description Interrupt enable set [15:0]
Reset 0x00000000

Name	RSVD	intenset

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
15:0	R/W	intenset	Write 1 Sets the interrupt enable bit. Write 0 No effect. Read back 0 Interrupt disabled. Read back 1 Interrupt enabled.	16'b0

Back to top

Name INTENCLR
Address 0x4001102c
Description Interrupt enable clear [15:0]
Reset 0x00000000

Name	RSVD	intenclr

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
15:0	R/W	intenclr	Write 1 Clears the interrupt enable bit. Write 0 No effect. Read back 0 Interrupt disabled. Read back 1 Interrupt enabled.	16'b0

Back to top

Name INTTYPESET
Address 0x40011030
Description Interrupt type set [15:0]
Reset 0x00000000

Name	RSVD	inttypeset

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

CONFIDENTIAL

Bits	SW	Name	Description	Reset
15:0	R/W	inttypeset	Write 1 Sets the interrupt type bit. Write 0 No effect. Read back 0 For LOW or HIGH level. Read back 1 For falling edge or rising edge.	16'b0

Back to top

Name INTTYPECLR
Address 0x40011034
Description Interrupt type clear [15:0]
Reset 0x00000000

Name	RSVD																inttypeclr															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	inttypeclr	Write 1 Clears the interrupt enable bit. Write 0 No effect. Read back 0 For LOW or HIGH level. Read back 1 For falling edge or rising edge.	16'b0

Back to top

Name INTPOLSET
Address 0x40011038
Description Polarity-level, edge IRQ configuration [15:0]
Reset 0x00000000

Name	RSVD																intpolset															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	intpolset	Write 1 Sets the interrupt polarity bit. Write 0 No effect. Read back 0 For LOW level or falling edge. Read back 1 For HIGH level or rising edge.	16'b0

Back to top

Name INTPOLCLR
Address 0x4001103c
Description Interrupt type clear [15:0]
Reset 0x00000000

Name	RSVD																inttyp															

CONFIDENTIAL

																														e c l r
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	inttypeclr	Write 1 Clears the interrupt polarity bit. Write 0 No effect. Read back 0 For LOW level or falling edge. Read back 1 For HIGH level or rising edge.	16'b0

Back to top

Name INTSTATUS_CLEAR
Address 0x40011040
Description Write one to clear interrupt request:
Reset 0x00000000

Name	R S V D																													i n t s t a t u s _ c l e a r
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	intstatus_clear	Write 1 clears the interrupt request. Write 0 No effect. Read back IRQ status Register.	16'b0

Back to top

Name MASKLOWBYTE
Address 0x40011400
Description Lower eight bits masked access. The register actually spans from 0x100 to 0x1FF
Reset 0x00000000

Name	R S V D																								m a s k l o w b y t e							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R/W	masklowbyte	Data for lower byte access, whith bits[9:2] of address value used as enable mask for each bit.	8'b0

Back to top

Name MASKHIGHBYTE
Address 0x40011800

CONFIDENTIAL

Description Higher eight bits masked access. The register actually spans from 0x200 to 0x2FF
Reset 0x00000000

Name	RSVD																maskhighbyte								RSVD							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:8	R/W	maskhighbyte	Data for higher byte access, whith bits[9:2] of address value used as enable mask for each bit.	8'b0

Back to top

SYSCTRL

Name MEM_CTRL
Address 0x4001f000
Description Contains register fields associated with MEM_CTRL.
Reset 0x00000001

Name	RSVD																														REMAP
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	REMAP	If 1, enable remap feature.	1'b1

Back to top

Name SYS_CTRL
Address 0x4001f008
Description Contains register fields associated with SYS_CTRL.
Reset 0x00000000

Name	RSVD																														LOCKUPRESETEN
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	LOCKUPRESETEN	If 1, system reset will be generated if processor is in lockup state.	1'b0

CONFIDENTIAL

Back to top

Name Reset_Information
Address 0x4001f010
Description Contains register fields associated with Reset_Information.
Reset 0x00000000

Name	RSVD																												LOCKUPRESET	WDOGRESETREQ	SYSRESETREQ
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
2	R/W	LOCKUPRESET	If 1, processor lockup caused the reset. Write 1 to clear.	1'b0
1	R/W	WDOGRESETREQ	If 1, watchdog caused the reset. Write 1 to clear.	1'b0
0	R/W	SYSRESETREQ	If 1, sysresetq caused the reset. Write 1 to clear.	1'b0

Back to top

I2C0

Name transaction_setup
Address 0x50000000
Description setup i2c transaction
Reset 0x00000000

Name	RSVD											go	RSVD		ack_value_to_drive	master_drives_ack	RSVD										tail		RSVD		head	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
20	R/W	go	Transaction lauches on rising edge. Not self reseting.	1'b0
17	R/W	ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'd0
16	R/W	master_drives_ack	Specifies that master is to drive ack.	1'd0
5:4	R/W	tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'd0
1:0	R/W	head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'd0

CONFIDENTIAL

Back to top

Name clock_control
Address 0x50000004
Description clock behavior
Reset 0x00000007

Name	RSVD	clk_pu	clkdiv
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11**	**10**	**9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
10	R/W	clk_pu	sck pin pullup enable	1'b0
9:0	R/W	clkdiv	freq_sclk = f_pclk/(4*(clkdiv+1))	10'd7

Back to top

Name outgoing_data
Address 0x50000008
Description outbound data
Reset 0x0000ff00

Name	RSVD	data_pu	data_oe	data_o
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17**	**16**	**15 14 13 12 11 10 9 8**	**7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
16	R/W	data_pu	sda pin pullup enable; if no external pullup resistor, this is needed to allow logic high when the slave device reponses.	1'b0
15:8	R/W	data_oe	sda output enable per bit	8'hff
7:0	R/W	data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'b0

Back to top

Name io_ctrl
Address 0x5000000c
Description input enable control
Reset 0x00000000

Name	RSVD	data_ie_ovrd
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	data_ie_ovrd	1 = ie is high	1'b0

CONFIDENTIAL

			0 = ie is ~oe	

Back to top

Name incoming_data
Address 0x50000010
Description inbound data
Reset 0x00000000

Name	RSVD																						data_i							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	data_i	inbound data	8'b0

Back to top

Name transaction_status
Address 0x50000014
Description transaction status
Reset 0x00000000

Name	RSVD																												running	ack_value
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R	running	Is transaction running?	1'b0
0	R	ack_value	What was most recent ack/nack value? Useless if mast_drives_ack is set.	1'b0

Back to top

Name interrupt_mask
Address 0x50000018
Description interrupt mask
Reset 0x00000000

Name	RSVD																													passthru0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	passthru0	1 = allow interrupt0 to be OR'ed into core interrupt	1'b0

Back to top

Name interrupt_status

CONFIDENTIAL

Address 0x5000001c
Description interrupt status
Reset 0x00000000

Name	RSVD	interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R	interrupt0	saw a rising edge on done; independent of mask	1'b0

Back to top

Name set_interrupt
Address 0x50000020
Description interrupt set
Reset 0x00000000

Name	RSVD	set_interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	set_interrupt0	set interrupt0; useful for software debug and testing; not auto cleared	1'b0

Back to top

Name reset_interrupt
Address 0x50000024
Description interrupt clear
Reset 0x00000000

Name	RSVD	reset_interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
0	R/W	reset_interrupt0	clear interrupt0; not auto cleared	1'b0

Back to top

Name id
Address 0x50000ffc
Description core ID
Reset 0x49324320

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	I2C in ASCII	32'h49324320

Back to top

I2C1

Name transaction_setup
Address 0x50001000
Description setup i2c transaction
Reset 0x00000000

Name	RSVD											go	RSVD		ack_value_to_drive	master_drives_ack	RSVD												tail		RSVD		head	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
20	R/W	go	Transaction lauches on rising edge. Not self reseting.	1'b0
17	R/W	ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'd0
16	R/W	master_drives_ack	Specifies that master is to drive ack.	1'd0
5:4	R/W	tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'd0
1:0	R/W	head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'd0

Back to top

Name clock_control
Address 0x50001004
Description clock behavior

CONFIDENTIAL

Reset 0x00000007

Name	RSVD																				clk_pu	clkdiv									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
10	R/W	clk_pu	sck pin pullup enable	1'b0
9:0	R/W	clkdiv	freq_sclk = f_pclk/(4*(clkdiv+1))	10'd7

Back to top

Name outgoing_data
Address 0x50001008
Description outbound data
Reset 0x0000ff00

Name	RSVD															data_pu	data_oe								data_o							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
16	R/W	data_pu	sda pin pullup enable; if no external pullup resistor, this is needed to allow logic high when the slave device reponses.	1'b0
15:8	R/W	data_oe	sda output enable per bit	8'hff
7:0	R/W	data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'b0

Back to top

Name io_ctrl
Address 0x5000100c
Description input enable control
Reset 0x00000000

Name	RSVD																														data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name incoming_data
Address 0x50001010

CONFIDENTIAL

Description inbound data
Reset 0x00000000

Name	RSVD																						data_i							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	data_i	inbound data	8'b0

Back to top

Name transaction_status
Address 0x50001014
Description transaction status
Reset 0x00000000

Name	RSVD																											running	ack_value
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R	running	Is transaction running?	1'b0
0	R	ack_value	What was most recent ack/nack value? Useless if mast_drives_ack is set.	1'b0

Back to top

Name interrupt_mask
Address 0x50001018
Description interrupt mask
Reset 0x00000000

Name	RSVD																												passthru0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	passthru0	1 = allow interrupt0 to be OR'ed into core interrupt	1'b0

Back to top

Name interrupt_status
Address 0x5000101c
Description interrupt status
Reset 0x00000000

Name	RS	in

CONFIDENTIAL

	VD																													interrupt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R	interrupt0	saw a rising edge on done; independent of mask	1'b0

Back to top

Name set_interrupt
Address 0x50001020
Description interrupt set
Reset 0x00000000

Name	RSVD																													set_interrupt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	set_interrupt0	set interrupt0; useful for software debug and testing; not auto cleared	1'b0

Back to top

Name reset_interrupt
Address 0x50001024
Description interrupt clear
Reset 0x00000000

Name	RSVD																													reset_interrupt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	reset_interrupt0	clear interrupt0; not auto cleared	1'b0

Back to top

CONFIDENTIAL

Name id
Address 0x50001ffc
Description core ID
Reset 0x49324320

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	I2C in ASCII	32'h49324320

Back to top

NVM

Name opmode
Address 0x50002000
Description Contains register fields associated with opmode.
Reset 0x00000000

Name	go	RSVD																									auto_inc	word_size		program	read
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R/W	go	Requested operation starts when rising edge on 'go' is detected.	1'b0
4	R/W	auto_inc	0 = read address is not auto-incremented 1 = read address is auto-inc	1'b0
3:2	R/W	word_size	0 = byte-read 1 = half-word read 2 = 3-bytes read 3 = word read	2'b0
1	R/W	program	Indicates that the 'go' request is to program to the NVM.	1'b0
0	R/W	read	Indicates that the 'go' request is to read from the NVM.	1'b0

Back to top

Name address
Address 0x50002004
Description Contains register fields associated with address.
Reset 0x00000000

Name	RSVD																address														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
14:0	R/W	address	The whole address space is made up of 4 blocks of 8kb efuse. If not programmed, fuse value is 0. Programming is not reversible. If opmode is program, specify per fuse using address bits [12:0] and specify the block with	15'b0

CONFIDENTIAL

			bits [14:13]. if opmode is read, specify per byte using address bits [9:0] and specify the block with bits [11:10].	

Back to top

Name status
Address 0x50002008
Description status information
Reset 0x00000000

Name	running	RSVD																												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R	running	core's busy indicator	1'b0

Back to top

Name read_data
Address 0x5000200c
Description read data buffer
Reset 0x00000000

Name	read_data																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	read_data	read data buffer	32'b0

Back to top

Name t_sp_pg_avdd
Address 0x50002010
Description Contains register fields associated with t_sp_pg_avdd.
Reset 0x0000000f

Name	RSVD																						cycles									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
9:0	R/W	cycles	want >= 1000 nsec; actual delay is programmed number plus 2	10'd15

Back to top

Name t_sp_pgm
Address 0x50002014

CONFIDENTIAL

Description Contains register fields associated with t_sp_pgm.
Reset 0x00000000

Name	RSVD																				cycles									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
9:0	R/W	cycles	want >= 100 nsec; actual delay is programmed number plus 2	10'd0

Back to top

Name t_pgm
Address 0x50002018
Description Contains register fields associated with t_pgm.
Reset 0x0000008f

Name	RSVD																				cycles									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
9:0	R/W	cycles	want >= 9000 nsec and <= 11000 nsec; actual delay is programmed number plus 2	10'd143

Back to top

Name t_hp_pgm
Address 0x5000201c
Description Contains register fields associated with t_hp_pgm.
Reset 0x00000000

Name	RSVD																				cycles									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
9:0	R/W	cycles	want >= 100 nsec; actual delay is programmed number plus 2	10'd0

Back to top

Name t_hp_pg_avdd
Address 0x50002020
Description Contains register fields associated with t_hp_pg_avdd.
Reset 0x0000000f

Name	RSVD																				cycles									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
9:0	R/W	cycles	want >= 1000 nsec; actual delay is programmed number plus 2	10'd15

Back to top

Name t_sr_rd
Address 0x50002024
Description Contains register fields associated with t_sr_rd.
Reset 0x00000001

Name	RSVD																					cycles									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
9:0	R/W	cycles	want >= 100 nsec; actual delay is programmed number plus 1	10'd1

Back to top

Name t_rd
Address 0x50002028
Description Contains register fields associated with t_rd.
Reset 0x00000000

Name	RSVD																					cycles									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
9:0	R/W	cycles	want >= 40 nsec; actual delay is programmed number + 1	10'd0

Back to top

Name t_hr_rd
Address 0x5000202c
Description Contains register fields associated with t_hr_rd.
Reset 0x00000001

Name	RSVD																					cycles									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
9:0	R/W	cycles	want >= 100 nsec; actual delay is programmed number plus 1	10'd1

Back to top

Name t_setup
Address 0x50002030
Description Contains register fields associated with t_setup.
Reset 0x00000000

CONFIDENTIAL

Name	RSVD																									cycles				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	cycles	want >= 10 nsec; actual delay is programmed number + 1	5'd0

Back to top

Name t_hold
Address 0x50002034
Description Contains register fields associated with t_hold.
Reset 0x00000000

Name	RSVD																									cycles				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	cycles	want >= 10 nsec; actual delay is programmed number + 1	5'd0

Back to top

Name t_addr
Address 0x50002038
Description Contains register fields associated with t_addr.
Reset 0x00000000

Name	RSVD																									cycles				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	cycles	want >= 10 nsec; actual delay is programmed number + 1	5'd0

Back to top

Name interrupt_mask
Address 0x5000203c
Description interrupt mask
Reset 0x00000000

Name	RSVD																													passthru0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
0	R/W	passthru0	1=allow interrupt0 to be OR'ed into core interrupt	1'b0

Back to top

Name interrupt_status
Address 0x50002040
Description interrupt status
Reset 0x00000000

Name	RSVD	interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R	interrupt0	saw a rising edge on done; independent of mask	1'b0

Back to top

Name set_interrupt
Address 0x50002044
Description Contains register fields associated with set_interrupt.
Reset 0x00000000

Name	RSVD	set_interrupt0
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	set_interrupt0	set the interrupt; not auto cleared; useful for software testing	1'b0

Back to top

Name reset_interrupt
Address 0x50002048
Description Contains register fields associated with reset_interrupt.
Reset 0x00000000

Name	RSVD	reset_inte

CONFIDENTIAL

																														reset_interrupt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	reset_interrupt0	clear the interrupt; not auto cleared	1'b0

Back to top

Name efuse_autoread
Address 0x5000204c
Description read data from last word of efuse; HW initiated after each PORESETn de-assertion
Reset 0x00000000

Name	reserved2			otp_batt_type		otp_harv_type	otp_gen_vddIO	otp_nabg_trim									RSVD			otp_brownout_thr			RSVD						swd_disable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R	no_32khz_xtal_on_board	no_32khz_xtal_on_board 0 = external 32.768kHz crystal present 1 = no external 32.768kHz crystal	3'b0
30		disable_otp_writes	disable_otp_writes 0 = OTP writes allowed 1 = OTP writes disabled	
29		otp_disable_vdd	otp_disable_vdd 0 = internal DVDD1 and VDD1A 1 = external DVDD1 and VDD1A	
28:27	R	otp_batt_type	battery type 0 = Battery <= 1.8V 1 = Battery > 1.8V 2 = No Battery 3 = Rechargeable Battery > 1.8V	2'b0
26	R	otp_harv_type	harvester type 0 = non-RF harvesting 1 = RF harvesting	1'b0
25	R	otp_disable_vddIO	VDDIO generation 0 = internal VDDIO/VDDIOA 1 = external VDDIO/VDDIOA	1'b0
24	R	ignore_32khz_xtal_check	ignore_32khz_xtal_check 0 = wait for external 32.768kHz crystal to stabilize before entering low power states 1 = do not wait for external 32.768kHz crystal to stabilize before entering low power states otp_nabg_trim = trim bits for native bandgap	10'b0
11:9	R	otp_brownout_thr	VBAT brownout threshold level 25mV*(38+otp_brownout_thr) for otp_batt_type = 0 50mV*(38+otp_brownout_thr) for otp_batt_type = 1 or 3	3'b0
0	R	swd_disable	Disable the SWD interface. 0 = enable SWD 1 = disable SWD	1'b0

CONFIDENTIAL

Back to top

Name id
Address 0x50002ffc
Description core ID
Reset 0x4e564d20

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	NVM in ASCII	32'h4e564d20

Back to top

WRPR1

Name apb0_ctrl
Address 0x50003000
Description Contains register fields associated with apb0_ctrl.
Reset 0x00000002

Name	RSVD																												sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 0 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 0 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb1_ctrl
Address 0x50003004
Description Contains register fields associated with apb1_ctrl.
Reset 0x00000002

Name	RSVD																												sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

1	R/W	sreset	reset peripheral in APB slot 1 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 1 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb2_ctrl
Address 0x50003008
Description Contains register fields associated with apb2_ctrl.
Reset 0x00000002

Name	RSVD																													sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 2 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 2 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb3_ctrl
Address 0x5000300c
Description Contains register fields associated with apb3_ctrl.
Reset 0x00000002

Name	RSVD																													sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 3 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 3 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb4_ctrl
Address 0x50003010
Description Contains register fields associated with apb4_ctrl.
Reset 0x00000002

CONFIDENTIAL

Name	RSVD	sreset	clk_enable
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 4 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 4 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb5_ctrl
Address 0x50003014
Description Contains register fields associated with apb5_ctrl.
Reset 0x00000002

Name	RSVD	sreset	clk_enable
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 5 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 5 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb6_ctrl
Address 0x50003018
Description Contains register fields associated with apb6_ctrl.
Reset 0x00000002

Name	RSVD	sreset	clk_enable
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 6 1 = reset 0 = not reset	1'b1

CONFIDENTIAL

0	R/W	clk_enable	enable clock to peripheral in APB slot 6 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb7_ctrl
Address 0x5000301c
Description Contains register fields associated with apb7_ctrl.
Reset 0x00000002

Name	RSVD																														sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 7 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 7 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb8_ctrl
Address 0x50003020
Description Contains register fields associated with apb8_ctrl.
Reset 0x00000002

Name	RSVD																														sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 8 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 8 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb9_ctrl
Address 0x50003024
Description Contains register fields associated with apb9_ctrl.
Reset 0x00000002

Name	RS																														sre	clk

CONFIDENTIAL

	VD																											reset	clk_enable	
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 9 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 9 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb10_ctrl
Address 0x50003028
Description Contains register fields associated with apb10_ctrl.
Reset 0x00000002

Name	RSVD																											reset	clk_enable	
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 10 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 10 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb11_ctrl
Address 0x5000302c
Description Contains register fields associated with apb11_ctrl.
Reset 0x00000002

Name	RSVD																											reset	clk_enable	
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 11 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 11 1 = clock enabled	1'b0

CONFIDENTIAL

			0 = clock not enabled	

Back to top

Name apb12_ctrl
Address 0x50003030
Description Contains register fields associated with apb12_ctrl.
Reset 0x00000002

Name	RSVD																													sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 12 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 12 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb13_ctrl
Address 0x50003034
Description Contains register fields associated with apb13_ctrl.
Reset 0x00000002

Name	RSVD																													sreset	clk_enable
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 13 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 13 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb14_ctrl
Address 0x50003038
Description Contains register fields associated with apb14_ctrl.
Reset 0x00000002

Name	RSVD																													sres	clk_e

CONFIDENTIAL

																														e t	n a b l e
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 14 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 14 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name apb15_ctrl
Address 0x5000303c
Description Contains register fields associated with apb15_ctrl.
Reset 0x00000002

Name	R S V D																													s r e s e t	c l k _ e n a b l e
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	sreset	reset peripheral in APB slot 15 1 = reset 0 = not reset	1'b1
0	R/W	clk_enable	enable clock to peripheral in APB slot 15 1 = clock enabled 0 = clock not enabled	1'b0

Back to top

Name core_id
Address 0x50003ffc
Description Contains register fields associated with core_id.
Reset 0x57525052

Name	i d																														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	'WRPR' in ASCII	32'h57525052

Back to top

RADIO

Name transaction_setup
Address 0x50005000
Description transaction setup
Reset 0x00000000

CONFIDENTIAL

Name	RSVD		dummy_cycles				csn_stays_low	loopback	opcode								clkdiv								rwb	num_data_bytes				start
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
29:26	R/W	dummy_cycles	number of extra shift cycles to help with turn-around	4'd0
25	R/W	csn_stays_low	at end of transaction, leave csn low	1'b0
24	R/W	loopback	if 1, mosi is internally connected to miso	1'b0
23:16	R/W	opcode	opcode	8'd0
15:6	R/W	clkdiv	freq_sclk = f_pclk/(2*(clkdiv+1))	10'd0
5	R/W	rwb	1=read 0=write; note it's possibly redundant with opcode	1'b0
4:1	R/W	num_data_bytes	just the data; don't add 1 for opcode	4'd0
0	R/W	start	FSM needs to see low to high transition to start; not auto cleared	1'b0

Back to top

Name transaction_status
Address 0x50005004
Description transaction status
Reset 0x00000000

Name	RSVD																opcode_status								RSVD						running	done
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:8	R	opcode_status	opcode status	8'b0
1	R	running	transaction is running	1'b0
0	R	done	transaction is done	1'b0

Back to top

Name data_bytes_lower
Address 0x50005008
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte3								byte2								byte1								byte0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
31:24	R/W	byte3	data byte3	8'b0
23:16	R/W	byte2	data byte2	8'b0
15:8	R/W	byte1	data byte1	8'b0
7:0	R/W	byte0	data byte0	8'b0

Back to top

Name data_bytes_upper
Address 0x5000500c
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte7								byte6								byte5								byte4							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte7	data byte7	8'b0
23:16	R/W	byte6	data byte6	8'b0
15:8	R/W	byte5	data byte5	8'b0
7:0	R/W	byte4	data byte4	8'b0

Back to top

Name interrupt_mask
Address 0x50005010
Description interrupt mask
Reset 0x00000000

Name	RSVD	passthru0
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	passthru0	1=allow interrupt0 to be OR'ed into core interrupt	1'b0

Back to top

Name interrupt_status
Address 0x50005014
Description interrupt status
Reset 0x00000000

Name	RSVD	interrupt0
Bits	**31:1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
0	R	interrupt0	saw a rising edge on done; independent of mask	1'b0

Back to top

Name set_interrupt
Address 0x50005018
Description interrupt set
Reset 0x00000000

Name	RSVD	set_interrupt0
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	set_interrupt0	set interrupt0; not auto cleared	1'b0

Back to top

Name reset_interrupt
Address 0x5000501c
Description interrupt clear
Reset 0x00000000

Name	RSVD	reset_interrupt0
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	reset_interrupt0	clear interrupt0; not auto cleared	1'b0

Back to top

Name core_id
Address 0x50005ffc
Description core ID
Reset 0x53504920

Name	id

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
31:0	R	id	SPI in ASCII	32'h53504920

Back to top

PMU

Name transaction_setup
Address 0x50007000
Description transaction setup
Reset 0x00000000

Name	RSVD	dummy_cycles	csn_stays_low	loopback	opcode	clkdiv	rwb	num_data_bytes	start
Bits	31 30	29 28 27 26	25	24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6	5	4 3 2 1	0

Bits	SW	Name	Description	Reset
29:26	R/W	dummy_cycles	number of extra shift cycles to help with turn-around	4'd0
25	R/W	csn_stays_low	at end of transaction, leave csn low	1'b0
24	R/W	loopback	if 1, mosi is internally connected to miso	1'b0
23:16	R/W	opcode	opcode	8'd0
15:6	R/W	clkdiv	freq_sclk = f_pclk/(2*(clkdiv+1))	10'd0
5	R/W	rwb	1=read 0=write; note it's possibly redundant with opcode	1'b0
4:1	R/W	num_data_bytes	just the data; don't add 1 for opcode	4'd0
0	R/W	start	FSM needs to see low to high transition to start; not auto cleared	1'b0

Back to top

Name transaction_status
Address 0x50007004
Description transaction status
Reset 0x00000000

Name	RSVD	opcode_status	RSVD	running	done
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2	1	0

Bits	SW	Name	Description	Reset
15:8	R	opcode_status	opcode status	8'b0
1	R	running	transaction is running	1'b0

CONFIDENTIAL

0	R	done	transaction is done	1'b0

Back to top

Name data_bytes_lower
Address 0x50007008
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte3								byte2								byte1								byte0							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte3	data byte3	8'b0
23:16	R/W	byte2	data byte2	8'b0
15:8	R/W	byte1	data byte1	8'b0
7:0	R/W	byte0	data byte0	8'b0

Back to top

Name data_bytes_upper
Address 0x5000700c
Description preload if write; postread if read; readback data will change if num_data_bytes changes post done
Reset 0x00000000

Name	byte7								byte6								byte5								byte4							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte7	data byte7	8'b0
23:16	R/W	byte6	data byte6	8'b0
15:8	R/W	byte5	data byte5	8'b0
7:0	R/W	byte4	data byte4	8'b0

Back to top

Name interrupt_mask
Address 0x50007010
Description interrupt mask
Reset 0x00000000

Name	RSVD																														passthru0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	passthru0	1=allow interrupt0 to be OR'ed into core interrupt	1'b0

Back to top

CONFIDENTIAL

Name interrupt_status
Address 0x50007014
Description interrupt status
Reset 0x00000000

Name	RSVD	interrupt0
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R	interrupt0	saw a rising edge on done; independent of mask	1'b0

Back to top

Name set_interrupt
Address 0x50007018
Description interrupt set
Reset 0x00000000

Name	RSVD	set_interrupt0
Bits	**31:1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	set_interrupt0	set interrupt0; not auto cleared	1'b0

Back to top

Name reset_interrupt
Address 0x5000701c
Description interrupt clear
Reset 0x00000000

Name	RSVD	reset_interrupt0
Bits	**31:1**	**0**

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
0	R/W	reset_interrupt0	clear interrupt0; not auto cleared	1'b0

Back to top

Name core_id
Address 0x50007ffc
Description core ID
Reset 0x53504920

Name	id																															
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
31:0	R	id	SPI in ASCII	32'h53504920

Back to top

PSEQ

Name ctrl0
Address 0x50008000
Description signals to prepare for and to request low power state
Reset 0x00280000

Name	RSVD				req_soc_off	gpio_latch_open	ksm_latch_open	pinsel_latch_open	RSVD	wurx_detl_sel	wurx_det0_sel	block_dbg_wake	RSVD														req_hibernate_state	req_retain_all_state	RSVD	manage_xtal
Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
27	R/W	req_soc_off	If set, requests SOC to go SOC Off state.	1'b0
26	R/W	gpio_latch_open	If set, opens all latches on gpio control signals.	1'b0
25	R/W	ksm_latch_open	If set, opens all latches on ksm control signals.	1'b0
24	R/W	pinsel_latch_open	If set, opens all latches all pin select signals.	1'b0
22:21	R/W	wurx_det1_sel	Is wurx1 detect level or edge? 0=level 1=rising 2,3=falling	2'd1
20:19	R/W	wurx_det0_sel	Is wurx0 detect level or edge? 0=level 1=rising 2,3=falling	2'd1
18	R/W	block_dbg_wake	If set, debugger cannot wake chip from low power state.	1'b0
3	R/W	req_hibernate_state	If set, requests SOC to go to Hibernate low power state.	1'b0

CONFIDENTIAL

2	R/W	req_retain_all_state	If set, requests SOC to go to Retain All low power state.	1'b0
0	R/W	manage_xtal	If set, permits SOC to power down clk_mpc crystal circuit during low power mode.	1'b0

Back to top

Name retain_all_wake_mask
Address 0x50008004
Description signals to select which events can generate a wake event while in Retain All
Reset 0x00000000

Name	RSVD	watch_shub	watch_ksm	watch_qdec	watch_wurx1	watch_wurx0	watch_cntdown	watch_gpio
Bits	**31:7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
6	R/W	watch_shub	valid to wake on sensor hub trigger	1'b0
5	R/W	watch_ksm	valid to wake on keyboard activity	1'b0
4	R/W	watch_qdec	valid to wake on qdec activity	1'b0
3	R/W	watch_wurx1	valid to wake on wurx1 detect	1'b0
2	R/W	watch_wurx0	valid to wake on wurx0 detect	1'b0
1	R/W	watch_cntdown	valid to wake on count down timer expiring	1'b0
0	R/W	watch_gpio	valid to wake on gpio activity	1'b0

Back to top

Name hib_wake_mask
Address 0x50008008
Description signals to select which events can generate a wake event while in Hiberbate
Reset 0x00000000

Name	RSVD	watch_shub	watch_ksm	watch_qdec	watch_wurx1	watch_wurx0	watch_cntdown	watch_gpio
Bits	**31:7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
6	R/W	watch_shub	valid to wake on sensor hub trigger	1'b0
5	R/W	watch_ksm	valid to wake on keyboard activity	1'b0
4	R/W	watch_qdec	valid to wake on qdec activity	1'b0
3	R/W	watch_wurx1	valid to wake on wurx1 detect	1'b0
2	R/W	watch_wurx0	valid to wake on wurx0 detect	1'b0
1	R/W	watch_cntdown	valid to wake on count down timer expiring	1'b0
0	R/W	watch_gpio	valid to wake on gpio activity	1'b0

Back to top

CONFIDENTIAL

Name gpio_wake_mask
Address 0x50008014
Description mask function to specify which gpio's can generate a gpio event
Reset 0x00000000

Name	upper																lower													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:16	R/W	upper	upper 16 gpio's	16'h0
15:0	R/W	lower	lower 16 gpio's	16'h0

Back to top

Name gpio_wake_type
Address 0x50008018
Description specifies how a gpio generates an event
Reset 0x00000000

Name	upper																lower													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:16	R/W	upper	(per bit) 1 = edge sensitive , 0 = level	16'h0
15:0	R/W	lower	(per bit) 1 = edge sensitive , 0 = level	16'h0

Back to top

Name gpio_wake_pol
Address 0x5000801c
Description specifies how a gpio generates an event
Reset 0xffffffff

Name	upper																lower													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:16	R/W	upper	(per bit) 1 = level high/rising edge , 0 = level low/falling edge	16'hffff
15:0	R/W	lower	(per bit) 1 = level high/rising edge , 0 = level low/falling edge	16'hffff

Back to top

Name gpio_wake_both_edges
Address 0x50008020
Description specifies how a gpio generates an event
Reset 0x00000000

Name	upper	lower

CONFIDENTIAL

	e r	e r
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
31:16	R/W	upper	(per bit) 1 = either edge will wake; higher precedence than gpio_wake_type and gpio_wake_pol	16'h0
15:0	R/W	lower	(per bit) 1 = either edge will wake; higher precedence than gpio_wake_type and gpio_wake_pol	16'h0

Back to top

Name wurx_config
Address 0x50008024
Description wurx configuration
Reset 0x00000004

Name	RSVD	wurx_cutvdd_b	wurx_rstb	wurx_en
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3	2	1	0

Bits	SW	Name	Description	Reset
2	R/W	wurx_cutvdd_b	if low, removes power from wurx	1'b1
1	R/W	wurx_rstb	if low, holds wurx in reset	1'b0
0	R/W	wurx_en	if low, disables wurx datapath	1'b0

Back to top

Name gadc_config
Address 0x50008028
Description gadc configuration
Reset 0x00000000

Name	RSVD	gadc_cutvdd_b	gadc_rstb	gadc_en
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3	2	1	0

Bits	SW	Name	Description	Reset
2	R/W	gadc_cutvdd_b	if low, removes power from gadc	1'b0
1	R/W	gadc_rstb	if low, holds gadc in reset	1'b0
0	R/W	gadc_en	if low, disables gadc datapath	1'b0

Back to top

Name count0

CONFIDENTIAL

Address 0x5000802c
Description specifies how long power sequencer FSM will stay in states referencing count0
Reset 0x00000000

Name	RSVD																											cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count1
Address 0x50008030
Description specifies how long power sequencer FSM will stay in states referencing count1
Reset 0x00000000

Name	RSVD																											cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count2
Address 0x50008034
Description specifies how long power sequencer FSM will stay in states referencing count2
Reset 0x00000000

Name	RSVD																											cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count3
Address 0x50008038
Description specifies how long power sequencer FSM will stay in states referencing count3
Reset 0x00000000

Name	RSVD																											cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

CONFIDENTIAL

Name count4
Address 0x5000803c
Description specifies how long power sequencer FSM will stay in states referencing count4
Reset 0x00000000

Name	RSVD	cnt
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4**	**3 2 1 0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count5
Address 0x50008040
Description specifies how long power sequencer FSM will stay in states referencing count5
Reset 0x00000000

Name	RSVD	cnt
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4**	**3 2 1 0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count6
Address 0x50008044
Description specifies how long power sequencer FSM will stay in states referencing count6
Reset 0x00000000

Name	RSVD	cnt
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4**	**3 2 1 0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count7
Address 0x50008048
Description specifies how long power sequencer FSM will stay in states referencing count7
Reset 0x00000000

Name	RSVD	cnt
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4**	**3 2 1 0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

CONFIDENTIAL

Name count8
Address 0x5000804c
Description specifies how long power sequencer FSM will stay in states referencing count8
Reset 0x00000000

Name	RSVD																											cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count9
Address 0x50008050
Description specifies how long power sequencer FSM will stay in states referencing count9
Reset 0x00000000

Name	RSVD																											cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count0_radio
Address 0x50008054
Description specifies how long power sequencer FSM will stay in states referencing radio count0
Reset 0x00000000

Name	RSVD																											cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count1_radio
Address 0x50008058
Description specifies how long power sequencer FSM will stay in states referencing radio count1
Reset 0x00000000

Name	RSVD																											cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

CONFIDENTIAL

Back to top

Name count2_radio
Address 0x5000805c
Description specifies how long power sequencer FSM will stay in states referencing radio count2
Reset 0x00000000

Name	RSVD																										cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count3_radio
Address 0x50008060
Description specifies how long power sequencer FSM will stay in states referencing radio count3
Reset 0x00000000

Name	RSVD																										cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count4_radio
Address 0x50008064
Description specifies how long power sequencer FSM will stay in states referencing radio count4
Reset 0x00000004

Name	RSVD																										cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd4

Back to top

Name count5_radio
Address 0x50008068
Description specifies how long power sequencer FSM will stay in states referencing radio count5
Reset 0x00000000

Name	RSVD																										cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count6_radio
Address 0x5000806c
Description specifies how long power sequencer FSM will stay in states referencing radio count6
Reset 0x00000000

Name	RSVD																										cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count7_radio
Address 0x50008070
Description specifies how long power sequencer FSM will stay in states referencing radio count7
Reset 0x00000000

Name	RSVD																										cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count8_radio
Address 0x50008074
Description specifies how long power sequencer FSM will stay in states referencing radio count8
Reset 0x00000000

Name	RSVD																										cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name count9_radio
Address 0x50008078
Description specifies how long power sequencer FSM will stay in states referencing radio count9
Reset 0x00000000

Name	RSVD																										cnt			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
3:0	R/W	cnt	will wait N+1 32kHz clock cycles	4'd0

Back to top

Name xtal_bits0
Address 0x5000807c
Description control signals for clk_mpc crystal circuit
Reset 0x0000035a

Name	RSVD																				xocapout					xocapin					RSVD
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
10:6	R/W	xocapout	16MHz crystal tuning cap on XTALO_16M pin, typically set to same value as xocapin	5'hd
5:1	R/W	xocapin	16MHz crystal tuning cap on XTALI_16M pin, typically set to same value as xocapout	5'hd

Back to top

Name xtal_bits1
Address 0x50008080
Description control signals for clk_mpc crystal circuit
Reset 0x0016c000

Name	RSVD										extclk	clkmpc_en	clkhpc_en	doublerdcc_en	rcoscfreq				RSVD												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
21	R/W	extclk	Set to 1 if supplying external clock source on XTALI_16M pin	1'b0
20	R/W	clkmpc_en	enable root clk_mpc buffer; chip will not be operational if cleared	1'b1
19	R/W	clkhpc_en	enable root clk_hpc buffer	1'b0
18	R/W	doublerdcc_en	enable duty cycle correction for clock doubler	1'b1
17:14	R/W	rcoscfreq	16MHz RC oscillator frequency tuning, 0 - slowest, 15 - fastest	4'hB

Back to top

Name overrides
Address 0x50008084
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD																											force	force	force
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

																													osc_on_req	precision_val	precision_req
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
2	R/W	force_osc_on_req	force that clk_mpc oscillator request always be on	1'b0
1	R/W	force_precision_val	when type of precision for clk_mpc is forced, what should it be? 1=high/xtal 0=low/osc	1'b0
0	R/W	force_precision_req	force the type of precision when clk_mpc is requested	1'b0

Back to top

Name overrides2
Address 0x50008088
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	block_efuse_vddcut	block_sysrom_vddcut	RSVD	override_cache_vddcut	override_cache_vddcut_val	override_cache_clken	override_cache_clken_val	override_cache_iso	override_cache_iso_val	override_cache_nap	override_cache_nap_val	override_cache_ret	override_cache_ret_val	override_cache_psd	override_cache_psd_val	block_cache_vddcut	block_cache_psd
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18**	**17**	**16**	**15 14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
17	R/W	block_efuse_vddcut	if set, this core's FSM cannot power down the e-fuse macros	1'b0
16	R/W	block_sysrom_vddcut	if set, this core's FSM cannot power down the ROM macros	1'b0
13	R/W	override_cache_vddcut	if set, head switch on cache is now software controllable	1'b0
12	R/W	override_cache_vddcut_val	if override is set, should head switch on cache be open or close? 1=open 0=close	1'b0
11	R/W	override_cache_clken	if set, clock gating to cache is now under software control	1'b0
10	R/W	override_cache_clken_val	if override is set, should the clock gate on the cache be enabled? 1=clock running 0=clock gated off	1'b0
9	R/W	override_cache_iso	if set, software now has direct control of the cache ISO pin	1'b0
8	R/W	override_cache_iso_val	if override is set, what value should be driven onto cache ISO pin?	1'b0
7	R/W	override_cache_nap	if set, software now has direct control of the cache NAP pin	1'b0

CONFIDENTIAL

6	R/W	override_cache_nap_val	if override is set, what value should be driven onto cache NAP pin?	1'b0
5	R/W	override_cache_ret	if set, software now has direct control of the cache RET pin	1'b0
4	R/W	override_cache_ret_val	if override is set, what value should be driven onto cache RET pin?	1'b0
3	R/W	override_cache_psd	if set, software now has direct control of the cache PSD pin	1'b0
2	R/W	override_cache_psd_val	if override is set, what value should be driven onto cache PSD pin?	1'b0
1	R/W	block_cache_vddcut	if set, this core's FSM cannot power off the cache	1'b0
0	R/W	block_cache_psd	if set, this core's FSM cannot assert the cache PSD pin	1'b0

Back to top

Name overrides3
Address 0x5000808c
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	override_sysrom_psd	override_sysrom_psd_val	override_sysrom_iso	override_sysrom_iso_val	override_sysrom_clken	override_sysrom_clken_val	RSVD
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12**	**11**	**10**	**9**	**8**	**7**	**6**	**5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
11	R/W	override_sysrom_psd	if set, software now has direct control of the rom PSD pin	1'b0
10	R/W	override_sysrom_psd_val	if override is set, what value should be driven onto rom PSD pin?	1'b0
9	R/W	override_sysrom_iso	if set, software now has direct control of the rom ISO pin	1'b0
8	R/W	override_sysrom_iso_val	if override is set, what value should be driven onto rom ISO pin?	1'b0
7	R/W	override_sysrom_clken	if set, software now has direct control of the rom clock enable pin	1'b0
6	R/W	override_sysrom_clken_val	if override is set, what value should be driven onto rom clock enable pin?	1'b0

Back to top

Name overrides4
Address 0x50008090
Description override portion of pmu control signals
Reset 0x00000000

Name	override_efuse_pnr_blee_

	e_vddcut	e_vddcut_val	e_iso	e_iso_val	retn	retn_val	loadb	loadb_val	sleep	sleep_val	kill	kill_val	iso	iso_val	first	first_val	clken	clken_val	retn	retn_val	loadb	loadb_val	sleep	sleep_val	kill	kill_val	iso	iso_val	first	first_val	clken	clken_val
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R/W	override_efuse_vddcut	if set, head switch on e-fuse is now software controllable	1'b0
30	R/W	override_efuse_vddcut_val	if override is set, should head switch on e-fuse be open or close? 1=open 0=close	1'b0
29	R/W	override_efuse_iso	if set, software now has direct control of the e-fuse ISO pin	1'b0
28	R/W	override_efuse_iso_val	if override is set, what value should be driven onto e-fuse ISO pin?	1'b0
27	R/W	override_pnr_retn	if set, software now has direct control of the retain signal going to the cpu domain	1'b0
26	R/W	override_pnr_retn_val	if override is set, what value should be driven onto the cpu domain's retain signal?	1'b0
25	R/W	override_pnr_loadb	if set, software now has direct control of the load signal going to the cpu domain	1'b0
24	R/W	override_pnr_loadb_val	if override is set, what value should be driven onto the cpu domain's load signal?	1'b0
23	R/W	override_pnr_sleep	if set, software now has direct control of the sleep signal going to the cpu domain	1'b0
22	R/W	override_pnr_sleep_val	if override is set, what value should be driven onto the cpu domain's sleep signal?	1'b0
21	R/W	override_pnr_kill	if set, software now has direct control of the kill signal going to the cpu domain	1'b0
20	R/W	override_pnr_kill_val	if override is set, what value should be driven onto the cpu domain's kill signal?	1'b0
19	R/W	override_pnr_iso	if set, software now has direct control of the iso signal going to the cpu domain	1'b0
18	R/W	override_pnr_iso_val	if override is set, what value should be driven onto the cpu domain's iso signal?	1'b0
17	R/W	override_pnr_frst	if set, software now has direct control of the forced reset signal going to the cpu domain	1'b0
16	R/W	override_pnr_frst_val	if override is set, what value should be driven onto the cpu domain's forced reset signal?	1'b0
15	R/W	override_pnr_clken	if set, software now has direct control of the clock enable signal going to the cpu domain	1'b0
14	R/W	override_pnr_clken_val	if override is set, what value should be driven onto the cpu domain's clock enable signal?	1'b0
13	R/W	override_ble_retn	if set, software now has direct control of the retain signal going to the ble domain	1'b0
12	R/W	override_ble_retn_val	if override is set, what value should be driven onto the ble domain's retain signal?	1'b0
11	R/W	override_ble_loadb	if set, software now has direct control of the load signal going to the ble domain	1'b0
10	R/W	override_ble_loadb_val	if override is set, what value should be driven onto the ble domain's load signal?	1'b0
9	R/W	override_ble_sleep	if set, software now has direct control of the sleep signal going to the ble domain	1'b0
8	R/W	override_ble_sleep_val	if override is set, what value should be driven onto the ble domain's sleep signal?	1'b0
7	R/W	override_ble_kill	if set, software now has direct control of the kill signal going to the ble domain	1'b0
6	R/W	override_ble_kill_val	if override is set, what value should be driven onto the ble domain's kill signal?	1'b0
5	R/W	override_ble_iso	if set, software now has direct control of the iso signal going to the ble domain	1'b0
4	R/W	override_ble_iso_val	if override is set, what value should be driven onto the ble domain's iso signal?	1'b0
3	R/W	override_ble_frst	if set, software now has direct control of the forced reset signal going to the	1'b0

CONFIDENTIAL

			ble domain	
2	R/W	override_ble_frst_val	if override is set, what value should be driven onto the ble domain's forced reset signal?	1'b0
1	R/W	override_ble_clken	if set, software now has direct control of the clock enable signal going to the ble domain	1'b0
0	R/W	override_ble_clken_val	if override is set, what value should be driven onto the ble domain's clock enable signal?	1'b0

Back to top

Name overrides5
Address 0x50008094
Description override portion of pmu control signals
Reset 0x00000000

Name	override_xtal	override_xtal_val	override_retv	override_retv_val	override_radio_en_o	override_radio_en_o_val	override_cutvdd_b	override_cutvdd_b_val	override_radio_frst_b	override_radio_frst_b_val	override_wreq	override_wreq_val	override_radio_iso	override_radio_iso_val	override_ble_frstdp	override_ble_frstdp_val	RSVD
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
31	R/W	override_xtal	If set, software now has direct control of clk_mpc's crystal circuit enable.	1'b0
30	R/W	override_xtal_val	If override is set, what value is clk_mpc's cystal circuit enable? 0=off 1=on	1'b0
29	R/W	override_retv	If set, software now has direct control of the memory retention voltage.	1'b0
28	R/W	override_retv_val	If override is set, what value is the memory retention voltage? 0=nominal 1=ultra low	1'b0
27	R/W	override_radio_en_o	If set, software now has direct control of radio enable.	1'b0
26	R/W	override_radio_en_o_val	If override is set, what value is the radio enable? 0=off 1=on	1'b0
25	R/W	override_cutvdd_b	If set, software now has direct control of radio head switches.	1'b0
24	R/W	override_cutvdd_b_val	If override is set, what value is the radio enable? 0=head switches are open 1=head switches are closed	1'b0
23	R/W	override_radio_frst_b	If set, software now has direct control of radio forced reset .	1'b0
22	R/W	override_radio_frst_b_val	If override is set, what value is the radio forced reset? 0=forced reset is enabled 1=forced reset is disabled	1'b0
21	R/W	override_wreq	If set, software now has direct control of wake up request.	1'b0
20	R/W	override_wreq_val	If override is set, what value is value of wake up request? Useful for debug/test of pseq interrupt. 0=no wake 1=wake	1'b0
19	R/W	override_radio_iso	If set, software now has direct control of radio isolation signal .	1'b0
18	R/W	override_radio_iso_val	If override is set, what value is the radio isolation signal?	1'b0
17	R/W	override_ble_frstdp	If set, software now has direct control of ble domain's datapath forced reset.	1'b0

CONFIDENTIAL

			Not all state is retained even in Retain All state.	
16	R/W	override_ble_frstdp_val	If override is set, what value is the ble domain's datapath forced reset.	1'b0

Back to top

Name sysram_overrides
Address 0x50008098
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	override_sysram_vddcut	override_sysram_vddcut_val
Bits	**31 30 29 28 27 26 25 24**	**23 22 21 20 19 18 17 16 15 14 13 12**	**11 10 9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
23:12	R/W	override_sysram_vddcut	If set, software now has direct control of sysram's head switches. [23:18] reserved [17] 16kB macro at high address space [16] 16kB macro [15] 16kB macro [14] 16kB macro [13] 16kB macro [12] 16kB macro at lowest address space	12'b0
11:0	R/W	override_sysram_vddcut_val	If override is set, what is state of sysram's head switches? [11:6] reserved [5] 16kB macro at high address space; [1] head switch open [0] head switch close [4] 16kB macro; [1] head switch open [0] head switch close [3] 16kB macro; [1] head switch open [0] head switch close [2] 16kB macro; [1] head switch open [0] head switch close [1] 16kB macro; [1] head switch open [0] head switch close [0] 16kB macro at lowest address space; [1] head switch open [0] head switch close	12'b0

Back to top

Name sysram_overrides2
Address 0x5000809c
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	override	override

CONFIDENTIAL

									override_sysram_clken												override_sysram_clken_val										
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
23:12	R/W	override_sysram_clken	If set, software now has direct control of sysram's clock enable signals. [23:18] reserved [17] 16kB macro at high address space [16] 16kB macro [15] 16kB macro [14] 16kB macro [13] 16kB macro [12] 16kB macro at lowest address space	12'b0
11:0	R/W	override_sysram_clken_val	If override is set, what is state of sysram's clock enable signals? [11:6] reserved [5] 16kB macro at high address space [4] 16kB macro [3] 16kB macro [2] 16kB macro [1] 16kB macro [0] 16kB macro at lowest address space	12'b0

Back to top

Name sysram_overrides3
Address 0x500080a0
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD								override_sysram_iso												override_sysram_iso_val										
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
23:12	R/W	override_sysram_iso	If set, software now has direct control of sysram's isolation signals. [23:18] reserved [17] 16kB macro at high address space [16] 16kB macro [15] 16kB macro	12'b0

CONFIDENTIAL

			[14] 16kB macro [13] 16kB macro [12] 16kB macro at lowest address space	
11:0	R/W	override_sysram_iso_val	If override is set, what is state of sysram's isolation signals? [11:6] reserved [5] 16kB macro at high address space [4] 16kB macro [3] 16kB macro [2] 16kB macro [1] 16kB macro [0] 16kB macro at lowest address space	12'b0

Back to top

Name sysram_overrides4
Address 0x500080a4
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	override_sysram_nap	override_sysram_nap_val
Bits	**31 30 29 28 27 26 25 24**	**23 22 21 20 19 18 17 16 15 14 13 12**	**11 10 9 8 7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
23:12	R/W	override_sysram_nap	If set, software now has direct control of sysram's nap signals. [23:18] reserved [17] 16kB macro at high address space [16] 16kB macro [15] 16kB macro [14] 16kB macro [13] 16kB macro [12] 16kB macro at lowest address space	12'b0
11:0	R/W	override_sysram_nap_val	If override is set, what is state of sysram's nap signals? [11:6] reserved [5] 16kB macro at high address space [4] 16kB macro [3] 16kB macro [2] 16kB macro [1] 16kB macro [0] 16kB macro at lowest address space	12'b0

Back to top

Name sysram_overrides5
Address 0x500080a8
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	over	over

CONFIDENTIAL

	override_sysram_ret	override_sysram_ret_val
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
23:12	R/W	override_sysram_ret	If set, software now has direct control of sysram's retain signals. [23:18] reserved [17] 16kB macro at high address space [16] 16kB macro [15] 16kB macro [14] 16kB macro [13] 16kB macro [12] 16kB macro at lowest address space	12'b0
11:0	R/W	override_sysram_ret_val	If override is set, what is state of sysram's retain signals? [11:6] reserved [5] 16kB macro at high address space [4] 16kB macro [3] 16kB macro [2] 16kB macro [1] 16kB macro [0] 16kB macro at lowest address space	12'b0

Back to top

Name sysram_overrides6
Address 0x500080ac
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	override_sysram_psd	override_sysram_psd_val
Bits	31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
23:12	R/W	override_sysram_psd	If set, software now has direct control of sysram's psd signals. [23:18] reserved [17] 16kB macro at high address space	12'b0

CONFIDENTIAL

			[16] 16kB macro [15] 16kB macro [14] 16kB macro [13] 16kB macro [12] 16kB macro at lowest address space	
11:0	R/W	override_sysram_psd_val	If override is set, what is state of sysram's psd signals? [11:6] reserved [5] 16kB macro at high address space [4] 16kB macro [3] 16kB macro [2] 16kB macro [1] 16kB macro [0] 16kB macro at lowest address space	12'b0

Back to top

Name sysram_overrides7
Address 0x500080b0
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD																				block_sysram_vddcut											
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11:0	R/W	block_sysram_vddcut	Block core's FSM from cutting the vdd supplies to the sysram. [11:6] reserved [5] 16kB macro at high address space [4] 16kB macro [3] 16kB macro [2] 16kB macro [1] 16kB macro [0] 16kB macro at lowest address space	12'b0

Back to top

Name sysram_overrides8
Address 0x500080b4
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD																				block_sysram_p											

CONFIDENTIAL

																														s d	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11:0	R/W	block_sysram_psd	Block core's FSM from setting the psd signal to the sysram. [11:6] reserved [5] 16kB macro at high address space [4] 16kB macro [3] 16kB macro [2] 16kB macro [1] 16kB macro [0] 16kB macro at lowest address space	12'b0

Back to top

Name emram_overrides
Address 0x500080bc
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	override_emram_vddcut	RSVD	override_emram_vddcut_val
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14**	**13 12**	**11 10 9 8 7 6 5 4 3 2**	**1 0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
13:12	R/W	override_emram_vddcut	If set, software now has direct control of exchange memory vddcut signals.	2'b0
1:0	R/W	override_emram_vddcut_val	If override is set, what are the states of exchange memory vddcut signals?	2'b0

Back to top

Name emram_overrides2
Address 0x500080c0
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD	override_emra	RSVD	override_emra

																				m_clken										m_clken_val	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
13:12	R/W	override_emram_clken	If set, software now has direct control of exchange memory clock enable signals.	2'b0
1:0	R/W	override_emram_clken_val	If override is set, what are the states of exchange memory clock enable signals?	2'b0

Back to top

Name emram_overrides3
Address 0x500080c4
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD																		override_emram_iso		RSVD									override_emram_iso_val	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
13:12	R/W	override_emram_iso	If set, software now has direct control of exchange memory isolation signals.	2'b0
1:0	R/W	override_emram_iso_val	If override is set, what are the states of exchange memory isolation signals?	2'b0

Back to top

Name emram_overrides4
Address 0x500080c8
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD																		override_e		RSVD									override_e	

CONFIDENTIAL

																				ram_psd										ram_psd_val	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
13:12	R/W	override_emram_psd	If set, software now has direct control of exchange memory psd signals.	2'b0
1:0	R/W	override_emram_psd_val	If override is set, what are the states of exchange memory psd signals?	2'b0

Back to top

Name emram_overrides7
Address 0x500080d4
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD																												block_emram_vddcut	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1:0	R/W	block_emram_vddcut	If set, core's FSM cannot open the exchange memory head switches.	2'b0

Back to top

Name emram_overrides8
Address 0x500080d8
Description override portion of pmu control signals
Reset 0x00000000

Name	RSVD																												block_emram_psd	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
1:0	R/W	block_emram_psd	If set, core's FSM cannot set the exchange memory psd signals.	2'b0

Back to top

Name counter_control
Address 0x500080dc
Description control signals for counters
Reset 0x00000000

Name	RSVD	load_count_down_timer	halt_real_time_counter
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0

Bits	SW	Name	Description	Reset
1	R/W	load_count_down_timer	loads on positive edge; hardware doesn't reset this bit	1'b0
0	R/W	halt_real_time_counter	stop the real time counter from advancing	1'b0

Back to top

Name current_real_time
Address 0x500080e0
Description counter value
Reset 0x00000000

Name	val
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
31:0	R	val	real time in clk_lpc clock cycles	32'd0

Back to top

Name current_count_down_time
Address 0x500080e4
Description counter value
Reset 0x00000000

Name	val
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset

CONFIDENTIAL

31:0	R	val	this value is on 32kHz domain; read twice and check value is the same; if it's not, reread	32'd0

Back to top

Name init_count_down
Address 0x500080e8
Description counter value
Reset 0x00000fff

Name	val																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	val	initial value for count down timer	32'hfff

Back to top

Name inst_pending
Address 0x500080ec
Description instructions from clk_mpc to clk_lpc still pending
Reset 0x00000000

Name	RSVD															cntdwn_timer_load	RSVD														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
16	R	cntdwn_timer_load	load of count down timer is still pending	1'b0

Back to top

Name status
Address 0x500080f0
Description status signals
Reset 0x00000000

Name	RSVD															shub_triggered	dbg_triggered	ksm_triggered	qdec_triggered	wurxl_triggered	wurx0_triggered	gpio_triggered	timer_triggered	RSVD			in_hibernate_st	in_cpu_active_s	in_active_state
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

																														ate	tate	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15	R	shub_triggered	wake triggered by sensor hub	1'b0
14	R	dbg_triggered	wake triggered by debugger	1'b0
13	R	ksm_triggered	wake triggered by ksm activity	1'b0
12	R	qdec_triggered	wake triggered by qdec activity	1'b0
11	R	wurx1_triggered	wake triggered by wurx1	1'b0
10	R	wurx0_triggered	wake triggered by wurx0	1'b0
9	R	gpio_triggered	wake triggered by gpio	1'b0
8	R	timer_triggered	wake triggered by count down timer	1'b0
2	R	in_hibernate_state	pseq is currently in hibernate state	1'b0
1	R	in_cpu_active_state	pseq is currently in cpu active state	1'b0
0	R	in_active_state	pseq is currently in active state	1'b0

Back to top

Name persistent0
Address 0x500080f4
Description persistent scratchpad register
Reset 0x00000000

Name	data																														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	data	holds persistent state; persists through hibernation; might be of use to sw	32'h0

Back to top

Name sensor_hub_control
Address 0x500080f8
Description control signals for sensor hub
Reset 0x00000010

Name	RSVD																									shub_vddcut	shub_iso	shub_clken	shub_running_ovrd	shub_running_val	shub_reset	enable_func
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
6	R/W	shub_vddcut	sensor hub head switch control 0 = head switch is closed 1 = head switch is open	1'b0
5	R/W	shub_iso	sensor hub isolation 0 = not isolated	1'b0

CONFIDENTIAL

			1 = isolated	
4	R/W	shub_clken	sensor hub clock enable 0 = clock gated off 1 = clock running	1'b1
3	R/W	shub_running_ovrd	1= override	1'b0
2	R/W	shub_running_val	value fo shub_running when override is set	1'b0
1	R/W	shub_reset	1= keep shub in reset state; 0= release shub from reset	1'b0
0	R/W	enable_func	1= sensor hub can trigger pseq out of hibernation early due to a threshold crossing event in sensor hub. 0= sensor hub cannot trigger pseq out of hibernatio This bit does not disable the trigger signal to start the sensor functionality at the onset of hibernation.	1'b0

Back to top

Name ksmqdec_control
Address 0x500080fc
Description keyboard and mouse power domain controls
Reset 0x00000001

Name	RSVD	ksmqdec_vddcut	ksmqdec_iso	ksmqdec_frst	ksmqdec_clken
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3	R/W	ksmqdec_vddcut	head switch control 0 = head switch is closed 1 = head switch is open	1'b0
2	R/W	ksmqdec_iso	isolation 0 = not isolated 1 = isolated	1'b0
1	R/W	ksmqdec_frst	forced reset 0 = reset 1 = not reset	1'b0
0	R/W	ksmqdec_clken	clock enable 0 = clock gated off 1 = clock running	1'b1

Back to top

Name pmu_status
Address 0x50008100
Description pmu status information
Reset 0x00000000

Name	RSVD	soc_off_wkup_d	brownout	endoflife	energy4TX	energy4CPU

CONFIDENTIAL

																											e t				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
6:4	R	soc_off_wkup_det	returned from SOC Off state	3'b0
3	R	brownout	brownout detected; go to low power state as quickly as possible	1'b0
2	R	endoflife	battery is detected to be at end of life; do not start flash writes or radio operation	1'b0
1	R	energy4TX	enough energy is present to do a transmit	1'b0
0	R	energy4CPU	enough energy is present to operation the CPU	1'b0

Back to top

Name flash_control
Address 0x50008104
Description low power control of external flash needs
Reset 0x00000002

Name	RSVD	opcode	mode	rpd_has_clock	power_cycle_en
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12**	**11 10 9 8 7 6 5 4**	**3 2**	**1**	**0**

Bits	SW	Name	Description	Reset
11:4	R/W	opcode	opcode that wakes up the external flash from deep power down mode	8'b0
3:2	R/W	mode	0,3=single 1=dual 2=quad	2'b0
1	R/W	rpd_has_clock	1=flash requires opcode to wakeup, 0=flash requires only CS to wakeup	1'b1
0	R/W	power_cycle_en	1'b1=trigger command to release flash from power down when waking up, 1'b0= no action	1'b0

Back to top

Name pmu_interrupt
Address 0x50008108
Description pmu interrupt source
Reset 0x00000000

Name	RSVD	clear_fe_indicator	clear_re_indicator	intmask
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18**	**17 16 15**	**14 13 12**	**11 10 9 8 7 6 5 4 3 2 1 0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
17:15	R/W	clear_fe_indicator	clear the rising edge capture blocks	3'b0
14:12	R/W	clear_re_indicator	clear the falling edge capture blocks	3'b0
11	R/W	energy4TX_falling	0 = pmu_status_brownout level 1 = pmu_status_brownout negated level 2 = pmu_status_brownout rising edge 3 = pmu_status_brownout falling edge 4 = pmu_status_energy4CPU level 5 = pmu_status_energy4CPU negated level 6 = pmu_status_energy4CPU rising edge 7 = pmu_status_energy4CPU falling edge 8 = pmu_status_energy4TX level 9 = pmu_status_energy4TX negated level 10 = pmu_status_energy4TX rising edge 11 = pmu_status_energy4TX falling edge	12'b0
10		energy4TX_rising		
9		energy4TX_low		
8		energy4TX_high		
7		energy4CPU_falling		
6		energy4CPU_rising		
5		energy4CPU_low		
4		energy4CPU_high		
3		brownout_falling		
2		brownout_rising		
1		brownout_low		
0		brownout_high		

Back to top

Name interrupt_status
Address 0x5000810c
Description interrupt status
Reset 0x00000000

CONFIDENTIAL

Name	RSVD	returned_from_low_power_state
Bits	31:1	0

Bits	SW	Name	Description	Reset
0	R	returned_from_low_power_state	indicates system returned from low power state (as oppposed to first power on)	1'b0

Back to top

Name interrupt_mask
Address 0x50008110
Description interrupt mask
Reset 0x00000001

Name	RSVD	intr

																														pt_mask
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	intrpt_mask	if nth bit set, the nth interrupt source is allowed to propogate to core's interrupt	1'b1

Back to top

Name reset_interrupt
Address 0x50008114
Description interrupt resets
Reset 0x00000000

Name	RSVD																													intrpt_reset
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	intrpt_reset	not self clearing	1'b0

Back to top

Name core_id
Address 0x50008ffc
Description core's id
Reset 0x50534551

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	PSEQ in ASCII	32'h50534551

Back to top

GADC

Name ctrl
Address 0x5000a000
Description control signals
Reset 0x0000603c

Name	watc	aver	mode	warm	wait	enab

CONFIDENTIAL

	watch_channels																averaging_amount			mode	warmup	wait_amount								enable_dp
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:16	R/W	watch_channels	Which channels are active? 0 = Vbatt 1 = Vstor 2 = core supply 3 = temperature 4 = external port 0; differential 6 = external port 0; single ended 7 = external port 0; single ended 9 = external port 1; single ended	16'h0
15:13	R/W	averaging_amount	average 2^averaging_amount samples	3'd3
12	R/W	mode	What is the mode of operation? 0 = round robin 1 = one-shot	1'b0
11	R/W	warmup	Do 8 clocks cycle of warmup at beginning of round. No data sampled during this period.	1'b0
10:1	R/W	wait_amount	Specifies how long to wait between round robin cycles in clk_mpc cycles.	10'd30
0	R/W	enable_dp	Enables digital datapath.	1'b0

Back to top

Name datapath_parameters
Address 0x5000a004
Description offset and gain
Reset 0x00002000

Name	RSVD		offset														RSVD		gain												
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
29:16	R/W	offset	Q<9,4>	14'h0000
13:0	R/W	gain	U<1,13>	14'h2000

Back to top

Name datapath_output
Address 0x5000a008
Description a read of this register pops the data FIFO
Reset 0x00000000

Name	empty		RSVD														data														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R	empty	if 1, ignore data sample because fifo has been drained	1'b0

CONFIDENTIAL

15:0	R	data	{4b for the channel, 12b for the processed ADC sample Q<9,2>}	16'b0

Back to top

Name final_inversion
Address 0x5000a00c
Description inversion and range per channel
Reset 0x00000000

Name	gext																ch															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:16	R/W	gext	voltage range	16'h0
15:0	R/W	ch	if bit [n] is set, then invert channel n's post processed output	16'h0

Back to top

Name interrupts
Address 0x5000a010
Description interrupt status
Reset 0x00000000

Name	RSVD																													intrpt2	intrpt1	intrpt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
2	R	intrpt2	one shot or one round of round robin has completed	1'b0
1	R	intrpt1	fifo overran	1'b0
0	R	intrpt0	hardware wrote to fifo	1'b0

Back to top

Name interrupt_mask
Address 0x5000a014
Description interupt masking
Reset 0x00000007

Name	RSVD																													mask_intrpt2	mask_intrpt1	mask_intrpt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
2	R/W	mask_intrpt2	allow intrpt2 to propogate to core's single output interrupt	1'b1
1	R/W	mask_intrpt1	allow intrpt1 to propogate to core's single output interrupt	1'b1
0	R/W	mask_intrpt0	allow intrpt0 to propogate to core's single output interrupt	1'b1

CONFIDENTIAL

Back to top

Name interrupt_clear
Address 0x5000a018
Description clear interrupts
Reset 0x00000000

Name	RSVD																											clear_intrpt2	clear_intrpt1	clear_intrpt0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
2	R/W	clear_intrpt2	clear interrupt 2; not self reseting	1'b0
1	R/W	clear_intrpt1	clear interrupt 1; not self reseting	1'b0
0	R/W	clear_intrpt0	clear interrupt 0; not self reseting	1'b0

Back to top

Name core_id
Address 0x5000affc
Description core ID
Reset 0x47414443

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	GADC in ASCII	32'h47414443

Back to top

TRNG

Name control
Address 0x5000b000
Description control
Reset 0x00010010

Name	RSVD	radio_warmup_c	block_force_ra	launch_on_radi	RSVD	lsb_source	adc_capture_ed	go

CONFIDENTIAL

																	n t									d i o _ i n t o _ t r n g _ m o d e	o _ u p			g e	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
16:6	R/W	radio_warmup_cnt	specifies how long to wait for radio to warmup in clk_mpc cycles before starting the random number generation process; a value of 0 is not supported	11'h400
5	R/W	block_force_radio_into_trng_mode	the hardware fsm, when generating the random number, will force the radio to disable the RF input and to maximize all gain stages; set this bit to block; if blocked, radio will operate normally but trng will still run	1'b0
4	R/W	launch_on_radio_up	when radio is first coming back from power down mode, generate automatically (without software intervension) a new random number; this core will assert the force_radio_into_trng_mode, wait 50usec, and then start the bit gathering process	1'b1
2	R/W	lsb_source	0 = rif.rif_i[0]; 1 = radio.adc_i[0]	1'b0
1	R/W	adc_capture_edge	0 = rising edge; 1 = falling edge	1'b0
0	R/W	go	start the process of generating a true random number ; not self reseting	1'b0

Back to top

Name status
Address 0x5000b004
Description status
Reset 0x00000000

Name	R S V D	r u n n i n g
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
0	R	running	generation of random number is in progress	1'b0

Back to top

Name interrupt_status
Address 0x5000b008
Description interrupt status
Reset 0x00000000

Name	R S V D	t r n g _ t r o u b	t r n g _ r e a d y

CONFIDENTIAL

																														le	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R	trng_trouble	random number generation timed out and random number is not available	1'b0
0	R	trng_ready	random number generation completed and random number is available	1'b0

Back to top

Name interrupt_mask
Address 0x5000b00c
Description interrupt mask
Reset 0x00000003

Name	RSVD																													intrpt_mask	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1:0	R/W	intrpt_mask	if nth bit set, the nth interrupt source is allowed to propogate to core's interrupt	2'h3

Back to top

Name set_interrupt
Address 0x5000b010
Description interrupt set
Reset 0x00000000

Name	RSVD																													intrpt_set	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1:0	R/W	intrpt_set	not self clearing	2'b0

Back to top

Name reset_interrupt
Address 0x5000b014
Description interrupt clear
Reset 0x00000000

Name	RSVD																													intrpt	

CONFIDENTIAL

																															_reset
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1:0	R/W	intrpt_reset	not self clearing	2'b0

Back to top

Name trng
Address 0x5000b018
Description the random number
Reset 0x00000000

Name	val																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	val	the generated true random number	32'b0

Back to top

Name core_id
Address 0x5000bffc
Description core ID
Reset 0x54524e47

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	'TRNG' in ASCII	32'h54524e47

Back to top

RCOS_CAL

Name cal_ctrl
Address 0x5000c000
Description Contains register fields associated with cal_ctrl.
Reset 0x00000000

Name	RSVD																												hw_cal_sel	sw_cal
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	hw_cal_sel	Is cal under HW or SW control?	1'b0

CONFIDENTIAL

			1 = choose cal from HW 0 = choose cal from SW	
0	R/W	sw_cal	Rising edge starts SW cal, falling edge terminates cal if cal is not done; best to leave it on once set.	1'b0

Back to top

Name slow_clk_set
Address 0x5000c004
Description Contains register fields associated with slow_clk_set.
Reset 0x00000064

Name	RSVD																				slow_clk_set											
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11:0	R/W	slow_clk_set	Set the number of cycles to run slow clock. Counting stops when this number is reached or cal is pulled down. Constraint: 1 <= slow_clk_set <= 63. If set to 0, hardware will use 1.	12'd100

Back to top

Name slow_clk_cnt
Address 0x5000c008
Description Contains register fields associated with slow_clk_cnt.
Reset 0x00000000

Name	RSVD																				slow_clk_cnt											
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11:0	R	slow_clk_cnt	Number of cycles counted by slow clock.	12'b0

Back to top

Name fast_clk_cnt
Address 0x5000c00c
Description Contains register fields associated with fast_clk_cnt.
Reset 0x00000000

Name	RSVD	fast_c

CONFIDENTIAL

Reset 0x00000000

Name	RSVD																												breakout	go
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1	R/W	breakout	useful for testing.	1'b0
0	R/W	go	not self reseting; for software manual use.	1'b0

Back to top

Name setup1
Address 0x5000d004
Description loop time setup determines the duration of one read/write cycle. In each cycle, the sensor hub read from one sensor connected to port0 and port1 and write to flash.
Reset 0x00003fff

Name	loop_time																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	loop_time	how long should each iteration (sensor read and flash write) be ? (800 = wr_delay*2 + 16bytes read + overhead). The multiplier of 2 is needed beause when there are mulitple data bytes straddling the page boundary, the shub issues the page program twice. The first one sends all bytes up to the at the page boundary, and the second sends the rest of the data bytes.	32'h3fff

Back to top

Name port0_setup
Address 0x5000d008
Description configuration bits for Port 0
Reset 0x00000000

Name	RSVD																				trig_en	RSVD				num_sensors			RSVD		spi_n_i2c	port_en
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11	R/W	trig_en	1= enable threshold comparison with measurement results.	1'd0
6:4	R/W	num_sensors	number of sensors connected to port = num_sensors + 1	3'd0
1	R/W	spi_n_i2c	1= use spi for port0, 0= use i2c	1'b0
0	R/W	port_en	enable port0	1'b0

CONFIDENTIAL

Back to top

Name port0_sensor0_setup
Address 0x5000d00c
Description number of bytes to read from this sensor
Reset 0x00000001

Name	RSVD	num_bytes
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5	4 3 2 1 0

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

Name port0_sensor1_setup
Address 0x5000d010
Description number of bytes to read from this sensor
Reset 0x00000001

Name	RSVD	num_bytes
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5	4 3 2 1 0

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

Name port0_sensor2_setup
Address 0x5000d014
Description number of bytes to read from this sensor
Reset 0x00000001

Name	RSVD	num_bytes
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5	4 3 2 1 0

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

CONFIDENTIAL

Name port0_sensor3_setup
Address 0x5000d018
Description number of bytes to read from this sensor
Reset 0x00000001

Name	RSVD																									num_bytes				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

Name port0_sensor4_setup
Address 0x5000d01c
Description number of bytes to read from this sensor
Reset 0x00000001

Name	RSVD																									num_bytes				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

Name port0_sensor5_setup
Address 0x5000d020
Description number of bytes to read from this sensor
Reset 0x00000001

Name	RSVD																									num_bytes				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

Name port0_sensor6_setup
Address 0x5000d024
Description number of bytes to read from this sensor

CONFIDENTIAL

Reset 0x00000001

Name	RSVD	num_bytes
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5**	**4 3 2 1 0**

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

Name port0_sensor7_setup
Address 0x5000d028
Description number of bytes to read from this sensor
Reset 0x00000001

Name	RSVD	num_bytes
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5**	**4 3 2 1 0**

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

Name port0_spi_eng0_ctrl
Address 0x5000d02c
Description config register for spi interface
Reset 0x00000000

Name	RSVD	sensor_opcode
Bits	**31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8**	**7 6 5 4 3 2 1 0**

Bits	SW	Name	Description	Reset
7:0	R/W	sensor_opcode	opcode used to communicate with sensor in spi mode	8'd0

Back to top

Name port0_i2c_idw_0
Address 0x5000d030
Description id(read) of sensor0 on port0 prior to pointer address write

CONFIDENTIAL

Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idw_1
Address 0x5000d034
Description id(read) of sensor1 on port0 prior to pointer address write
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idw_2
Address 0x5000d038
Description id(read) of sensor2 on port0 prior to pointer address write
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	31 30 29 28 27 26 25	24	23	22 21	20 19	18	17	16 15 14 13 12 11 10 9	8 7 6 5 4 3 2 1	0

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

CONFIDENTIAL

Back to top

Name port0_i2c_idw_3
Address 0x5000d03c
Description id(read) of sensor3 on port0 prior to pointer address write
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idw_4
Address 0x5000d040
Description id(read) of sensor4 on port0 prior to pointer address write
Reset 0x0001fe00

Name	RSVD	ext_ack_value_	ext_master_dri	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ov

CONFIDENTIAL

								to_drive	ves_ack																						rd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idw_5
Address 0x5000d044
Description id(read) of sensor5 on port0 prior to pointer address write
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0

CONFIDENTIAL

16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idw_6
Address 0x5000d048
Description id(read) of sensor6 on port0 prior to pointer address write
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idw_7
Address 0x5000d04c
Description id(read) of sensor7 on port0 prior to pointer address write
Reset 0x0001fe00

Name	RSVD	ext_ack_	ext_mast	ext_tail	ext_head	ext_clk_	ext_data	ext_data	ext_data	ext_data

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								ext_ack_value_to_drive	ext_master_drives_ack			ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe							ext_data_o							ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_addr_0
Address 0x5000d050
Description pointer addr of sensor0 on port0
Reset 0x0001fe00

Name	RSVD							ext_ack_value_to_drive	ext_master_drives_ack	ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe							ext_data_o							ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0

CONFIDENTIAL

20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_addr_1
Address 0x5000d054
Description pointer addr of sensor1 on port0
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_addr_2
Address 0x5000d058
Description pointer addr of sensor2 on port0
Reset 0x0001fe00

Name	RS	ex	ex	ex	ex	ex	ex	ex	ex	ex

CONFIDENTIAL

	VD							t_ack_value_to_drive	t_master_drives_ack	t_tail		t_head		t_clk_pu	t_data_pu	t_data_oe								t_data_o							t_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_addr_3
Address 0x5000d05c
Description pointer addr of sensor3 on port0
Reset 0x0001fe00

Name	RSVD							ext_ack_value_to_drive	ext_master_drives_ack	ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe								ext_data_o							ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_addr_4
Address 0x5000d060
Description pointer addr of sensor4 on port0
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_addr_5
Address 0x5000d064

Description pointer addr of sensor5 on port0
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_addr_6
Address 0x5000d068
Description pointer addr of sensor6 on port0
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_dri	ext_master_drives_a	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd

CONFIDENTIAL

								ve	ck																					
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_addr_7
Address 0x5000d06c
Description pointer addr of sensor7 on port0
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

CONFIDENTIAL

Back to top

Name port0_i2c_idr_0
Address 0x5000d070
Description id(read) of sensor0 on port0 prior to data read
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idr_1
Address 0x5000d074
Description id(read) of sensor1 on port0 prior to data read
Reset 0x0001fe00

Name	RSVD	ext_ack_value_	ext_master_dri	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ov

CONFIDENTIAL

								ext_ack_value_to_drive	ext_master_drives_ack																							rd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idr_2
Address 0x5000d078
Description id(read) of sensor2 on port0 prior to data read
Reset 0x0001fe00

Name	RSVD							ext_ack_value_to_drive	ext_master_drives_ack	ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe								ext_data_o								ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0

CONFIDENTIAL

16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idr_3
Address 0x5000d07c
Description id(read) of sensor3 on port0 prior to data read
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idr_4
Address 0x5000d080
Description id(read) of sensor4 on port0 prior to data read
Reset 0x0001fe00

Name	RSVD	ext_ack_	ext_mast	ext_tail	ext_head	ext_clk_	ext_data	ext_data	ext_data	ext_data

CONFIDENTIAL

								ext_ack_value_to_drive	ext_master_drives_ack					ext_clk_pu	ext_data_pu	ext_data_oe								ext_data_o								ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idr_5
Address 0x5000d084
Description id(read) of sensor5 on port0 prior to data read
Reset 0x0001fe00

Name	RSVD							ext_ack_value_to_drive	ext_master_drives_ack	ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe								ext_data_o								ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0

CONFIDENTIAL

20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idr_6
Address 0x5000d088
Description id(read) of sensor6 on port0 prior to data read
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_idr_7
Address 0x5000d08c
Description id(read) of sensor7 on port0 prior to data read
Reset 0x0001fe00

Name	RS	ex	ex	ex	ex	ex	ex	ex	ex	ex

	VD							t_ack_value_to_drive	t_master_drives_ack	t_tail		t_head		t_clk_pu	t_data_pu	t_data_oe								t_data_o							t_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_dat_ctrl
Address 0x5000d090
Description slave(port0) data byte read (used in not the last byte)
Reset 0x0001fe00

Name	RSVD							ext_ack_value_to_drive	ext_master_drives_ack	ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe								ext_data_o							ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_i2c_dat_last_ctrl
Address 0x5000d094
Description slave(port0) data byte read (used in the last byte)
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port0_alm0_quan_ctrl
Address 0x5000d098

Description alarm0 quantity's definition
Reset 0x05500000

Name	quan_dat_is_signed	lsb_first	RSVD		quan3_size		quan2_size		quan1_size		quan0_size		quan3_pos					quan2_pos					quan1_pos					quan0_pos				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R/W	quan_dat_is_signed	quantity data is a signed data	1'b0
30	R/W	lsb_first	1= least significant byte comes in first ; 0= most significant byte comes in first	1'b0
27:26	R/W	quan3_size	number of bytes making up the quantity data = programmed quan3_size + 1; must be < 2 (3 bytes max)	2'd1
25:24	R/W	quan2_size	number of bytes making up the quantity data = programmed quan2_size + 1; must be < 2 (3 bytes max)	2'd1
23:22	R/W	quan1_size	number of bytes making up the quantity data = programmed quan1_size + 1; must be < 2 (3 bytes max)	2'd1
21:20	R/W	quan0_size	number of bytes making up the quantity data = programmed quan0_size + 1; must be < 2 (3 bytes max)	2'd1
19:15	R/W	quan3_pos	starting position of the quantity 3	5'd0
14:10	R/W	quan2_pos	starting position of the quantity 2	5'd0
9:5	R/W	quan1_pos	starting position of the quantity 1	5'd0
4:0	R/W	quan0_pos	starting position of the quantity 0	5'd0

Back to top

Name port0_alm0_trig_ctrl
Address 0x5000d0a0
Description alarm0 trigger function. trigger = trig_func_en(out_of_range(quan3), out_of_range(quan2),out_of_range(quan1),out_of_range(quan0))
Reset 0x00000000

Name	RSVD																trig_func_en															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	trig_func_en	trigger function.	16'b0

Back to top

Name port0_alm0_thrhld_max_0
Address 0x5000d0a8

CONFIDENTIAL

Description 4 bytes out of 12 used to construct reference quantities
Reset 0xffffffff

Name	byte_max_03								byte_max_02								byte_max_01								byte_max_00							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_max_03	4th byte to be contructed as a ceiling reference quantity	8'hff
23:16	R/W	byte_max_02	3rd byte to be contructed as a ceiling reference quantity	8'hff
15:8	R/W	byte_max_01	2nd byte to be contructed as a ceiling reference quantity	8'hff
7:0	R/W	byte_max_00	1st byte to be contructed as a ceiling reference quantity	8'hff

Back to top

Name port0_alm0_thrhld_max_1
Address 0x5000d0ac
Description 4 bytes out of 12 used to construct reference quantities
Reset 0xffffffff

Name	byte_max_07								byte_max_06								byte_max_05								byte_max_04							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_max_07	8th byte to be contructed as a ceiling reference quantity	8'hff
23:16	R/W	byte_max_06	7th byte to be contructed as a ceiling reference quantity	8'hff
15:8	R/W	byte_max_05	6th byte to be contructed as a ceiling reference quantity	8'hff
7:0	R/W	byte_max_04	4th byte to be contructed as a ceiling reference quantity	8'hff

Back to top

Name port0_alm0_thrhld_max_2
Address 0x5000d0b0
Description 4 bytes out of 12 used to construct reference quantities
Reset 0xffffffff

Name	byte_max_11								byte_max_10								byte_max_09								byte_max_08							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
31:24	R/W	byte_max_11	12th byte to be contructed as a ceiling reference quantity	8'hff
23:16	R/W	byte_max_10	11th byte to be contructed as a ceiling reference quantity	8'hff
15:8	R/W	byte_max_09	10th byte to be contructed as a ceiling reference quantity	8'hff
7:0	R/W	byte_max_08	9th byte to be contructed as a ceiling reference quantity	8'hff

Back to top

Name port0_alm0_thrhld_min_0
Address 0x5000d0b4
Description 4 bytes out of 12 used to construct reference quantities
Reset 0x00000000

Name	byte_min_03								byte_min_02								byte_min_01								byte_min_00							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_min_03	4th byte to be contructed as a floor reference quantity	8'h00
23:16	R/W	byte_min_02	3rd byte to be contructed as a floor reference quantity	8'h00
15:8	R/W	byte_min_01	2nd byte to be contructed as a floor reference quantity	8'h00
7:0	R/W	byte_min_00	1st byte to be contructed as a floor reference quantity	8'h00

Back to top

Name port0_alm0_thrhld_min_1
Address 0x5000d0b8
Description threshold of sensor results
Reset 0x00000000

Name	byte_min_07								byte_min_06								byte_min_05								byte_min_04							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_min_07	8th byte to be contructed as a floor reference quantity	8'h00
23:16	R/W	byte_min_06	7th byte to be contructed as a floor reference quantity	8'h00
15:8	R/W	byte_min_05	6th byte to be contructed as a floor reference quantity	8'h00
7:0	R/W	byte_min_04	4th byte to be contructed as a floor reference quantity	8'h00

Back to top

Name port0_alm0_thrhld_min_2
Address 0x5000d0bc
Description threshold of sensor results
Reset 0x00000000

CONFIDENTIAL

Name	byte_min_11								byte_min_10								byte_min_09								byte_min_08							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_min_11	12th byte to be contructed as a floor reference quantity	8'h00
23:16	R/W	byte_min_10	11th byte to be contructed as a floor reference quantity	8'h00
15:8	R/W	byte_min_09	10th byte to be contructed as a floor reference quantity	8'h00
7:0	R/W	byte_min_08	9th byte to be contructed as a floor reference quantity	8'h00

Back to top

Name port0_status
Address 0x5000d0d8
Description trigger status due to out of range quantities
Reset 0x00000000

Name	RSVD							triggered	RSVD																				out_of_range_stat			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R	triggered	triggerred by port 0.	1'b0
3:0	R	out_of_range_stat	each bit corresponds to out-of-range indicator of the sensor quantities. {quantity3,quantity2, quantity1,quantity0}	4'd0

Back to top

Name port1_setup
Address 0x5000d0dc
Description configuration bits for Port 1
Reset 0x00000000

Name	RSVD																				trig_en	RSVD							num_sensors			RSVD		spi_n_i2c	port_en
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
11	R/W	trig_en	1= enable threshold comparison with measurement results.	1'd0
6:4	R/W	num_sensors	number of sensors connected to port = num_sensors + 1	3'd0
1	R/W	spi_n_i2c	1= use spi for port0, 0= use i2c	1'b0
0	R/W	port_en	port 1 enable	1'b0

Back to top

Name port1_sensor0_setup
Address 0x5000d0e0
Description number of bytes to read from this sensor
Reset 0x00000001

Name	RSVD																									num_bytes				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:0	R/W	num_bytes	total number of bytes in sensor data = num_bytes+1	5'd1

Back to top

Name port1_spi_eng0_ctrl
Address 0x5000d0e4
Description config register for spi interface
Reset 0x00000000

Name	RSVD																								sensor_opcode							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R/W	sensor_opcode	opcode used to communicate with sensor in spi mode	8'd0

Back to top

Name port1_i2c_idw_0
Address 0x5000d0e8
Description id(read) of sensor0 on port0 prior to pointer address write
Reset 0x0001fe00

Name	RSVD	ext_ack_	ext_mast	ext_tail	ext_head	ext_clk_	ext_data	ext_data	ext_data	ext_data

CONFIDENTIAL

								ext_ack_value_to_drive	ext_master_drives_ack			ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe							ext_data_o							ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port1_i2c_addr_0
Address 0x5000d0ec
Description pointer addr of sensor0 on port0
Reset 0x0001fe00

Name	RSVD							ext_ack_value_to_drive	ext_master_drives_ack	ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe									ext_data_o							ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0

CONFIDENTIAL

20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port1_i2c_idr_0
Address 0x5000d0f0
Description id(read) of sensor0 on port0 prior to data read
Reset 0x0001fe00

Name	RSVD	ext_ack_value_to_drive	ext_master_drives_ack	ext_tail	ext_head	ext_clk_pu	ext_data_pu	ext_data_oe	ext_data_o	ext_data_ie_ovrd
Bits	**31 30 29 28 27 26 25**	**24**	**23**	**22 21**	**20 19**	**18**	**17**	**16 15 14 13 12 11 10 9**	**8 7 6 5 4 3 2 1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port1_i2c_dat_ctrl
Address 0x5000d0f4
Description slave(port0) data byte read (used in not the last byte)
Reset 0x0001fe00

Name	RS	ex	ex	ex	ex	ex	ex	ex	ex	ex

CONFIDENTIAL

	RSVD							t_ack_value_to_drive	t_master_drives_ack	t_tail		t_head		t_clk_pu	t_data_pu	t_data_oe								t_data_o							t_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port1_i2c_dat_last_ctrl
Address 0x5000d0f8
Description slave(port0) data byte read (used in the last byte)
Reset 0x0001fe00

Name	RSVD							ext_ack_value_to_drive	ext_master_drives_ack	ext_tail		ext_head		ext_clk_pu	ext_data_pu	ext_data_oe								ext_data_o							ext_data_ie_ovrd
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset

CONFIDENTIAL

24	R/W	ext_ack_value_to_drive	If ack is to be driven by master, what value to drive?	1'b0
23	R/W	ext_master_drives_ack	Specifies that master is to drive ack.	1'b0
22:21	R/W	ext_tail	At end of 8 bits plus ack/nack, what should tail be? 0 = stop 1 = stall 2 = restart	2'h0
20:19	R/W	ext_head	Before data transmission starts, what should head be? 0 = start 1 = stall	2'h0
18	R/W	ext_clk_pu	sck pin pullup enable	1'b0
17	R/W	ext_data_pu	sda pin pullup enable	1'b0
16:9	R/W	ext_data_oe	sda output enable per bit	8'hff
8:1	R/W	ext_data_o	outbound data; can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator	8'h0
0	R/W	ext_data_ie_ovrd	1 = ie is high 0 = ie is ~oe	1'b0

Back to top

Name port1_alm0_quan_ctrl
Address 0x5000d0fc
Description alarm0 quantity's definition
Reset 0x05500000

Name	quan_dat_is_signed	lsb_first	RSVD		quan3_size		quan2_size		quan1_size		quan0_size		quan3_pos					quan2_pos					quan1_pos					quan0_pos				
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
31	R/W	quan_dat_is_signed	quantity data is a signed data	1'b0
30	R/W	lsb_first	1= least significant byte comes in first ; 0= most significant byte comes in first	1'b0
27:26	R/W	quan3_size	number of bytes making up the quantity data = programmed quan3_size + 1; must be < 2 (3 bytes max)	2'd1
25:24	R/W	quan2_size	number of bytes making up the quantity data = programmed quan2_size + 1; must be < 2 (3 bytes max)	2'd1
23:22	R/W	quan1_size	number of bytes making up the quantity data = programmed quan1_size + 1; must be < 2 (3 bytes max)	2'd1
21:20	R/W	quan0_size	number of bytes making up the quantity data = programmed quan0_size + 1; must be < 2 (3 bytes max)	2'd1
19:15	R/W	quan3_pos	starting position of the quantity 3	5'd0
14:10	R/W	quan2_pos	starting position of the quantity 2	5'd0
9:5	R/W	quan1_pos	starting position of the quantity 1	5'd0
4:0	R/W	quan0_pos	starting position of the quantity 0	5'd0

Back to top

Name port1_alm0_trig_ctrl
Address 0x5000d100
Description alarm0 trigger function. trigger = trig_func_en(out_of_range(quan3), out_of_range(quan2),out_of_range(quan1),out_of_range(quan0))
Reset 0x00000000

Name	RSVD																trig_func_en															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
15:0	R/W	trig_func_en	trigger function.	16'b0

Back to top

Name port1_alm0_thrhld_max_0
Address 0x5000d104
Description 4 bytes out of 12 used to construct reference quantities
Reset 0xffffffff

Name	byte_max_03								byte_max_02								byte_max_01								byte_max_00							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_max_03	4th byte to be contructed as a ceiling reference quantity	8'hff
23:16	R/W	byte_max_02	3rd byte to be contructed as a ceiling reference quantity	8'hff
15:8	R/W	byte_max_01	2nd byte to be contructed as a ceiling reference quantity	8'hff
7:0	R/W	byte_max_00	1st byte to be contructed as a ceiling reference quantity	8'hff

Back to top

Name port1_alm0_thrhld_max_1
Address 0x5000d108
Description 4 bytes out of 12 used to construct reference quantities
Reset 0xffffffff

Name	byte_max_07								byte_max_06								byte_max_05								byte_max_04							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_max_07	8th byte to be contructed as a ceiling reference quantity	8'hff
23:16	R/W	byte_max_06	7th byte to be contructed as a ceiling reference quantity	8'hff
15:8	R/W	byte_max_05	6th byte to be contructed as a ceiling reference quantity	8'hff

CONFIDENTIAL

7:0	R/W	byte_max_04	4th byte to be contructed as a ceiling reference quantity	8'hff

Back to top

Name port1_alm0_thrhld_max_2
Address 0x5000d10c
Description 4 bytes out of 12 used to construct reference quantities
Reset 0xffffffff

Name	byte_max_11								byte_max_10								byte_max_09								byte_max_08							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_max_11	12th byte to be contructed as a ceiling reference quantity	8'hff
23:16	R/W	byte_max_10	11th byte to be contructed as a ceiling reference quantity	8'hff
15:8	R/W	byte_max_09	10th byte to be contructed as a ceiling reference quantity	8'hff
7:0	R/W	byte_max_08	9th byte to be contructed as a ceiling reference quantity	8'hff

Back to top

Name port1_alm0_thrhld_min_0
Address 0x5000d110
Description 4 bytes out of 12 used to construct reference quantities
Reset 0x00000000

Name	byte_min_03								byte_min_02								byte_min_01								byte_min_00							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_min_03	4th byte to be contructed as a floor reference quantity	8'h00
23:16	R/W	byte_min_02	3rd byte to be contructed as a floor reference quantity	8'h00
15:8	R/W	byte_min_01	2nd byte to be contructed as a floor reference quantity	8'h00
7:0	R/W	byte_min_00	1st byte to be contructed as a floor reference quantity	8'h00

Back to top

Name port1_alm0_thrhld_min_1
Address 0x5000d114
Description threshold of sensor results
Reset 0x00000000

Name	byte_m	byte_m	byte_m	byte_m

CONFIDENTIAL

	in_07								in_06								in_05								in_04							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_min_07	8th byte to be contructed as a floor reference quantity	8'h00
23:16	R/W	byte_min_06	7th byte to be contructed as a floor reference quantity	8'h00
15:8	R/W	byte_min_05	6th byte to be contructed as a floor reference quantity	8'h00
7:0	R/W	byte_min_04	4th byte to be contructed as a floor reference quantity	8'h00

Back to top

Name port1_alm0_thrhld_min_2
Address 0x5000d118
Description threshold of sensor results
Reset 0x00000000

Name	byte_min_11								byte_min_10								byte_min_09								byte_min_08							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	byte_min_11	12th byte to be contructed as a floor reference quantity	8'h00
23:16	R/W	byte_min_10	11th byte to be contructed as a floor reference quantity	8'h00
15:8	R/W	byte_min_09	10th byte to be contructed as a floor reference quantity	8'h00
7:0	R/W	byte_min_08	9th byte to be contructed as a floor reference quantity	8'h00

Back to top

Name port1_status
Address 0x5000d11c
Description trigger status due to out of range quantities
Reset 0x00000000

Name	RSVD							triggered	RSVD																				out_of_range_stat			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
24	R	triggered	triggerred by port 1.	1'b0
3:0	R	out_of_range_stat	each bit corresponds to out-of-range indicator of the sensor quantities.	4'd0

CONFIDENTIAL

			{quantity3,quantity2, quantity1,quantity0}	

Back to top

Name flash_ctrl0
Address 0x5000d120
Description flash control opcodes
Reset 0xabb90206

Name	opcode_rpd								opcode_pd								opcode_pp								opcode_wren							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:24	R/W	opcode_rpd	release_power_down (not needed for macronix)	8'hab
23:16	R/W	opcode_pd	power down opcode	8'hb9
15:8	R/W	opcode_pp	page program opcode	8'h02
7:0	R/W	opcode_wren	write_enable opcode	8'h06

Back to top

Name flash_ctrl1
Address 0x5000d124
Description start address register
Reset 0x00000000

Name	RSVD								wr_start_address																							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
23:0	R/W	wr_start_address	The starting address in the flash reserved for sensor hub's data	24'h0

Back to top

Name flash_ctrl2
Address 0x5000d128
Description end address register
Reset 0x00000000

Name	RSVD	wr_ma

CONFIDENTIAL

									x_address																						
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
23:0	R/W	wr_max_address	The last address in the flash reserved for sensor hub's data	24'h0

Back to top

Name flash_ctrl3
Address 0x5000d12c
Description miscellaneous control bits for flash controller
Reset 0x00000082

Name	RSVD																								rpd_has_clock	serialize_pp_address	clear_lost_write_cnt	clear_flash_overflow	qspi_mode		is_macronix	power_cycle_en
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
7	R/W	rpd_has_clock	for Macronix flash, rpd has no clock	1'b1
6	R/W	serialize_pp_address	1 = address is serialized in quad mode ; 0 = address bits are shifted out 4 bits per cycle in quad mode.	1'b0
5	R/W	clear_lost_write_cnt	write 1 to clear lost_write_cnt; The bit is not self-clearing.	1'b0
4	R/W	clear_flash_overflow	write 1 to clear wr_saturated; The bit is not sefl-clearing.	1'b0
3:2	R/W	qspi_mode	0,1,3 = single spi mode; 2 = qspi mode	2'b0
1	R/W	is_macronix	1= flash device from Macronix	1'b1
0	R/W	power_cycle_en	1= cycle flash into and out of deep power down mode.	1'h0

Back to top

Name flash_ctrl4
Address 0x5000d130
Description delay parameters for flash
Reset 0x00200140

Name	RSVD				pd_delay								RSVD								pp_delay											
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
27:20	R/W	pd_delay	delay cycles (pd_delay+1) after flash power down / release from power down	8'd2
11:0	R/W	pp_delay	delay cycles (pp_delay+1) after flash write. (320 cycles of 32k clock = 10ms pp time from Macronix)	12'd320

Back to top

Name flash_status0
Address 0x5000d134
Description saturation and next address pointer to the flash.
Reset 0x00000000

Name	RSVD	wr_saturated	wr_current_address
Bits	31 30 29 28 27 26 25	24	23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
24	R	wr_saturated	indicates whether allocated flash area is overflowed.	1'b0
23:0	R	wr_current_address	indicates the last flash address + 1.	24'h0

Back to top

Name flash_status1
Address 0x5000d138
Description number of lost bytes
Reset 0x00000000

Name	lost_write_cnt
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
31:0	R	lost_write_cnt	indicates the number of bytes lost due to flash address saturation.	32'h0

Back to top

Name shub_status0
Address 0x5000d13c
Description state of main FSM

CONFIDENTIAL

Reset 0x00000000

Name	RSVD																										state			
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
3:0	R	state	indicates the state of the sensor hub FSM. useful for testing	4'b0

Back to top

Name id
Address 0x5000dffc
Description core id
Reset 0x53485542

Name	id																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	SHUB in ASCII	32'h53485542

Back to top

PDM

Name control
Address 0x5000f000
Description control
Reset 0x00000008

Name	RSVD																											pdm_clk_sel		capture_on_rising	bypass_iir	enable_dp
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
4:3	R/W	pdm_clk_sel	pdm clock frequency 0 = 500kHz 1 = 1MHz 2,3 = 2MHz	2'd1
2	R/W	capture_on_rising	capture pdm data on rising edge of pdm clock	1'b0
1	R/W	bypass_iir	bypass the iir	1'b0
0	R/W	enable_dp	enable the datapath	1'b0

Back to top

CONFIDENTIAL

Name iir_config
Address 0x5000f004
Description iir configuration
Reset 0x00000005

Name	RSVD	bit_shift_amount
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4	3 2 1 0

Bits	SW	Name	Description	Reset
3:0	R/W	bit_shift_amount	specifies bit shift amount	4'd5

Back to top

Name gain_control_config
Address 0x5000f008
Description gain configuration
Reset 0x00000040

Name	RSVD	code
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7	6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
6:0	R/W	code	gain code for LUT	7'd64

Back to top

Name pcm_sample
Address 0x5000f00c
Description a read of this register pops data FIFO
Reset 0x00000000

Name	empty	RSVD	data
Bits	31	30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
31	R	empty	if 1, ignore data sample because fifo has been drained	1'b0
15:0	R	data	signed, twos complement	16'b0

Back to top

Name interrupts
Address 0x5000f014
Description interrupt status

CONFIDENTIAL

Reset 0x00000000

Name	RSVD	intrpt5	intrpt4	intrpt3	intrpt2	intrpt1	intrpt0
Bits	**31:6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R	intrpt5	error in buffer B; hw wants to write to buffer but lock hasn't been released by sw	1'b0
4	R	intrpt4	error in buffer A; hw wants to write to buffer but lock hasn't been released by sw	1'b0
3	R	intrpt3	ping pong buffer B is locked and needs to be emptied	1'b0
2	R	intrpt2	ping pong buffer A is locked and needs to be emptied	1'b0
1	R	intrpt1	fifo overran	1'b0
0	R	intrpt0	hardware wrote to fifo	1'b0

Back to top

Name interrupt_mask
Address 0x5000f018
Description interrupt mask
Reset 0x0000003f

Name	RSVD	mask_intrpt5	mask_intrpt4	mask_intrpt3	mask_intrpt2	mask_intrpt1	mask_intrpt0
Bits	**31:6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R/W	mask_intrpt5	allow intrpt5 to propogate to core's single output interrupt	1'b1
4	R/W	mask_intrpt4	allow intrpt4 to propogate to core's single output interrupt	1'b1
3	R/W	mask_intrpt3	allow intrpt3 to propogate to core's single output interrupt	1'b1
2	R/W	mask_intrpt2	allow intrpt2 to propogate to core's single output interrupt	1'b1
1	R/W	mask_intrpt1	allow intrpt1 to propogate to core's single output interrupt	1'b1
0	R/W	mask_intrpt0	allow intrpt0 to propogate to core's single output interrupt	1'b1

Back to top

Name interrupt_clear
Address 0x5000f01c
Description interrupt clear
Reset 0x00000000

Name	RSVD	clear_intrp	clear_intrp	clear_intrp	clear_intrp	clear_intrp	clear_intrp

CONFIDENTIAL

																										t5	t4	t3	t2	t1	t0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
5	R/W	clear_intrpt5	clear interrupt 5; not self reseting	1'b0
4	R/W	clear_intrpt4	clear interrupt 4; not self reseting	1'b0
3	R/W	clear_intrpt3	clear interrupt 3; not self reseting	1'b0
2	R/W	clear_intrpt2	clear interrupt 2; not self reseting	1'b0
1	R/W	clear_intrpt1	clear interrupt 1; not self reseting	1'b0
0	R/W	clear_intrpt0	clear interrupt 0; not self reseting	1'b0

Back to top

Name core_id
Address 0x5000fffc
Description core id
Reset 0x50444d20

Name	id																														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	'PDM ' in ASCII	32'h50444d20

Back to top

AT_DMA

Name opmode
Address 0x70000000
Description operation mode control register
Reset 0x00000000

Name	go	RSVD																											dat_inv	const_trans
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R/W	go	op code start when rising edge on 'go' is detected.	1'b0
1	R/W	dat_inv	0= non-inverted write ; 1= inverted write (write data are bit-wise inverted)	1'b0
0	R/W	const_trans	0= source of transfer comes from a location specified by src_addr; 1= source data is a constant value specified by const_wdata.	1'b0

Back to top

Name const_wdata
Address 0x70000004
Description constant value register
Reset 0x00000000

Name	c

CONFIDENTIAL

	const_wdata
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
31:0	R/W	const_wdata	constant value to be written to target when the operation is in const_trans mode.	32'b0

Back to top

Name src_addr
Address 0x70000008
Description source address register
Reset 0x00000000

Name	src_addr
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
31:0	R/W	src_addr	source address	32'b0

Back to top

Name tar_addr
Address 0x7000000c
Description target address register
Reset 0x00000000

Name	tar_addr
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits	SW	Name	Description	Reset
31:0	R/W	tar_addr	target address	32'b0

Back to top

Name size
Address 0x70000010
Description size of payload
Reset 0x00000000

Name	RSVD	size

CONFIDENTIAL

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits	SW	Name	Description	Reset
15:0	R/W	size	payload size in bytes.	16'b0

Back to top

Name interrupt_status
Address 0x70000014
Description interrupt status register
Reset 0x00000000

Name	RSVD	dma_err	dma_done
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0

Bits	SW	Name	Description	Reset
1	R	dma_err	interrupt status error	1'b0
0	R	dma_done	interrupt status normal	1'b0

Back to top

Name interrupt_mask
Address 0x70000018
Description interrupt mask
Reset 0x00000001

Name	RSVD	intrpt_mask
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0

Bits	SW	Name	Description	Reset
1:0	R/W	intrpt_mask	if nth bit set, the nth interrupt is enabled	2'b01

Back to top

Name set_interrupt
Address 0x7000001c
Description interrupt set force
Reset 0x00000000

Name	RSVD	intrpt_set
Bits	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0

CONFIDENTIAL

Bits	SW	Name	Description	Reset
1:0	R/W	intrpt_set	if nth bit set, the nth interrupt is set; not self reseting	2'b0

Back to top

Name reset_interrupt
Address 0x70000020
Description interrupt reset force
Reset 0x00000000

Name	RSVD																													intrpt_reset	
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
1:0	R/W	intrpt_reset	if nth bit set, the nth interrupt is reset; not self reseting	2'b0

Back to top

Name err_stat
Address 0x70000024
Description The status is cleared when the err interrupt is cleared
Reset 0x00000000

Name	RSVD																													bus_err	addr_ovlp	size_0
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
2	R	bus_err	0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'.	1'b0
1	R	addr_ovlp	0= no overlap between src and tar addr range, 1= there is an overlap	1'b0
0	R	size_0	0= size is not 0 when go is launched, 1= size is 0 when go is launched.	1'b0

Back to top

Name id
Address 0x70000ffc
Description core id
Reset 0x444d4120

Name	id																														
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	'DMA '	32'h444d4120

CONFIDENTIAL

Back to top

PMU_TOP

Name pmu0
Address 0xe1090000
Description timer_target_lsb - written by software
Reset 0x00000001

Name	timer_target_lsb																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	timer_target_lsb	written by software, target value to wake up from in soc-off mode pmu starts from 0 and counts 32KHz clock cycles until it reaches this value to come out of soc-off mode if how2wkup reg is set to 1xx 31 lsb bits	32'd1

Back to top

Name pmu1
Address 0xe1090004
Description software controlled registers
Reset 0x00000400

Name	RSVD																				lpcomp_in_sel	how2wkup			timer_target_msb							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11	R/W	lpcomp_in_sel	0: P12, 1: P11	1'd0
10	R/W	how2wkup_timer	written by software - xx1: pin, x1x: lpcomp, 1xx: timer	3'd4
9		how2wkup_lpcomp		
8		how2wkup_pin		
7:0	R/W	timer_target_msb	written by software, target value to wake up from in soc-off mode pmu starts from 0 and counts 32KHz clock cycles until it reaches this value to come out of soc-off mode if how2wkup reg is set to 1xx 8 msb bits	8'd0

CONFIDENTIAL

Back to top

Name pmu11
Address 0xe1090030
Description atb and clocks
Reset 0x00000800

Name	RSVD																				lpcomp_ref				RSVD							
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11:8	R/W	lpcomp_ref	reference select for lpcomp 25*[65,61,57,53,49,45,41,37,33,29,25,21,17,13,9,5] for i=[15:0]	4'd8

Back to top

Name core_id
Address 0xe109004c
Description Core ID
Reset 0x504d5520

Name	id																															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R	id	PMU in ASCII	32'h504D5520

Back to top

PMU_WURX

Name wurx_general
Address 0xe10a0000
Description Wake-up RX general registers
Reset 0x00000000

Name	RSVD																															wurx_select
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
0	R/W	wurx_select	which algorithm to use: 0: algorithm 0, 1: algorithm 1	1'd0

Back to top

CONFIDENTIAL

Name wurx0_0
Address 0xe10a0004
Description Wake-up RX unique ID, algorithm 0
Reset 0xaaaaaaaa

Name	wurx0_uid																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	wurx0_uid	User ID, upper 16 bits for group ID, lower 16 bits for user ID	32'hAAAAAAAA

Back to top

Name wurx0_1
Address 0xe10a0008
Description Wake-up RX global ID, algorithm 0
Reset 0xffffffff

Name	wurx0_gid																													
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	wurx0_gid	Global ID	32'hFFFFFFFF

Back to top

Name wurx0_2
Address 0xe10a000c
Description Wake-up RX settings, algorithm 0
Reset 0x0c00001f

Name	RSVD	wurx0_thr_length				wurx0_corr_length						RSVD														wurx0_err_threshold					
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

CONFIDENTIAL

Bits	SW	Name	Description	Reset
30:27	R/W	wurx0_thr_length	how many sybmols to adjust threshold on, thr_length = 1: 2 symbols	4'd1
26:21	R/W	wurx0_corr_length	number of symbols in one ID	6'd32
5:0	R/W	wurx0_err_threshold	number of bits of error between RX ID and detected ID	6'd31

Back to top

Name wurx0_3
Address 0xe10a0010
Description Wake-up RX settings, algorithm 0
Reset 0x20100c04

Name	RSVD		wurx0_clk_freq	wurx0_sym1_threshold								wurx0_energy_toohigh									wurx0_energy_toolow								
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
29	R/W	wurx0_clk_freq	0 - 16KHz, 1- 32KHz	1'b1
28:19	R/W	wurx0_sym1_threshold	number of counts above this means symbol 1 is detected	10'd2
18:9	R/W	wurx0_energy_toohigh	received energy more than this --> increase threshold	10'd6
8:0	R/W	wurx0_energy_toolow	received energy less than this --> decrease threshold	9'd4

Back to top

Name wurx0_4
Address 0xe10a0014
Description Wake-up RX settings, algorithm 0
Reset 0x0000000f

Name	RSVD																						wurx0_timer_max									
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
9:0	R/W	wurx0_timer_max	number of cycles in one symbol period - 1	10'd15

Back to top

CONFIDENTIAL

Name wurx1_0
Address 0xe10a001c
Description Wake-up RX settings, algorithm 1
Reset 0x010c8001

Name	wurxl_clk_freq	RSVD			wurxl_thr_step				wurxl_thr_max												wurxl_thr_min											
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31	R/W	wurx1_clk_freq	0 - 16KHz, 1- 32KHz	1'b0
27:24	R/W	wurx1_thr_step	step for changing threshold	4'd1
23:12	R/W	wurx1_thr_max	increase thr if energy more than this in a thr period	12'd200
11:0	R/W	wurx1_thr_min	decrease thr if energy less than this in a thr period	12'd1

Back to top

Name wurx1_1
Address 0xe10a0020
Description Wake-up RX settings, algorithm 1
Reset 0x0000010b

Name	RSVD																				wurxl_thr_timer_max											
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
11:0	R/W	wurx1_thr_timer_max	determines thr check window	12'd267

Back to top

Name wurx1_rb
Address 0xe10a0024
Description Wake-up RX read-out bits, algorithm 1
Reset 0x00000000

Name	RSVD	anal

CONFIDENTIAL

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Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
7:0	R	analog_cal	read back analog cal value --> make it common for both algorithms	8'd0

Back to top

Name wurx_spare
Address 0xe10a0028
Description Wake-up RX spare bits
Reset 0x00000000

Name																s p a r e s															
Bits	**31**	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Bits	SW	Name	Description	Reset
31:0	R/W	spares	wurx local register spare bits	32'd0

Back to top

CONFIDENTIAL



CONFIDENTIAL

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